

JESS Thermodynamic Database v8.9

Silicon

24-Jan-23

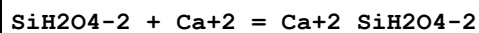
Reaction No. 1423							
H+1 + SiH2O4-2 = H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:13.2(0.05SD)	0	CMN	447
3	25	0	Inf. Dilution	lgK:13.1(3SF)	0	ENS	6405
Note (3): p. 161							
4	25	0	Inf. Dilution	lgK:13.06(4SF)	0	CRV	8740
5	25	0	Inf. Dilution	lgK:13.10(4SF)	0	CRV	9402
6	25	0	Inf. Dilution	lgK:13.45(0.07SD)	4	EOD	30201
7	25	0	UCT_Sea	lgK:13.10(0.01SD)	0	EDH	5390
8	25	0.1	UCT_Sea	lgK:12.72(0.01SD)	0	EDH	5390
9	25	0.1	Unknown	lgK:12.96(4SF)	0	CRV	552
10	25	0.2	UCT_Sea	lgK:12.65(0.01SD)	0	EDH	5390
11	25	0.3	UCT_Sea	lgK:12.61(0.01SD)	0	EDH	5390
12	25	0.4	UCT_Sea	lgK:12.58(0.01SD)	0	EDH	5390
13	25	0.5	NaCl	lgK:12.7(0.05SD)	0	MIS	430
14	25	0.5	NaClO4	lgK:12.6(0.05SD)	0	MIS	430
15	25	0.5	UCT_Sea	lgK:12.56(0.01SD)	0	EDH	5390
16	25	0.5	Unknown	lgK:12.71(4SF)	0	ENS	6405
Note (16): p. 161							
17	25	0.6	UCT_Sea	lgK:12.55(0.01SD)	0	EDH	5390
18	25	0.7	UCT_Sea	lgK:12.54(0.01SD)	0	EDH	5390
19	25	3	NaClO4	lgK:12.7(0.05SD)	0	MIS	430
20	37	0.15	Blood Plasma	lgK:11.4(0.05SD)	0	ENS	94
21	50	0	Inf. Dilution	lgK:12.6(0.05SD)	0	CMN	447
22	50	0	Inf. Dilution	lgK:12.95(0.25SD)	4	EOD	30201
23	75	0	Inf. Dilution	lgK:12.56(0.25SD)	4	EOD	30201
24	100	0	Inf. Dilution	lgK:12.28(0.25SD)	4	EOD	30201
25	0:50	0	Inf. Dilution	dH:-14.(1.SD)	0	EVH	12
26	20:50	0.5	Unknown	dH:-10.00(4SF)	0	CRV	552

27	25	0	Inf. Dilution	dH:-10.00 (4SF)	0	CRV	9402
28	25:50	0.5	Unknown	dH:-11. (1.SD)	0	EVH	430
29	25	0	Inf. Dilution	Cp:0.0 (1SF)	0	CRV	9402

Reaction No. 1424							
SiH2O4-2 + 2<H+1> = H+1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.8 (3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:23.0 (0.1SD)	4	ENS	557
3	25	0	Inf. Dilution	lgK:23.0 (3SF)	4	ENS	6496
4	25	0	Inf. Dilution	lgK:22.96 (4SF)	4	CRV	9402
5	25	0	Inf. Dilution	lgK:23.140 (0.090SD)	7	CRV	14923
6	25	0	Inf. Dilution	lgK:23.140 (0.045SD)	5	CRV	27292
7	25	0	Inf. Dilution	lgK:22.90 (4SF)	4	CRV	32224
8	25	0	CHEMVAL1	lgK:21.47 (0.01SD)	0	ENS	5156
9	25	0	CHEMVAL2	lgK:21.47 (0.01SD)	0	ENS	5156
10	25	0	CHEMVAL3	lgK:21.47 (0.01SD)	0	ENS	5156
11	25	0	CHEMVAL4	lgK:21.50 (0.01SD)	0	ENS	5156
12	25	0	UCT_Sea	lgK:22.96 (0.01SD)	4	EDH	5390
13	25	0.05	Na (Cl)	lgK:22.74 (0.06MD)	3	MHY	10238
14	25	0.1	Na (Cl)	lgK:22.55 (0.03MD)	5	MHY	10238
15	25	0.1	UCT_Sea	lgK:22.36 (0.01SD)	4	EDH	5390
16	25	0.2	Na (Cl)	lgK:22.39 (0.02MD)	6	MHY	10238
17	25	0.2	UCT_Sea	lgK:22.23 (0.01SD)	4	EDH	5390
18	25	0.3	UCT_Sea	lgK:22.14 (0.01SD)	4	EDH	5390
19	25	0.4	Na (Cl)	lgK:22.25 (0.02MD)	6	MHY	10238
20	25	0.4	UCT_Sea	lgK:22.06 (0.01SD)	4	EDH	5390
21	25	0.5	UCT_Sea	lgK:22.02 (0.01SD)	4	EDH	5390
22	25	0.6	Na (Cl)	lgK:22.07 (0.01MD)	4	MHY	10238
23	25	0.6	NaCl	lgK:22.07 (0.01SD)	4	MIS	438
24	25	0.6	NaCl	lgK:22.12 (4SF)	5	CRV	30909
25	25	0.6	UCT_Sea	lgK:22.01 (0.01SD)	4	EDH	5390
26	25	0.7	UCT_Sea	lgK:22.00 (0.01SD)	4	EDH	5390
27	25	2	Na (Cl)	lgK:22.11 (0.05MD)	6	MHY	10238
28	37	0.15	Blood Plasma	lgK:20.5 (0.05SD)	0	ENS	94
29	25	0	Inf. Dilution	dG:-132.080 (0.26SD) kJ	5	CRV	27292

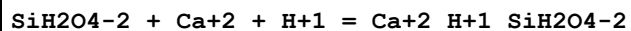
30	25	0	Inf. Dilution	dH:-17.6(3SF)	4	ENS	6496
31	25	0	Inf. Dilution	dH:-15.00(4SF)	0	CRV	9402
32	25	0	Inf. Dilution	dH:-75.000(15.000SD)kJ	6	CRV	14923
33	25	0	Inf. Dilution	dH:-75.000(7.5SD)kJ	5	CRV	27292
34	25	0	Inf. Dilution	Cp:0.0(1SF)	3	CRV	9402

Reaction No. 1425



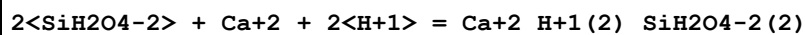
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:4.36(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:3.58(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:3.41(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:3.32(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:3.25(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:3.21(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:3.17(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:3.14(0.01SD)	4	EDH	5390
9	25	1	NaClO ₄	lgK:3.1(0.05SD)	5	MIS	455
10	37	0.15	Blood Plasma	lgK:3.0(0.05SD)	4	ENS	94

Reaction No. 1426



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.3(3SF)	1	ELC	
2	25	0	UCT_Sea	lgK:14.08(0.01SD)	0	EDH	5390
3	25	0.1	UCT_Sea	lgK:13.31(0.01SD)	0	EDH	5390
4	25	0.2	UCT_Sea	lgK:13.13(0.01SD)	0	EDH	5390
5	25	0.3	UCT_Sea	lgK:13.08(0.01SD)	0	EDH	5390
6	25	0.4	UCT_Sea	lgK:13.02(0.01SD)	0	EDH	5390
7	25	0.5	UCT_Sea	lgK:12.98(0.01SD)	0	EDH	5390
8	25	0.6	UCT_Sea	lgK:12.96(0.01SD)	0	EDH	5390
9	25	0.7	UCT_Sea	lgK:12.94(0.01SD)	0	EDH	5390
10	37	0.15	Blood Plasma	lgK:11.8(0.05SD)	0	ENS	94

Reaction No. 1427



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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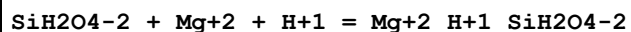
1	25	0	UCT_Sea	lgK:30.02(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:28.65(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:28.39(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:28.24(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:28.13(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:28.06(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:28.02(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:27.99(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:25.7(0.05SD)	4	ENS	94

Reaction No. 1428							
SiH2O4-2 + Fe+3 + H+1 = Fe+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:22.44(0.01SD)	4	EDH	5390
2	25	0.1	ClO4-1 salt	lgK:9.3(0.05SD)	0	MSP	5293
3	25	0.1	UCT_Sea	lgK:21.62(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:21.48(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:21.41(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:21.36(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:21.33(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:21.32(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:21.30(0.01SD)	4	EDH	5390
10	37	0.15	Blood Plasma	lgK:20.4(0.05SD)	3	ENS	94

Reaction No. 1429							
SiH2O4-2 + Mg+2 = Mg+2_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:5.32(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:4.58(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:4.43(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:4.35(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:4.30(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:4.26(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:4.23(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:4.21(0.01SD)	4	EDH	5390
9	25	1	NaClO4	lgK:4.2(0.05SD)	5	MIS	455

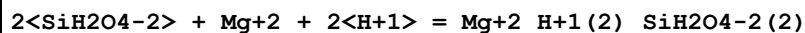
10	37	0.15	Blood Plasma	lgK:4.2(0.05SD)	4	ENS	94
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Reaction No. 1430



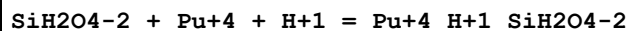
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:14.20(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:13.48(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:13.35(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:13.28(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:13.24(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:13.21(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:13.19(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:13.18(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:12.0(0.05SD)	4	ENS	94

Reaction No. 1431



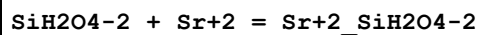
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:30.82(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:29.50(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:29.26(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:29.12(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:29.03(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:28.96(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:28.93(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:28.90(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:26.6(0.05SD)	4	ENS	94

Reaction No. 1432



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.1(3SF)	1	EES	
2	37	0.15	Blood Plasma	lgK:20.9(0.05SD)	2	ENS	94

Reaction No. 1433



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:3.1(2SF)	1	ESC	
2	37	0.15	Blood Plasma	lgK:2.9(0.05SD)	0	ENS	94

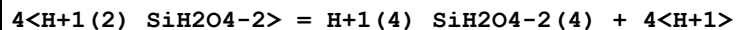
Reaction No. 2614							
H+1_SiH2O4-2 + H+1 = H+1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.8(0.05SD)	4	CMN	447
2	25	0	Inf. Dilution	lgK:9.77(0.05SD)	4	MGL	450
3	25	0	Inf. Dilution	lgK:9.86(3SF)	4	ENS	6405
4	25	0	Inf. Dilution	lgK:9.825(0.033SD)	4	CRV	6478
5	25	0	Inf. Dilution	lgK:9.83(3SF)	4	ENS	6496
6	25	0	Inf. Dilution	lgK:9.82(3SF)	4	CRV	8740
7	25	0	Inf. Dilution	lgK:9.810(0.020SD)	7	CRV	14923
8	25	0	Inf. Dilution	lgK:9.810(0.01SD)	5	CRV	27292
9	25	0	Inf. Dilution	lgK:9.68(0.14SD)	4	EOD	30201
10	25	0	Inf. Dilution	lgK:9.80(3SF)	4	CRV	32224
11	25	0	CHEMVAL1	lgK:9.77(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL2	lgK:9.77(0.01SD)	0	ENS	5156
13	25	0	CHEMVAL3	lgK:9.77(0.01SD)	0	ENS	5156
14	25	0	CHEMVAL4	lgK:9.77(0.01SD)	4	ENS	5156
15	25	0	Na(Cl)	lgK:9.84(3SF)	6	MHY	10238
16	25	0	Unknown	lgK:9.929(4SF)	3	ENS	7373
17	25	0.05	Na(Cl)	lgK:9.654(0.005MD)	6	MHY	10238
18	25	0.1	Na(Cl)	lgK:9.587(0.008MD)	6	MHY	10238
19	25	0.1	Unknown	lgK:9.58(0.01SD)	6	CRV	552
20	25	0.2	Na(Cl)	lgK:9.547(0.005MD)	6	MHY	10238
21	25	0.4	Na(Cl)	lgK:9.495(0.011MD)	6	MHY	10238
22	25	0.5	NaClO4	lgK:9.5(0.05SD)	5	MIS	430
23	25	0.5	Unknown	lgK:9.61(3SF)	5	ENS	6405
Note (23): p. 161							
24	25	0.6	Na(Cl)	lgK:9.472(0.002MD)	6	MHY	10238
25	25	0.6	NaCl	lgK:9.5(0.05SD)	5	MIS	438
26	25	0.6	NaCl	lgK:9.473(4SF)	5	CRV	30909
27	25	1	NaClO4	lgK:9.5(0.05SD)	5	MIS	449
28	25	2	Na(Cl)	lgK:9.460(0.008MD)	4	MHY	10238
29	25	2	NaCl	lgK:9.46(3SF)	3	CRV	552

30	25	3	NaCl	lgK:9.46 (3SF)	0	CRV	552
31	25	3	NaClO4	lgK:9.4 (0.05SD)	2	MIS	430
32	25	4.5	NaCl	lgK:9.54 (3SF)	0	CRV	552
33	35	0	Inf. Dilution	lgK:9.70 (0.05SD)	4	MGL	450
34	37	0.15	Blood Plasma	lgK:9.6 (2SF)	0	ENS	1086
35	50	0	Inf. Dilution	lgK:9.4 (0.05SD)	4	CMN	447
36	50	0	Inf. Dilution	lgK:9.34 (0.16SD)	4	EOD	30201
37	75	0	Inf. Dilution	lgK:9.10 (0.17SD)	4	EOD	30201
38	90	0	Inf. Dilution	lgK:9.18 (3SF)	4	CRV	8740
39	100	0	Inf. Dilution	lgK:8.94 (0.16SD)	4	EOD	30201
40	100	0	Inf. Dilution/m	lgK:9.10 (3SF)	5	MPT	4914
41	100	0.1	NaCl/m	lgK:8.80 (3SF)	5	MPT	4914
42	100	1	NaCl/m	lgK:8.52 (3SF)	4	MPT	4914
43	100	3	NaCl/m	lgK:8.53 (3SF)	5	MPT	4914
44	100	5	NaCl/m	lgK:8.64 (3SF)	5	MPT	4914
45	125	0	Inf. Dilution/m	lgK:8.98 (3SF)	5	MPT	4914
46	125	0.1	NaCl/m	lgK:8.66 (3SF)	5	MPT	4914
47	125	1	NaCl/m	lgK:8.35 (3SF)	5	MPT	4914
48	125	3	NaCl/m	lgK:8.32 (3SF)	5	MPT	4914
49	125	5	NaCl/m	lgK:8.40 (3SF)	4	MPT	4914
50	130	0	Inf. Dilution/m	lgK:8.88 (3SF)	5	MPT	4914
51	150	0	Inf. Dilution/m	lgK:8.90 (3SF)	5	MPT	4914
52	150	0.1	NaCl/m	lgK:8.55 (3SF)	5	MPT	4914
53	150	1	NaCl/m	lgK:8.20 (3SF)	5	MPT	4914
54	150	3	NaCl/m	lgK:8.13 (3SF)	5	MPT	4914
55	150	5	NaCl/m	lgK:8.18 (3SF)	4	MPT	4914
56	175	0	Inf. Dilution/m	lgK:8.85 (3SF)	5	MPT	4914
57	175	0.1	NaCl/m	lgK:8.48 (3SF)	5	MPT	4914
58	175	1	NaCl/m	lgK:8.09 (3SF)	5	MPT	4914
59	175	3	NaCl/m	lgK:7.97 (3SF)	5	MPT	4914
60	175	5	NaCl/m	lgK:7.97 (3SF)	4	MPT	4914
61	200	0	Inf. Dilution/m	lgK:8.85 (3SF)	5	MPT	4914
62	200	0.1	NaCl/m	lgK:8.45 (3SF)	5	MPT	4914
63	200	1	NaCl/m	lgK:8.00 (3SF)	5	MPT	4914
64	200	3	NaCl/m	lgK:7.82 (3SF)	5	MPT	4914
65	200	5	NaCl/m	lgK:7.77 (3SF)	5	MPT	4914

66	225	0	Inf. Dilution/m	lgK:8.89 (3SF)	5	MPT	4914
67	225	0.1	NaCl/m	lgK:8.44 (3SF)	4	MPT	4914
68	225	1	NaCl/m	lgK:7.94 (3SF)	5	MPT	4914
69	225	3	NaCl/m	lgK:7.68 (3SF)	5	MPT	4914
70	225	5	NaCl/m	lgK:7.58 (3SF)	5	MPT	4914
71	250	0	Inf. Dilution/m	lgK:8.96 (3SF)	5	MPT	4914
72	250	0.1	NaCl/m	lgK:8.47 (3SF)	3	MPT	4914
73	250	1	NaCl/m	lgK:7.88 (3SF)	5	MPT	4914
74	250	3	NaCl/m	lgK:7.55 (3SF)	5	MPT	4914
75	250	5	NaCl/m	lgK:7.39 (3SF)	5	MPT	4914
76	275	0	Inf. Dilution/m	lgK:9.07 (3SF)	5	MPT	4914
77	275	0.1	NaCl/m	lgK:8.51 (3SF)	4	MPT	4914
78	275	1	NaCl/m	lgK:7.84 (3SF)	5	MPT	4914
79	275	3	NaCl/m	lgK:7.41 (3SF)	3	MPT	4914
80	275	5	NaCl/m	lgK:7.17 (3SF)	3	MPT	4914
81	300	0	Inf. Dilution/m	lgK:9.22 (3SF)	4	MPT	4914
82	300	0.1	NaCl/m	lgK:8.57 (3SF)	4	MPT	4914
83	300	1	NaCl/m	lgK:7.78 (3SF)	0	MPT	4914
84	300	3	NaCl/m	lgK:7.24 (3SF)	0	MPT	4914
85	300	5	NaCl/m	lgK:6.92 (3SF)	0	MPT	4914
86	25	0	Inf. Dilution	dG:-55.996 (0.06SD) kJ	5	CRV	27292
87	20:50	0.5	Unknown	dH:-5. (1SF)	3	CRV	552
88	25	0	Inf. Dilution	dH:-6.12 (0.3SD)	4	CRV	6478
89	25	0	Inf. Dilution	dH:-6.12 (3SF)	4	ENS	6496
90	25	0	Inf. Dilution	dH:-25.600 (2.000SD) kJ	6	CRV	14923
91	25	0	Inf. Dilution	dH:-25.600 (1.0SD) kJ	5	CRV	27292
92	25	0	Unknown	dH:-8.936 (4SF)	0	ENS	7373

Note (92): Low wgt for internal consistency

Reaction No. 2615



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.7 (3SF)	1	EOR	
2	25	3	NaClO4	lgK:-32.5 (0.05SD)	3	MIS	430

Reaction No. 2616

$2\text{<H+1(2)_SiH2O4-2>} = \text{H+1(2)_SiH2O4-2(2)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-18.0(0.06SD)	5	MGC	5521
2	25	0.6	NaCl	lgK:-18.00(4SF)	3	CRV	30909
Note (2): Corrected misprint in source							
3	25	3	NaClO4	lgK:-18.1(0.05SD)	5	MIS	430

Reaction No. 2617							
$4\text{<H+1(2)_SiH2O4-2>} = \text{H+1(6)_SiH2O4-2(4)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.6(3SF)	1	ELC	
2	25	0.5	NaClO4	lgK:-12.6(0.05SD)	0	MIS	430
3	50	0.5	NaClO4	lgK:-12.6(0.05SD)	3	MIS	430

Reaction No. 2618							
$2\text{<H+1>} + 4\text{<H+1_SiH2O4-2>} = \text{H+1(6)_SiH2O4-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.7(0.05SD)	5	CMN	447
2	50	0	Inf. Dilution	lgK:24.5(0.05SD)	5	CMN	447

Reaction No. 2619							
$\text{SiO2(am.,s)} + 2\text{<H2O>} = \text{H+1(2)_SiH2O4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:-2.947(4SF)	0	CRV	25435
2	25	0	Inf. Dilution	lgK:-2.74(3SF)	5	CRV	403
3	25	0	Inf. Dilution	lgK:-2.89(3SF)	5	CRV	403
4	25	0	Inf. Dilution	lgK:-2.7(0.05SD)	4	ENS	557
5	25	0	Inf. Dilution	lgK:-2.71(3SF)	5	CRV	6405
Note (5): p. 245							
6	25	0	Inf. Dilution	lgK:-2.71(3SF)	4	ENS	6496
7	25	0	Inf. Dilution	lgK:-2.74(3SF)	5	CRV	8740
8	25	0	Inf. Dilution	lgK:-2.714(4SF)	0	CRV	25435
Note (8): All log Ks in source should be ln Ks							
9	25	0	Inf. Dilution	lgK:-2.714(4SF)	4	EOD	29487
10	25	0	Inf. Dilution	lgK:-2.71(3SF)	4	CRV	32224
11	25	0	CHEMVAL2	lgK:-2.74(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL3	lgK:-2.74(0.01SD)	5	ENS	5156

13	25	0	CHEMVAL4	lgK:-1.46(0.01SD)	0	ENS	5156
14	25	0	KCl/m	lgK:-2.654(4SF)	5	MSL	8747
15	25	0	KNO3/m	lgK:-2.658(4SF)	5	MSL	8747
16	25	0	LiCl/m	lgK:-2.674(4SF)	5	MSL	8747
17	25	0	LiNO3/m	lgK:-2.676(4SF)	5	MSL	8747
18	25	0	NaCl/m	lgK:-2.640(4SF)	5	MSL	8747
19	25	0	NaNO3/m	lgK:-2.662(4SF)	0	MSL	8747
20	25	0.28	KNO3/m	lgK:-2.693(4SF)	7	MSL	8747
21	25	0.35	KCl/m	lgK:-2.701(4SF)	7	MSL	8747
22	25	0.41	LiNO3/m	lgK:-2.738(4SF)	7	MSL	8747
23	25	0.51	LiCl/m	lgK:-2.742(4SF)	7	MSL	8747
24	25	0.51	NaCl/m	lgK:-2.708(4SF)	7	MSL	8747
25	25	0.51	NaNO3/m	lgK:-2.730(4SF)	8	MSL	8747
26	25	0.56	KNO3/m	lgK:-2.699(4SF)	7	MSL	8747
27	25	0.72	KCl/m	lgK:-2.714(4SF)	7	MSL	8747
28	25	0.82	LiNO3/m	lgK:-2.796(4SF)	7	MSL	8747
29	25	0.85	KNO3/m	lgK:-2.712(4SF)	7	MSL	8747
30	25	1	NaClO4	lgK:-2.9(0.05SD)	3	MSL	454
31	25	1.02	LiCl/m	lgK:-2.818(4SF)	7	MSL	8747
32	25	1.02	NaCl/m	lgK:-2.754(4SF)	7	MSL	8747
33	25	1.03	NaNO3/m	lgK:-2.767(4SF)	8	MSL	8747
34	25	1.09	KCl/m	lgK:-2.726(4SF)	7	MSL	8747
35	25	1.15	KNO3/m	lgK:-2.708(4SF)	7	MSL	8747
36	25	1.25	LiNO3/m	lgK:-2.848(4SF)	7	MSL	8747
37	25	1.46	KCl/m	lgK:-2.726(4SF)	7	MSL	8747
38	25	1.47	KNO3/m	lgK:-2.726(4SF)	7	MSL	8747
39	25	1.55	LiCl/m	lgK:-2.876(4SF)	3	MSL	8747
40	25	1.55	NaCl/m	lgK:-2.796(4SF)	7	MSL	8747
41	25	1.57	NaNO3/m	lgK:-2.807(4SF)	8	MSL	8747
42	25	1.68	LiNO3/m	lgK:-2.907(4SF)	4	MSL	8747
43	25	1.77	KNO3/m	lgK:-2.724(4SF)	7	MSL	8747
44	25	1.85	KCl/m	lgK:-2.730(4SF)	7	MSL	8747
45	25	2.09	LiCl/m	lgK:-2.959(4SF)	4	MSL	8747
46	25	2.09	NaCl/m	lgK:-2.842(4SF)	7	MSL	8747
47	25	2.1	KNO3/m	lgK:-2.738(4SF)	5	MSL	8747
48	25	2.12	NaNO3/m	lgK:-2.845(4SF)	8	MSL	8747

49	25	2.13	LiNO3/m	lgK:-2.963 (4SF)	4	MSL	8747
50	25	2.25	KCl/m	lgK:-2.742 (4SF)	5	MSL	8747
51	25	2.43	KNO3/m	lgK:-2.742 (4SF)	5	MSL	8747
52	25	2.59	LiNO3/m	lgK:-3.029 (4SF)	3	MSL	8747
53	25	2.63	LiCl/m	lgK:-3.030 (4SF)	3	MSL	8747
54	25	2.64	NaCl/m	lgK:-2.886 (4SF)	7	MSL	8747
55	25	2.66	KCl/m	lgK:-2.752 (4SF)	5	MSL	8747
56	25	2.76	NaNO3/m	lgK:-2.900 (4SF)	6	MSL	8747
57	25	2.77	KNO3/m	lgK:-2.752 (4SF)	5	MSL	8747
58	25	3.07	KCl/m	lgK:-2.767 (4SF)	3	MSL	8747
59	25	3.12	KNO3/m	lgK:-2.759 (4SF)	3	MSL	8747
60	25	3.2	LiCl/m	lgK:-3.100 (4SF)	3	MSL	8747
61	25	3.2	NaCl/m	lgK:-2.928 (4SF)	7	MSL	8747
62	25	3.36	NaNO3/m	lgK:-2.932 (4SF)	8	MSL	8747
63	25	3.5	KCl/m	lgK:-2.772 (4SF)	3	MSL	8747
64	25	3.55	LiNO3/m	lgK:-3.137 (4SF)	2	MSL	8747
65	25	3.76	KNO3/m	lgK:-2.757 (4SF)	3	MSL	8747
66	25	3.78	NaCl/m	lgK:-2.971 (4SF)	7	MSL	8747
67	25	3.94	KCl/m	lgK:-2.775 (4SF)	3	MSL	8747
68	25	4.36	LiCl/m	lgK:-3.245 (4SF)	2	MSL	8747
69	25	4.37	NaCl/m	lgK:-3.026 (4SF)	7	MSL	8747
70	25	4.57	LiNO3/m	lgK:-3.273 (4SF)	2	MSL	8747
71	25	4.61	NaNO3/m	lgK:-2.979 (4SF)	8	MSL	8747
72	25	4.81	KCl/m	lgK:-2.790 (4SF)	3	MSL	8747
73	25	4.98	NaCl/m	lgK:-3.071 (4SF)	5	MSL	8747
74	25	5.61	NaCl/m	lgK:-3.116 (4SF)	7	MSL	8747
75	25	6.12	NaNO3/m	lgK:-3.068 (4SF)	8	MSL	8747
76	25	6.14	NaCl/m	lgK:-3.170 (4SF)	7	MSL	8747
77	25	6.79	LiNO3/m	lgK:-3.491 (4SF)	7	MSL	8747
78	25	6.86	LiCl/m	lgK:-3.569 (4SF)	7	MSL	8747
79	25	8.01	LiNO3/m	lgK:-3.604 (4SF)	7	MSL	8747
80	25	8.2	LiCl/m	lgK:-3.693 (4SF)	7	MSL	8747
81	25	9.29	LiNO3/m	lgK:-3.706 (4SF)	7	MSL	8747
82	25	9.62	LiCl/m	lgK:-3.889 (4SF)	7	MSL	8747
83	25	10.65	LiNO3/m	lgK:-3.818 (4SF)	7	MSL	8747
84	25	11.11	LiCl/m	lgK:-3.955 (4SF)	7	MSL	8747

85	25	12.08	LiNO ₃ /m	lgK:-3.928 (4SF)	7	MSL	8747
86	25	12.68	LiCl/m	lgK:-4.071 (4SF)	7	MSL	8747
87	25	19.19	LiCl/m	lgK:-4.237 (4SF)	7	MSL	8747
88	35	0	Inf. Dilution	lgK:-2.64 (3SF)	5	CRV	403
89	60	0	Inf. Dilution	lgK:-2.445 (4SF)	0	CRV	25435
90	90	0	Inf. Dilution	lgK:-2.26 (3SF)	4	EOD	29487
91	100	0	Inf. Dilution	lgK:-2.202 (4SF)	0	CRV	25435
92	100	0	NaNO ₃ /m	lgK:-2.167 (4SF)	5	MSL	8747
93	100	0.51	NaNO ₃ /m	lgK:-2.153 (4SF)	6	MSL	8747
94	100	1.03	NaNO ₃ /m	lgK:-2.248 (4SF)	8	MSL	8747
95	100	2.12	NaNO ₃ /m	lgK:-2.299 (4SF)	8	MSL	8747
96	100	5.98	NaNO ₃ /m	lgK:-2.463 (4SF)	8	MSL	8747
97	150	0	Inf. Dilution	lgK:-1.971 (4SF)	0	CRV	25435
98	150	0	NaNO ₃ /m	lgK:-1.969 (4SF)	5	MSL	8747
99	150	0.51	NaNO ₃ /m	lgK:-2.008 (4SF)	8	MSL	8747
100	150	1.03	NaNO ₃ /m	lgK:-2.011 (4SF)	8	MSL	8747
101	150	2.12	NaNO ₃ /m	lgK:-2.109 (4SF)	8	MSL	8747
102	150	3.31	NaNO ₃ /m	lgK:-2.140 (4SF)	8	MSL	8747
103	150	4.6	NaNO ₃ /m	lgK:-2.203 (4SF)	8	MSL	8747
104	150	5.98	NaNO ₃ /m	lgK:-2.244 (4SF)	8	MSL	8747
105	200	0	Inf. Dilution	lgK:-1.802 (4SF)	0	CRV	25435
106	200	0	NaNO ₃ /m	lgK:-1.801 (4SF)	5	MSL	8747
107	200	0.507	NaNO ₃ /m	lgK:-1.860 (4SF)	8	MSL	8747
108	200	1.027	NaNO ₃ /m	lgK:-1.854 (4SF)	8	MSL	8747
109	200	2.124	NaNO ₃ /m	lgK:-1.903 (4SF)	8	MSL	8747
110	200	3.309	NaNO ₃ /m	lgK:-1.943 (4SF)	8	MSL	8747
111	200	4.6	NaNO ₃ /m	lgK:-1.987 (4SF)	8	MSL	8747
112	200	5.98	NaNO ₃ /m	lgK:-2.036 (4SF)	8	MSL	8747
113	250	0	Inf. Dilution	lgK:-1.684 (4SF)	0	CRV	25435
114	250	0	NaNO ₃ /m	lgK:-1.674 (4SF)	5	MSL	8747
115	250	0.507	NaNO ₃ /m	lgK:-1.690 (4SF)	8	MSL	8747
116	250	1.027	NaNO ₃ /m	lgK:-1.706 (4SF)	8	MSL	8747
117	250	2.124	NaNO ₃ /m	lgK:-1.740 (4SF)	8	MSL	8747
118	250	3.309	NaNO ₃ /m	lgK:-1.762 (4SF)	8	MSL	8747
119	250	4.6	NaNO ₃ /m	lgK:-1.812 (4SF)	8	MSL	8747
120	250	5.98	NaNO ₃ /m	lgK:-1.833 (4SF)	8	MSL	8747

121	300	0	Inf. Dilution	lgK:-1.605 (4SF)	0	CRV	25435
122	300	0	NaNO3/m	lgK:-1.570 (4SF)	5	MSL	8747
123	300	0.507	NaNO3/m	lgK:-1.580 (4SF)	8	MSL	8747
124	300	1.027	NaNO3/m	lgK:-1.600 (4SF)	8	MSL	8747
125	300	2.124	NaNO3/m	lgK:-1.600 (4SF)	8	MSL	8747
126	300	3.309	NaNO3/m	lgK:-1.636 (4SF)	8	MSL	8747
127	300	5.98	NaNO3/m	lgK:-1.708 (4SF)	8	MSL	8747
128	25	0	Inf. Dilution	dH:3.34 (3SF)	4	ENS	6496
129	25	0	Inf. Dilution	dH:14.594 (5SF) kJ	4	EOD	29487

Reaction No. 2796							
Ni+2 + H+1_SiH2O4-2 = Ni+2_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	NaClO4	lgK:5.73 (0.03SD)	3	MSE	29656
2	25	1	NaClO4	lgK:5.34 (0.03SD)	3	MSE	29656

Reaction No. 3025							
H+1(2)_SiH2O4-2 + Mn+2 = MnO.SiO2(s) + 2<H+1> + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.3 (0.05SD)	3	ENS	557
2	25	0	Inf. Dilution	lgK:-9.72 (3SF)	3	CRV	32224

Reaction No. 9803							
H+1(2)_SiH2O4-2 + 3<H+1_Tropolone-1> + H+1 = Si+4_Tropolone-1(3) + 4<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:7.1 (0.03SD)	4	MGL	1134

Reaction No. 17164							
H+1(2)_SiH2O4-2 - 2<H+1> + 3<H+1(2)_Cat-2> = Si+4_Cat-2(3) + 4<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.0 (3SF)	5	MSL	8740
2	25	0	Inf. Dilution	lgK:-11.5 (3SF)	4	EOD	8740
3	25	0	Unknown	lgK:-11.32 (4SF)	3	ENS	8687
4	25	0.1	Unknown	lgK:-10.89 (4SF)	3	ENS	8687
5	25	0.1	Unknown	lgK:-12.42 (4SF)	0	MMR	8688
Note (5): Paper argues against other values but unconvincingly (NMR,SiO2 sat)							
6	25	0.6	NaCl	lgK:-10.44 (0.03MD)	6	MPT	1364

Reaction No. 26572							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{Dopamine-2}> = H+1(9)_{Dopamine-2(3)_{SiH2O4-2} + 2<H+1 >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-9.70(0.03MD)	6	MGL	2279

Reaction No. 26573							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{LDopa-3}> = H+1(9)_{SiH2O4-2_{LDopa-3(3)} + 2<H+1 >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-10.08(0.05MD)	6	MGL	2279

Reaction No. 26574							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{Dopamine-2}> = H+1(8)_{Dopamine-2(3)_{SiH2O4-2} + 3<H+1 >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-19.33(0.03MD)	6	MGL	2279

Reaction No. 26575							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{LDopa-3}> = H+1(8)_{SiH2O4-2_{LDopa-3(3)} + 3<H+1 >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-19.35(0.07MD)	6	MGL	2279

Reaction No. 29643							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{Pyrogallol-3}> = \text{Si+4}_{\text{H+1(3)_{Pyrogallol-3(3)}} + 2<H+1> + 4<H2O >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-10.02(0.01SD)	6	MMR	1845

Reaction No. 29644							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(3)_{34DHB-3}> = \text{Si+4}_{34DHB-3(3)} + 5<H+1> + 4<H2O >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-21.9(0.05SD)	6	MMR	1845

Reaction No. 29645							
$\text{H+1(2)_{SiH2O4-2} + 3<H+1(4)_{Gallic-4}> = \text{Si+4}_{\text{H+1(3)_{Gallic-4(3)}} + 5<H+1> + 4<H2O >}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-21.3(0.03SD)	6	MMR	1845

Reaction No. 47006							
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H+1 + Ca+2 + Citric-3 + SiH2O4-2 = Ca+2_H+1_SiH2O4-2_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.15(0.01SD)	3	ETS	94

Reaction No. 47007							
H+1 + Ca+2 + Lactic-1 + SiH2O4-2 = Ca+2_H+1_Lactic-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.15(0.01SD)	3	ETS	94

Reaction No. 47008							
H+1 + Ca+2 + Malic-2 + SiH2O4-2 = Ca+2_H+1_Malic-2_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.35(0.01SD)	3	ETS	94

Reaction No. 47009							
H+1 + Ca+2 + Oxalic-2 + SiH2O4-2 = Ca+2_H+1_Oxalic-2_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.35(0.01SD)	3	ETS	94

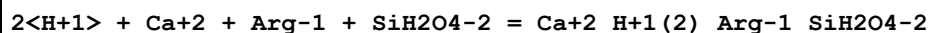
Reaction No. 47010							
H+1 + Ca+2 + Succinic-2 + SiH2O4-2 = Ca+2_H+1_SiH2O4-2_Succinic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.30(0.01SD)	3	ETS	94

Reaction No. 47011							
2<H+1> + Ca+2 + Ala-1 + SiH2O4-2 = Ca+2_H+1(2)_Ala-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.10(0.01SD)	3	ETS	94

Reaction No. 47012							
2<H+1> + Ca+2 + bAla-1 + SiH2O4-2 = Ca+2_H+1(2)_bAla-1_SiH2O4-2							

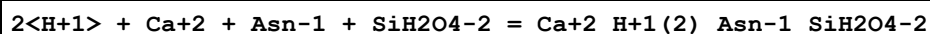
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.4 (3SF)	1	EAE	
2	37	0.15	Blood Plasma	lgK:22.75 (0.01SD)	3	ETS	94

Reaction No. 47013



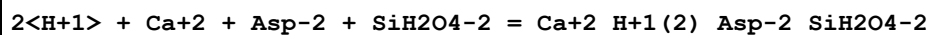
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.7 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.55 (0.01SD)	3	ETS	94

Reaction No. 47014



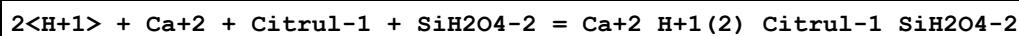
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.35 (0.01SD)	3	ETS	94

Reaction No. 47015



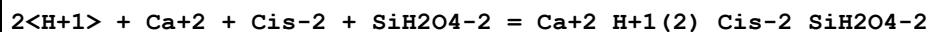
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.00 (0.01SD)	3	ETS	94

Reaction No. 47016



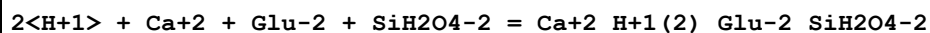
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.29 (0.01SD)	3	ETS	94

Reaction No. 47017



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.6 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.90 (0.01SD)	3	ETS	94

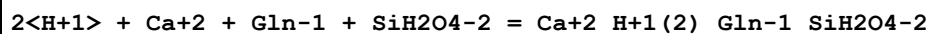
Reaction No. 47018



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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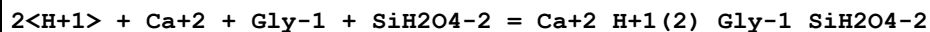
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.20(0.01SD)	3	ETS	94

Reaction No. 47019



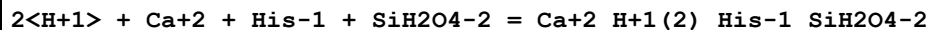
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.35(0.01SD)	3	ETS	94

Reaction No. 47020



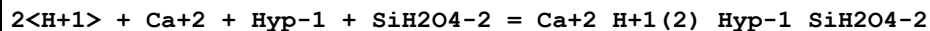
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.05(0.01SD)	3	ETS	94

Reaction No. 47021



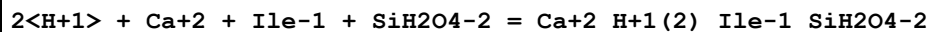
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.60(0.01SD)	3	ETS	94

Reaction No. 47022



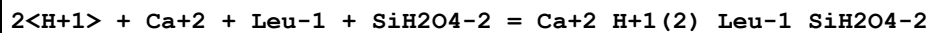
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.85(0.01SD)	3	ETS	94

Reaction No. 47023



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.94(0.01SD)	3	ETS	94

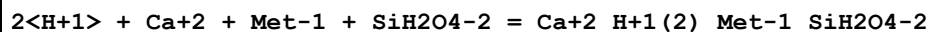
Reaction No. 47024



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(3SF)	2	EAE	

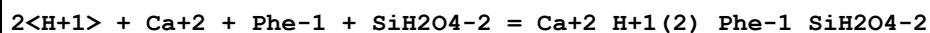
2	37	0.15	Blood Plasma	lgK:22.94 (0.01SD)	3	ETS	94
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Reaction No. 47025



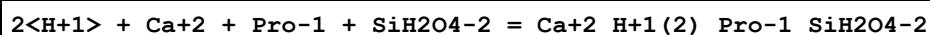
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.7 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.50 (0.01SD)	3	ETS	94

Reaction No. 47026



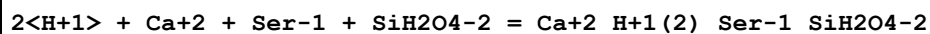
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.44 (0.01SD)	3	ETS	94

Reaction No. 47027



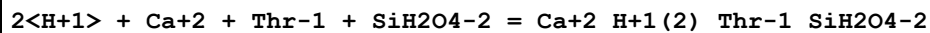
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.2 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:24.05 (0.01SD)	3	ETS	94

Reaction No. 47028



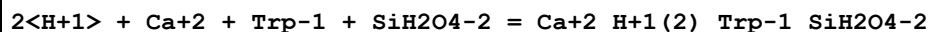
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.44 (0.01SD)	3	ETS	94

Reaction No. 47029



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.40 (0.01SD)	3	ETS	94

Reaction No. 47030



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.9 (3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.70 (0.01SD)	3	ETS	94

Reaction No. 47031							
$2\text{<H+1>} + \text{Ca}^{+2} + \text{Val}^{-1} + \text{SiH}_2\text{O}_4^{-2} = \text{Ca}^{+2}_{\text{H}+1(2)}_{\text{Val}^{-1}}\text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.85(0.01SD)	3	ETS	94

Reaction No. 47602							
$\text{H}^{+1} + \text{Mg}^{+2} + \text{Citric}^{-3} + \text{SiH}_2\text{O}_4^{-2} = \text{Mg}^{+2}_{\text{H}+1}\text{SiH}_2\text{O}_4^{-2}_{\text{Citric}^{-3}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.66(0.01SD)	3	ETS	94

Reaction No. 47603							
$\text{H}^{+1} + \text{Mg}^{+2} + \text{Lactic}^{-1} + \text{SiH}_2\text{O}_4^{-2} = \text{Mg}^{+2}_{\text{H}+1}\text{Lactic}^{-1}\text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.51(0.01SD)	3	ETS	94

Reaction No. 47604							
$\text{H}^{+1} + \text{Mg}^{+2} + \text{Malic}^{-2} + \text{SiH}_2\text{O}_4^{-2} = \text{Mg}^{+2}_{\text{H}+1}\text{Malic}^{-2}\text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.61(0.01SD)	3	ETS	94

Reaction No. 47605							
$\text{H}^{+1} + \text{Mg}^{+2} + \text{Oxalic}^{-2} + \text{SiH}_2\text{O}_4^{-2} = \text{Mg}^{+2}_{\text{H}+1}\text{Oxalic}^{-2}\text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.51(0.01SD)	3	ETS	94

Reaction No. 47606							
$\text{H}^{+1} + \text{Mg}^{+2} + \text{Succinic}^{-2} + \text{SiH}_2\text{O}_4^{-2} = \text{Mg}^{+2}_{\text{H}+1}\text{SiH}_2\text{O}_4^{-2}_{\text{Succinic}^{-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.1(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.66(0.01SD)	3	ETS	94

Reaction No. 47607							
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2<H+1> + Mg+2 + Ala-1 + SiH2O4-2 = Mg+2_H+1(2)_Ala-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.71(0.01SD)	3	ETS	94

Reaction No. 47608							
2<H+1> + Mg+2 + bAla-1 + SiH2O4-2 = Mg+2_H+1(2)_bAla-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	1	EAE	
2	37	0.15	Blood Plasma	lgK:23.36(0.01SD)	3	ETS	94

Reaction No. 47609							
2<H+1> + Mg+2 + Arg-1 + SiH2O4-2 = Mg+2_H+1(2)_Arg-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.11(0.01SD)	3	ETS	94

Reaction No. 47610							
2<H+1> + Mg+2 + Asn-1 + SiH2O4-2 = Mg+2_H+1(2)_Asn-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.96(0.01SD)	3	ETS	94

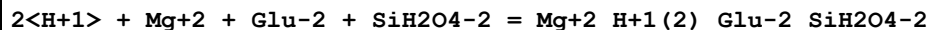
Reaction No. 47611							
2<H+1> + Mg+2 + Asp-2 + SiH2O4-2 = Mg+2_H+1(2)_Asp-2_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47612							
2<H+1> + Mg+2 + Citrul-1 + SiH2O4-2 = Mg+2_H+1(2)_Citrul-1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.01(0.01SD)	3	ETS	94

Reaction No. 47613							
2<H+1> + Mg+2 + Cis-2 + SiH2O4-2 = Mg+2_H+1(2)_Cis-2_SiH2O4-2							

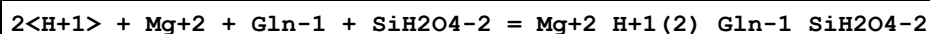
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.61(0.01SD)	3	ETS	94

Reaction No. 47614



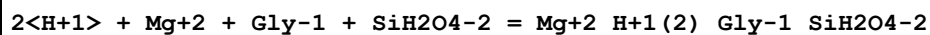
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.4(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.76(0.01SD)	3	ETS	94

Reaction No. 47615



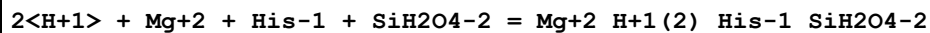
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.06(0.01SD)	3	ETS	94

Reaction No. 47616



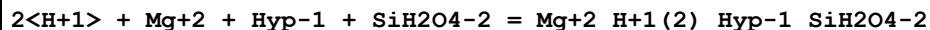
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.76(0.01SD)	3	ETS	94

Reaction No. 47617



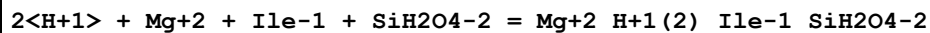
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.26(0.01SD)	3	ETS	94

Reaction No. 47618



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.51(0.01SD)	3	ETS	94

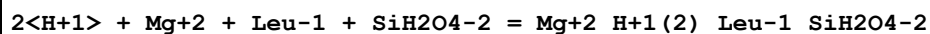
Reaction No. 47619



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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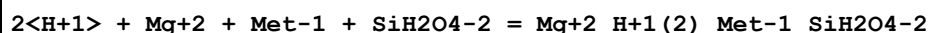
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47620



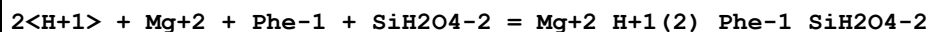
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47621



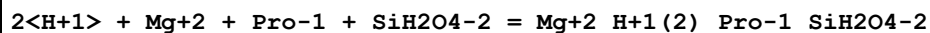
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.4(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.16(0.01SD)	3	ETS	94

Reaction No. 47622



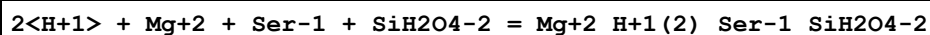
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.11(0.01SD)	3	ETS	94

Reaction No. 47623



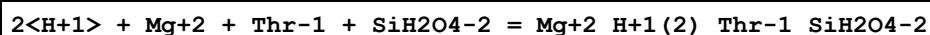
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:24.71(0.01SD)	3	ETS	94

Reaction No. 47624



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.06(0.01SD)	3	ETS	94

Reaction No. 47625



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	

2	37	0.15	Blood Plasma	lgK:22.96(0.01SD)	3	ETS	94
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Reaction No. 47626							
$2\text{<H+1>} + \text{Mg+2} + \text{Trp-1} + \text{SiH2O4-2} = \text{Mg+2_H+1(2)_Trp-1_SiH2O4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.36(0.01SD)	3	ETS	94

Reaction No. 47627							
$2\text{<H+1>} + \text{Mg+2} + \text{Val-1} + \text{SiH2O4-2} = \text{Mg+2_H+1(2)_Val-1_SiH2O4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.51(0.01SD)	3	ETS	94

Reaction No. 49887							
$\text{CaO.Al2O3.2SiO2(s)} + 8\text{<H2O>} = \text{Ca+2} + 2\text{<Al+3_OH-1(4)>} + 2\text{<H+1(2)_SiH2O4-2>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-21.071(5SF)	0	CRV	8798
2	25	0	Inf. Dilution	lgK:-19.1(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-20.484(5SF)	0	CRV	8798
4	25	0	Inf. Dilution	lgK:-54.579(5SF)	0	CRV	25435
Note (4): Given as log K, v. prob. should be ln K but -54.6/2.3 = -23.7 still bad							
5	50	0	Inf. Dilution	lgK:-19.857(5SF)	0	CRV	8798
6	100	0	Inf. Dilution	lgK:-18.960(5SF)	0	CRV	8798
7	150	0	Inf. Dilution	lgK:-18.514(5SF)	0	CRV	8798
8	200	0	Inf. Dilution	lgK:-18.485(5SF)	0	CRV	8798
9	250	0	Inf. Dilution	lgK:-18.839(5SF)	0	CRV	8798
10	300	0	Inf. Dilution	lgK:-19.547(5SF)	0	CRV	8798
11	350	0	Inf. Dilution	lgK:-20.586(5SF)	0	CRV	8798

Reaction No. 49888							
$\text{KAlSi3O8(San.,s)} + 8\text{<H2O>} = \text{K+1} + \text{Al+3_OH-1(4)} + 3\text{<H+1(2)_SiH2O4-2>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-23.764(5SF)	7	CRV	8798
2	25	0	Inf. Dilution	lgK:-22.338(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-20.774(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-18.267(5SF)	7	CRV	8798

5	150	0	Inf. Dilution	lgK:-16.441(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-15.166(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-14.351(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-13.930(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-13.855(5SF)	7	CRV	8798

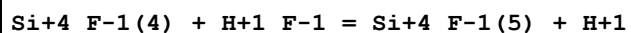
Reaction No. 49901							
KAlSi3O8(Micr.,s) + 8<H2O> = K+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-24.446(5SF)	0	CRV	8798
2	25	0	Inf. Dilution	lgK:-21.6(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-22.912(5SF)	0	CRV	8798
4	50	0	Inf. Dilution	lgK:-21.231(5SF)	0	CRV	8798
5	100	0	Inf. Dilution	lgK:-18.535(5SF)	2	CRV	8798
6	150	0	Inf. Dilution	lgK:-16.562(5SF)	0	CRV	8798
7	200	0	Inf. Dilution	lgK:-15.170(5SF)	0	CRV	8798
8	250	0	Inf. Dilution	lgK:-14.260(5SF)	0	CRV	8798
9	300	0	Inf. Dilution	lgK:-13.759(5SF)	0	CRV	8798
10	350	0	Inf. Dilution	lgK:-13.616(5SF)	0	CRV	8798

Reaction No. 50248							
NaAlSi3O8(AlbiteHI,s) + 8<H2O> = Na+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-19.816(5SF)	7	CRV	8798
2	25	0	Inf. Dilution	lgK:-18.779(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-17.622(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-15.737(5SF)	7	CRV	8798
5	150	0	Inf. Dilution	lgK:-14.359(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-13.417(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-12.858(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-12.640(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-12.730(5SF)	7	CRV	8798

Reaction No. 50272							
NaAlSi3O8(AlbiteLO,s) + 8<H2O> = Na+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-21.337(5SF)	7	CRV	8798

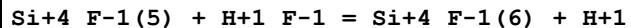
2	25	0	Inf. Dilution	lgK:-20.179(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-18.848(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-16.686(5SF)	7	CRV	8798
5	150	0	Inf. Dilution	lgK:-15.096(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-13.987(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-13.293(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-12.963(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-12.959(5SF)	7	CRV	8798

Reaction No. 53218



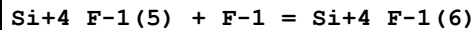
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10			lgK:2.46(3SF)	3	MMR	5398
2	25			lgK:2.40(3SF)	3	MMR	5398
3	40			lgK:2.35(3SF)	3	MMR	5398
4	54			lgK:2.30(3SF)	3	MMR	5398
5	10:54			dH:-1.5(2SF)	3	MTD	5398

Reaction No. 53219



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.4	LiClO4	lgK:2.10(3SF)	5	MKN	5398
2	10			lgK:1.92(3SF)	3	MMR	5398
3	25	0.4	LiClO4	lgK:1.89(3SF)	5	MKN	5398
4	25			lgK:1.60(3SF)	3	MMR	5398
5	40			lgK:1.33(3SF)	3	MMR	5398
6	54			lgK:1.10(3SF)	3	MMR	5398
7	10:54			dH:-7.8(2SF)	3	MTD	5398

Reaction No. 54154



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	HCl	lgK:3.96(3SF)	0	MAX	931
2	25	0	Inf. Dilution	lgK:4.5(2SF)	1	ELC	

Reaction No. 54549

SiO2(Quartz,s) + 2<H2O> = H+1(2)_SiH2O4-2							
Note: Quartz solubility at low temperature has been well examined by Rimstidt, GCA, 1997, 61, 2553 [9033]. His results indicate that the solubility at 25C is 10.8 ppm, about 80% higher than the widely accepted value of 6 ppm due to Fournier & Potter, GCA, 1982, 46, 1969 [9045] and Walther & Helgeson, 1979, 279, 1078 [9046]. The data are thus being modified (gradually) to reflect this.							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.0(0.05SD)	0	ENS	440
Note (1): See reaction comment							
2	25	0	Inf. Dilution	lgK:-4.00(3SF)	0	CRV	6405
Note (2): p. 245							
3	25	0	Inf. Dilution	lgK:-4.0(0.1SD)	0	CRV	6478
4	25	0	Inf. Dilution	lgK:-3.98(3SF)	0	ENS	6496
5	25	0	Inf. Dilution	lgK:-3.72(3SF)	7	MSL	9033
Note (5): See reaction comment							
6	25	0	Inf. Dilution	lgK:-4.000(0.100SD)	0	CRV	14923
7	25	0	Inf. Dilution	lgK:-4.01(3SF)	0	EOD	23102
8	25	0	Inf. Dilution	lgK:-4.000(0.05SD)	0	CRV	27292
9	25	0	Inf. Dilution	lgK:-3.746(4SF)	4	EOD	29487
10	25	0	Inf. Dilution	lgK:-4.18(3SF)	0	CRV	32224
11	25	0	CHEMVAL1	lgK:-3.98(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL2	lgK:-3.98(0.01SD)	0	ENS	5156
13	25	0	CHEMVAL3	lgK:-3.98(0.01SD)	0	ENS	5156
14	25	0	CHEMVAL4	lgK:-3.95(0.01SD)	0	ENS	5156
Note (14): Times are achanging							
15	90	0	Inf. Dilution	lgK:-3.10(3SF)	4	EOD	29487
16	25	0	Inf. Dilution	dG:22.36(4SF)kJ	3	CRV	6421
Note (16): Cited by Von Damm et al, Am. J. Sci., 1991, 291, 977 (p. 997) [8869]							
17	25	0	Inf. Dilution	dG:16.14(4SF)kJ	0	ENS	8869
Note (17): Successful prediction by the density model but not useful data							
18	25	0	Inf. Dilution	dG:22.832(0.3SD)kJ	0	CRV	27292
19	25	0	Inf. Dilution	dH:7.0(0.5SD)	0	CRV	6478
20	25	0	Inf. Dilution	dH:5.99(3SF)	4	ENS	6496
21	25	0	Inf. Dilution	dH:25.400(3.000SD)kJ	4	CRV	14923
22	25	0	Inf. Dilution	dH:25.400(1.5SD)kJ	2	CRV	27292
23	25	0	Inf. Dilution	dH:20.647(5SF)kJ	4	EOD	29487

Reaction No. 54550

SiO2(Cristob.,s) + 2<H2O> = H+1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-3.35(3SF)	0	CRV	403
3	25	0	Inf. Dilution	lgK:-3.5(0.05SD)	0	ENS	440
4	25	0	Inf. Dilution	lgK:-3.45(3SF)	0	CRV	6405
Note (4): p. 245							
5	25	0	Inf. Dilution	lgK:-4.18(3SF)	0	CRV	32224
6	25	0	CHEMVAL1	lgK:-3.55(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL2	lgK:-3.55(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL3	lgK:-3.55(0.01SD)	0	ENS	5156
9	25	0	CHEMVAL4	lgK:-3.52(0.01SD)	0	ENS	5156

Reaction No. 54551							
Mg+2 + H+1_SiH2O4-2 = Mg+2_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EOR	
2	25	1	NaClO4	lgK:0.6(0.05SD)	3	MIS	455

Reaction No. 54552							
Mg+2 + 2<H+1_SiH2O4-2> = Mg+2_H+1(2)_SiH2O4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.3(2SF)	1	EOR	
2	25	1	NaClO4	lgK:3.8(0.05SD)	3	MIS	455

Reaction No. 54554							
Ca+2 + H+1_SiH2O4-2 = Ca+2_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	2	EDH	
2	25	1	NaClO4	lgK:0.4(0.05SD)	4	MIS	455

Reaction No. 54555							
Ca+2 + 2<H+1_SiH2O4-2> = Ca+2_H+1(2)_SiH2O4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.5(2SF)	1	EOR	
2	25	1	NaClO4	lgK:2.9(0.05SD)	2	MIS	455

Reaction No. 54556							
$\text{Ca}+2_ \text{SiH}2\text{O}4-2_ (\text{s}) = \text{Ca}+2 + \text{SiH}2\text{O}4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -7.2 (0.1\text{SD})$	4	MSL	456

Reaction No. 54557							
$\text{Ca}+2 + \text{H}+1 (2)_ \text{SiH}2\text{O}4-2 - \text{H}2\text{O} = \text{CaO}.\text{SiO}2 (\text{s}) + 2\text{<H}+1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -13.4 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -13.6 (0.05\text{SD})$	0	ENS	440
3	25	0	Inf. Dilution	$\lg K: -13.64 (4\text{SF})$	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	$\lg K: -12.99 (4\text{SF})$	0	EOD	23102
5	25	0	Inf. Dilution	$\lg K: -13.73 (4\text{SF})$	2	CRV	32224
6	25	0	CHEMVAL1	$\lg K: -14.61 (0.01\text{SD})$	0	ENS	5156
7	25	0	CHEMVAL2	$\lg K: -14.61 (0.01\text{SD})$	0	ENS	5156
8	25	0	CHEMVAL3	$\lg K: -14.61 (0.01\text{SD})$	0	ENS	5156
9	25	0	CHEMVAL4	$\lg K: -13.10 (0.01\text{SD})$	0	ENS	5156

Reaction No. 54558							
$\text{Ca}+2 + \text{Mg}+2 + 2\text{<H}+1 (2)_ \text{SiH}2\text{O}4-2\text{>} - 2\text{<H}2\text{O}> = \text{CaO}.\text{MgO}.\text{SiO}2 (\text{s}) + 4\text{<H}+1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -21.7 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -21.2 (0.1\text{SD})$	0	ENS	440
3	25	0	Inf. Dilution	$\lg K: -21.20 (4\text{SF})$	0	CRV	6405
Note (3): p. 246							
4	25	0	Inf. Dilution	$\lg K: -21.07 (4\text{SF})$	0	CRV	32224
5	25	0	CHEMVAL1	$\lg K: -23.70 (0.01\text{SD})$	0	ENS	5156
6	25	0	CHEMVAL2	$\lg K: -23.70 (0.01\text{SD})$	0	ENS	5156
7	25	0	CHEMVAL3	$\lg K: -23.70 (0.01\text{SD})$	0	ENS	5156
8	25	0	CHEMVAL4	$\lg K: -19.90 (0.01\text{SD})$	0	ENS	5156

Reaction No. 54559							
$2\text{CaO}.\text{SiO}2.\text{H}2\text{O} (\text{s}) + 14\text{<H}+1\text{>} = 2\text{<Ca}+2\text{>} + 5\text{<Mg}+2\text{>} + 8\text{<H}+1 (2)_ \text{SiH}2\text{O}4-2\text{>} - 8\text{<H}2\text{O}>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 66.1 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 62.4 (0.1\text{SD})$	0	ENS	440

3	25	0	Inf. Dilution	lgK:62.44 (4SF)	0	CRV	6405
Note (3): p. 246							
4	25	0	Inf. Dilution	lgK:61.85 (4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:67.61 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:67.61 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:67.61 (0.01SD)	0	ENS	5156

Reaction No. 54560							
4<H+1> + 6<F-1> + H+1(2)_SiH2O4-2 = SiF6-2 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	1	NaCl	lgK:31.6 (0.05SD)	4	MIS	439
2	25	0	Inf. Dilution	lgK:28.7 (3SF)	1	EOR	
3	25	0	CHEMVAL1	lgK:30.18 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:30.18 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:30.18 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:27.80 (0.01SD)	0	ENS	5156
Note (6): Low wgt for internal consistency							
7	25	0.1	Na+1 salt	lgK:30.2 (0.1SD)	0	MIS	453
8	25	1	NaCl	lgK:30.0 (0.05SD)	4	MIS	439
9	60	1	NaCl	lgK:28.2 (0.05SD)	4	MIS	439
10	25	0	Inf. Dilution	dH:-16.3 (0.1SD)	2	ENS	766

Reaction No. 54561							
2<Mg+2> + 4<H2O> + 2<H+1> + 3<SiH2O4-2> = Mg+2(2)_H+1(6)_OH-1(4)_SiH2O4-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40. (0.3SD)	3	MSL	441
2	51	0	Inf. Dilution	lgK:-39. (0.3SD)	3	MSL	441
3	70	0	Inf. Dilution	lgK:-38. (0.3SD)	3	MSL	441

Reaction No. 54562							
2<Fe+2> + H+1(2)_SiH2O4-2 = 2FeO.SiO2(s) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.8 (0.05SD)	0	ENS	440
3	25	0	Inf. Dilution	lgK:-16.81 (4SF)	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	lgK:14.67 (4SF)	0	CRV	32224

5	25	0	CHEMVAL1	lgK:-16.71(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:-16.71(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-16.71(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-16.70(0.01SD)	0	ENS	5156

Reaction No. 54563							
$\text{H+1(2)}_{\text{SiH2O4-2}} + 2\text{<H+1>} + 2\text{<F-1>} = \text{Si(OH)2F2} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:11.1(0.05SD)	5	MIS	439

Reaction No. 54564							
$\text{H+1(2)}_{\text{SiH2O4-2}} + 3\text{<H+1>} + 4\text{<F-1>} = \text{SiOHF4-1} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	1	NaCl	lgK:21.4(0.05SD)	5	MIS	439
2	25	0	Inf. Dilution	lgK:22.2(3SF)	2	EAE	

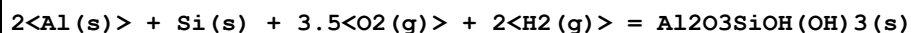
Reaction No. 54594							
$2\text{<Mn+2>} + \text{H+1(2)}_{\text{SiH2O4-2}} = 2\text{MnO.SiO2(s)} + 4\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-24.5(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:-23.47(4SF)	4	CRV	32224

Reaction No. 54682							
$\text{Si(s)} + 4\text{<e-1>} + 4\text{<H+1>} = \text{SiH4(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.8(0.05SD)	0	ENS	140
2	25	0	Inf. Dilution	E:-0.147(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	E:-0.143(3SF)	4	CRV	22551
4	25	0	Inf. Dilution	dH:34.2(3SF) kJ	5	CRV	7761

Reaction No. 54838							
$2\text{<Al(s)>} + 2\text{<H2(g)>} + 4.5\text{<O2(g)>} + 2\text{<Si(s)>} = \text{Al2Si2O5(OH)4(Hall.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:662.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:657.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:479.3(4SF)	3	ENS	6422

4	227	0	Inf. Dilution	lgK:372.2 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:300.8 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-903. (0.4SD)	2	ENS	2171
7	25	0	Inf. Dilution	dG:-903.6 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-898.5 (4SF)	0	CRV	7382
9	25	0	Inf. Dilution	dG:-3780.5 (5SF) kJ	4	CRV	9015
10	25	0	Inf. Dilution	dG:-898.5 (4SF)	0	CRV	12011
11	25	0	Inf. Dilution	dH:-980. (0.3SD)	4	ENS	2171
12	25	0	Inf. Dilution	dH:-980.2 (4SF)	4	MTD	6422
13	25	0	Inf. Dilution	dH:-975.0 (4SF)	0	CRV	7382
14	25	0	Inf. Dilution	dH:-4101.2 (5SF) kJ	4	CRV	9015
15	25	0	Inf. Dilution	dH:-975.1 (4SF)	0	CRV	12011
16	127	0	Inf. Dilution	dH:-980.5 (4SF)	3	MTD	6422
17	227	0	Inf. Dilution	dH:-980.2 (4SF)	2	MTD	6422
18	327	0	Inf. Dilution	dH:-979.6 (4SF)	0	MTD	6422

Reaction No. 54839



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-700. (0.5SD)	2	MCO	5479
2	25	0	Inf. Dilution	dG:-2921.80 (6SF) kJ	3	CRV	25435
3	25	0	Inf. Dilution	dH:-763. (0.5SD)	0	MCO	5479
4	25	0	Inf. Dilution	dH:-3170.40 (6SF) kJ	3	CRV	25435

Reaction No. 54840

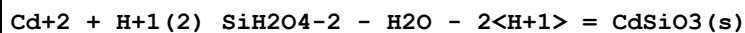


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:665.7 (4SF)	0	ENS	6422
Note (2): Changed from 655.7							
3	25	0	Inf. Dilution	lgK:665.3 (4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:661.2 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:660.8 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:481.8 (4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:481.5 (4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:374.2 (4SF)	0	ENS	6422

9	227	0	Inf. Dilution	lgK:373.9 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:302.5 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:302.2 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-3789.10 (6SF) kJ	0	CRV	440
13	25	0	Inf. Dilution	dG:-908.1 (0.2SD)	0	ENS	2171
14	25	0	Inf. Dilution	dG:-3799.40 (6SF) kJ	0	CRV	5619
15	25	0	Inf. Dilution	dG:-3777.60 (6SF) kJ	0	CRV	6421
16	25	0	Inf. Dilution	dG:-908.1 (4SF)	0	EFE	6422
17	25	0	Inf. Dilution	dG:-908.17 (5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dG:-907.6 (4SF)	0	EFE	6494
19	25	0	Inf. Dilution	dG:-3789.5 (5SF) kJ	0	MSL	7380
20	25	0	Inf. Dilution	dG:-903.0 (4SF)	0	CRV	7382
21	25	0	Inf. Dilution	dG:-3789.40 (6SF) kJ	0	CRV	8872
22	25	0	Inf. Dilution	dG:-3778.20 (6SF) kJ	0	EOD	8872
Note (22): Data from Naumov, Ryzhenko & Khodakovsky, 1971; low wgt for consistency							
23	25	0	Inf. Dilution	dG:-3776.90 (6SF) kJ	0	CRV	8873
24	25	0	Inf. Dilution	dG:-3780.00 (6SF) kJ	0	CRV	8874
25	25	0	Inf. Dilution	dG:-3799.7 (5SF) kJ	0	CRV	9015
26	25	0	Inf. Dilution	dG:-908.50 (5SF)	0	CRV	10174
27	25	0	Inf. Dilution	dG:-903.0 (4SF)	0	CRV	12011
28	25	0	Inf. Dilution	dG:-3801.70 (6SF) kJ	0	CRV	25435
29	25	0	Inf. Dilution	dG:-907.88 (5SF)	5	CRV	29553
30	25	0	GAMSPATH	dG:-3789. (4SF) kJ	0	CRV	12638
31	50	0	Inf. Dilution	dG:-909.76 (5SF)	0	CRV	10174
32	75	0	Inf. Dilution	dG:-911.15 (5SF)	0	CRV	10174
33	100	0	Inf. Dilution	dG:-912.64 (5SF)	0	CRV	10174
34	125	0	Inf. Dilution	dG:-914.26 (5SF)	0	CRV	10174
35	150	0	Inf. Dilution	dG:-915.97 (5SF)	0	CRV	10174
36	175	0	Inf. Dilution	dG:-917.80 (5SF)	0	CRV	10174
37	200	0	Inf. Dilution	dG:-919.72 (5SF)	0	CRV	10174
38	225	0	Inf. Dilution	dG:-921.73 (5SF)	0	CRV	10174
39	250	0	Inf. Dilution	dG:-923.83 (5SF)	0	CRV	10174
40	300	0	Inf. Dilution	dG:-928.28 (5SF)	0	CRV	10174
41	25	0	Inf. Dilution	dH:-985. (0.3SD)	0	ENS	2171
42	25	0	Inf. Dilution	dH:-984.6 (4SF)	0	MTD	6422
43	25	0	Inf. Dilution	dH:-984.8 (4SF)	0	ENS	6437

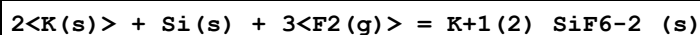
44	25	0	Inf. Dilution	dH:-984.78 (5SF)	0	EFE	6482
45	25	0	Inf. Dilution	dH:-984.5 (4SF)	0	MTD	6494
46	25	0	Inf. Dilution	dH:-979.6 (4SF)	0	CRV	7382
47	25	0	Inf. Dilution	dH:-4119.6 (5SF) kJ	0	CRV	9015
48	25	0	Inf. Dilution	dH:-976.6 (4SF)	0	CRV	12011
49	25	0	Inf. Dilution	dH:-4122.30 (6SF) kJ	0	CRV	25435
50	25	0	Inf. Dilution	dH:-984.53 (5SF)	5	CRV	29553
51	25	0	GAMSPATH	dH:-4109.60 (6SF) kJ	0	CRV	12638
52	127	0	Inf. Dilution	dH:-984.9 (4SF)	0	MTD	6422
53	127	0	Inf. Dilution	dH:-984.8 (4SF)	0	MTD	6494
54	227	0	Inf. Dilution	dH:-984.6 (4SF)	0	MTD	6422
55	227	0	Inf. Dilution	dH:-984.6 (4SF)	0	MTD	6494
56	327	0	Inf. Dilution	dH:-983.9 (4SF)	0	MTD	6422
57	327	0	Inf. Dilution	dH:-984.1 (4SF)	0	MTD	6494

Reaction No. 54916



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.1 (0.05SD)	0	ENS	766
3	25	0	Inf. Dilution	lgK:-7.49 (3SF)	4	CRV	32224
4	25	0	Inf. Dilution	dH:16.6 (0.1SD)	2	ENS	766

Reaction No. 55055



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-668.9 (0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	dH:-706.5 (0.1SD)	4	ENS	802

Reaction No. 55080



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.5 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:271.4 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:269.7 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:269.6 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:198.6 (4SF)	2	ENS	6422

6	127	0	Inf. Dilution	lgK:198.5 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:155.9 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:155.8 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:127.4 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:127.3 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-370.4 (0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-370.4 (4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-369.53 (5SF)	2	EFE	6482
14	25	0	Inf. Dilution	dG:-370.2 (4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1549.0 (5SF) kJ	5	CRV	8695
16	25	0	Inf. Dilution	dG:-370.22 (5SF)	4	EFE	32152
17	25	0	Inf. Dilution	dH:-390.8 (0.05SD)	4	ENS	802
18	25	0	Inf. Dilution	dH:-390.8 (4SF)	4	MTD	6422
19	25	0	Inf. Dilution	dH:-390.0 (4SF)	3	ENS	6437
20	25	0	Inf. Dilution	dH:-389.94 (5SF)	2	EFE	6482
21	25	0	Inf. Dilution	dH:-390.7 (4SF)	4	MTD	6494
22	25	0	Inf. Dilution	dH:-1634.90 (6SF) kJ	5	CRV	8695
23	127	0	Inf. Dilution	dH:-390.7 (4SF)	1	MTD	6422
24	127	0	Inf. Dilution	dH:-390.7 (4SF)	1	MTD	6494
25	227	0	Inf. Dilution	dH:-390.5 (4SF)	0	MTD	6422
26	227	0	Inf. Dilution	dH:-390.5 (4SF)	0	MTD	6494
27	327	0	Inf. Dilution	dH:-390.2 (4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-390.3 (4SF)	0	MTD	6494

Reaction No. 55081							
2<Ca(s)> + 2<O2(g)> + Si(s) = 2CaO.SiO2(beta,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:383.7 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:383.9 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:381.2 (4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:381.4 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:280.9 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:281.0 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:220.8 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:220.8 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:180.7 (4SF)	4	ENS	6422

10	327	0	Inf. Dilution	lgK:180.7 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-524. (0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-523.5 (4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-523.7 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-551. (0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-550.9 (4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-551.3 (4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-550.7 (4SF)	4	MTD	6422
18	127	0	Inf. Dilution	dH:-551.2 (4SF)	4	MTD	6494
19	227	0	Inf. Dilution	dH:-550.4 (4SF)	4	MTD	6422
20	227	0	Inf. Dilution	dH:-550.9 (4SF)	4	MTD	6494
21	327	0	Inf. Dilution	dH:-550.0 (4SF)	4	MTD	6422
22	327	0	Inf. Dilution	dH:-550.6 (4SF)	4	MTD	6494

Reaction No. 55082							
2<Ca(s)> + 2<O2(g)> + Si(s) = 2CaO.SiO2 (gamma,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:385.2 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:385.2 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:382.7 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:382.7 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:281.9 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:281.9 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:221.5 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:221.4 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:181.2 (4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:181.1 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-526. (0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-525.5 (4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-525.5 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-554. (0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-553.3 (4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-553.7 (4SF)	4	MTD	6494
17	27	0	Inf. Dilution	dH:-553.4 (4SF)	0	MTD	6422
18	127	0	Inf. Dilution	dH:-553.3 (4SF)	4	MTD	6422
19	127	0	Inf. Dilution	dH:-553.6 (4SF)	4	MTD	6494

20	227	0	Inf. Dilution	dH:-553.1 (4SF)	4	MTD	6422
21	227	0	Inf. Dilution	dH:-553.4 (4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-552.8 (4SF)	4	MTD	6422
23	327	0	Inf. Dilution	dH:-553.2 (4SF)	4	MTD	6494

Reaction No. 55083							
3<Ca(s)> + 2.5<O2(g)> + Si(s) = 3CaO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:487.7 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:488.2 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:484.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:485.0 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:357.1 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:357.3 (4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:280.6 (4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:280.8 (4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:229.7 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:229.8 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-665. (0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-665.4 (4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-666.0 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-700. (0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-700.1 (4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-701.0 (4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-699.8 (4SF)	2	MTD	6422
18	127	0	Inf. Dilution	dH:-700.8 (4SF)	2	MTD	6494
19	227	0	Inf. Dilution	dH:-699.2 (4SF)	0	MTD	6422
20	227	0	Inf. Dilution	dH:-700.4 (4SF)	0	MTD	6494
21	327	0	Inf. Dilution	dH:-698.6 (4SF)	0	MTD	6422
22	327	0	Inf. Dilution	dH:-699.8 (4SF)	0	MTD	6494

Reaction No. 55084							
3<Ca(s)> + 3.5<O2(g)> + 2<Si(s)> = 3CaO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:659.1 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:659.0 (4SF)	0	ENS	6422

3	25	0	Inf. Dilution	lgK:656.6 (4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:654.7 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:652.4 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:482.3 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:480.5 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:378.9 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:377.4 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:310.0 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:308.7 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-899. (0.5SD)	0	ENS	802
13	25	0	Inf. Dilution	dG:-899.0 (4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-895.8 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-947. (0.5SD)	0	ENS	802
16	25	0	Inf. Dilution	dH:-946.7 (4SF)	0	MTD	6422
17	25	0	Inf. Dilution	dH:-943.8 (4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-946.5 (4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-943.7 (4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-945.9 (4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-943.3 (4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-945.3 (4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-942.9 (4SF)	0	MTD	6494

Reaction No. 55097							
Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Clinoen.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:256.1 (4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:255.4 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:254.5 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:253.8 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:187.0 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:186.5 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:146.6 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:146.2 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:119.6 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:119.3 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-349.5 (0.05SD)	0	ENS	802

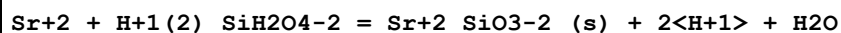
12	25	0	Inf. Dilution	dG:-349.4 (4SF)	2	EFE	6422
13	25	0	Inf. Dilution	dG:-348.61 (5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-348.5 (4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1458.1 (5SF) kJ	3	EOD	12843
16	25	0	Inf. Dilution	dH:-370.2 (0.05SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-370.2 (4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-369.49 (5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-369.3 (4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-1545.0 (5SF) kJ	3	EOD	12843
21	127	0	Inf. Dilution	dH:-370.3 (4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-369.3 (4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-370.2 (4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-369.3 (4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-370.1 (4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-369.2 (4SF)	0	MTD	6494

Reaction No. 55098							
2<Mg+2> + H+1 (2) _SiH2O4-2 = 2MgO.SiO2(s) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-28.9 (0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:-28.44 (4SF)	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	lgK:-28.14 (4SF)	4	CRV	32224
5	25	0	CHEMVAL1	lgK:-29.98 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:-29.98 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-29.98 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-29.10 (0.01SD)	0	ENS	5156

Reaction No. 55099							
Ca(s) + Si(s) + 1.5<O2(g)> = CaO.SiO2(Pseudo.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:270.6 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:270.4 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:268.9 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:268.6 (4SF)	0	ENS	6494

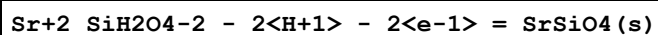
5	127	0	Inf. Dilution	lgK:198.0 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:197.8 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:155.5 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:155.3 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:127.2 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:127.0 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-369. (0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-369.2 (4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-368.81 (5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-368.9 (4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-389. (0.5SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-389.2 (4SF)	4	MTD	6422
17	25	0	Inf. Dilution	dH:-388.96 (5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-389.0 (4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-389.1 (4SF)	1	MTD	6422
20	127	0	Inf. Dilution	dH:-388.9 (4SF)	1	MTD	6494
21	227	0	Inf. Dilution	dH:-388.9 (4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-388.8 (4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-388.7 (4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-388.6 (4SF)	0	MTD	6494

Reaction No. 55158



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.2 (3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-14.1 (0.05SD)	3	ENS	5458
3	25	0	Inf. Dilution	lgK:-14.06 (4SF)	4	CRV	32224
4	25	0	CHEMVAL1	lgK:-15.64 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:-15.64 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:-15.64 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:-15.60 (0.01SD)	0	ENS	5156

Reaction No. 55159



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.8 (0.05SD)	0	ENS	5458

Reaction No. 55240							
$\text{Mn(s)} + \text{Si(s)} + 1.5\text{O}_2\text{(g)} = \text{MnO.SiO}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:217.3(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:218.1(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:215.9(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:216.6(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:158.4(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:159.1(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:123.9(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:124.6(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:101.0(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:101.6(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-296.5(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-296.5(0.1SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-296.5(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-297.5(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-315.7(0.1SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-315.7(0.1SD)	5	ENS	802
17	25	0	Inf. Dilution	dH:-315.7(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-315.9(4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-315.7(4SF)	4	MTD	6422
20	127	0	Inf. Dilution	dH:-315.8(4SF)	4	MTD	6494
21	227	0	Inf. Dilution	dH:-315.5(4SF)	4	MTD	6422
22	227	0	Inf. Dilution	dH:-315.6(4SF)	4	MTD	6494
23	327	0	Inf. Dilution	dH:-315.3(4SF)	4	MTD	6422
24	327	0	Inf. Dilution	dH:-315.4(4SF)	4	MTD	6494

Reaction No. 55241							
$2\text{Mn(s)} + \text{Si(s)} + 2\text{O}_2\text{(g)} = 2\text{MnO.SiO}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:285.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:285.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:284.1(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:283.9(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:208.8(4SF)	4	ENS	6422

6	127	0	Inf. Dilution	lgK:208.5(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:163.6(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:163.3(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:133.5(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:133.2(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-390.1(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-390.1(0.1SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-390.1(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-389.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-413.6(0.1SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-413.6(0.1SD)	5	ENS	802
17	25	0	Inf. Dilution	dH:-413.6(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-413.8(4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-413.5(4SF)	4	MTD	6422
20	127	0	Inf. Dilution	dH:-413.8(4SF)	4	MTD	6494
21	227	0	Inf. Dilution	dH:-413.3(4SF)	4	MTD	6422
22	227	0	Inf. Dilution	dH:-413.6(4SF)	4	MTD	6494
23	327	0	Inf. Dilution	dH:-413.0(4SF)	4	MTD	6422
24	327	0	Inf. Dilution	dH:-413.4(4SF)	4	MTD	6494

Reaction No. 55374

$$\text{K}^{+1} + 3\text{Mg}^{+2} + \text{Al}^{+3} + 3\text{H}^{+1}(2)_{\text{SiH}_2\text{O}_4-2} + 2\text{F}^{-1} = \text{KMg}_3\text{AlSi}_3\text{O}_{10}\text{F}_2(\text{s}) + 8\text{H}^{+1} + 2\text{H}_2\text{O}$$

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.8(0.05SD)	0	ENS	557

Reaction No. 55511

$$7\text{Mn}(\text{s}) + \text{Si}(\text{s}) + 6\text{O}_2(\text{g}) = \text{Mn}_7\text{SiO}_{12}(\text{s})$$

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:691.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:686.5(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:501.1(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:389.9(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:315.9(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-940.(5.SD)	0	ENT	5182
7	25	0	Inf. Dilution	dG:-942.8(4SF)	4	EFE	6494

8	25	0	Inf. Dilution	dH:-1010.(5SD)	4	ENT	5182
9	25	0	Inf. Dilution	dH:-1018.(4SF)	0	MTD	6494
10	127	0	Inf. Dilution	dH:-1018.(4SF)	1	MTD	6494
11	227	0	Inf. Dilution	dH:-1017.(4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1016.(4SF)	0	MTD	6494

Reaction No. 55530							
3<F2(g)> + Si(s) + 2<e-1> = SiF6-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-511.0(0.05SD)	0	ENS	3006
Note (1): Low wgt for inter-reaction consistency							
2	25	0	Inf. Dilution	dG:-525.7(4SF)	0	CRV	6488
3	25	0	Inf. Dilution	dG:-2138.(4SF)kJ	0	CRV	7421
4	25	0	Inf. Dilution	dG:-525.70(5SF)	1	CRV	10174
5	50	0	Inf. Dilution	dG:-526.39(5SF)	2	CRV	10174
6	75	0	Inf. Dilution	dG:-527.00(5SF)	2	CRV	10174
7	100	0	Inf. Dilution	dG:-527.54(5SF)	1	CRV	10174
8	125	0	Inf. Dilution	dG:-528.02(5SF)	1	CRV	10174
9	150	0	Inf. Dilution	dG:-528.41(5SF)	0	CRV	10174
10	175	0	Inf. Dilution	dG:-528.72(5SF)	0	CRV	10174
11	200	0	Inf. Dilution	dG:-528.93(5SF)	0	CRV	10174
12	225	0	Inf. Dilution	dG:-529.00(5SF)	0	CRV	10174
13	250	0	Inf. Dilution	dG:-528.91(5SF)	0	CRV	10174
14	300	0	Inf. Dilution	dG:-527.95(5SF)	0	CRV	10174
15	25	0	Inf. Dilution	dH:-558.3(0.1SD)	3	ENS	3006
16	25	0	Inf. Dilution	dH:-571.0(4SF)	0	CRV	6488

Reaction No. 55531							
H2(g) + 2<O2(g)> + Si(s) + 2<e-1> = SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:206.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-281.3(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	dG:-1175.7(1.265SD)kJ	0	CRV	14923
4	25	0	Inf. Dilution/m	dG:-1176.80(6SF)kJ	0	CRV	24953
5	25	0	Inf. Dilution	dH:-1382.0(15.330SD)kJ	1	CRV	14923
6	25	0	Inf. Dilution/m	dH:-1396.60(6SF)kJ	1	CRV	24953

Reaction No. 55734							
2<Fe(s)> + 2<O2(g)> + Si(s) = 2FeO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:241.6(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:241.6(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:240.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:240.0(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:175.6(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:175.7(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:137.0(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:137.1(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:111.3(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:111.5(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-329.6(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-329.6(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-329.86(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-329.6(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1379.1(5SF) kJ	5	CRV	8695
Note (15): p. 169							
16	25	0	Inf. Dilution	dG:-1379.1(5SF) kJ	3	EOD	12843
17	25	0	Inf. Dilution	dG:-1377.70(6SF) kJ	4	CRV	27292
18	25	0	Inf. Dilution	dH:-353.7(0.1SD)	4	ENS	802
19	25	0	Inf. Dilution	dH:-353.7(4SF)	4	MTD	6422
20	25	0	Inf. Dilution	dH:-353.3(4SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-353.58(5SF)	4	EFE	6482
22	25	0	Inf. Dilution	dH:-353.3(4SF)	4	MTD	6494
23	25	0	Inf. Dilution	dH:-1478.2(5SF) kJ	3	EOD	12843
24	25	0	Inf. Dilution	dH:-1476.80(6SF) kJ	4	CRV	27292
25	127	0	Inf. Dilution	dH:-353.5(4SF)	2	MTD	6422
26	127	0	Inf. Dilution	dH:-353.0(4SF)	2	MTD	6494
27	227	0	Inf. Dilution	dH:-353.1(4SF)	1	MTD	6422
28	227	0	Inf. Dilution	dH:-352.7(4SF)	1	MTD	6494
29	327	0	Inf. Dilution	dH:-352.7(4SF)	0	MTD	6422
30	327	0	Inf. Dilution	dH:-352.2(4SF)	0	MTD	6494
31	25	0	Inf. Dilution	Cp:132.030(6SF) J/K	4	CRV	27292

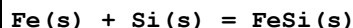
Reaction No. 55735							
$3\text{Fe(s)} + \text{Si(s)} = \text{Fe}_3\text{Si(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-22.6(0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	dH:-22.4(0.1SD)	4	ENS	802

Reaction No. 55736							
$\text{Fe(s)} + 2.33\text{Si(s)} = \text{FeSi}_{2.33}\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.25(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.19(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:7.619(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:6.078(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:5.051(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-14.(2SF)	4	ENS	802
7	25	0	Inf. Dilution	dG:-13.98(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-14.(2SF)	4	ENS	802
9	25	0	Inf. Dilution	dH:-14.1(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-14.1(3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-14.1(3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-14.1(3SF)	0	MTD	6422

Reaction No. 55737							
$\text{Fe(s)} + 2\text{Si(s)} = \text{FeSi}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.74(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:13.65(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:10.11(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:7.977(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:6.543(4SF)	3	ENS	6422
6	25	0	Inf. Dilution	dG:-18.7(0.1SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-18.74(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-19.4(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-19.4(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-19.5(3SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-19.6(3SF)	3	MTD	6422

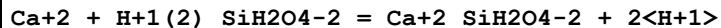
12	327	0	Inf. Dilution	dH:-19.7(3SF)	3	MTD	6422
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Reaction No. 55738



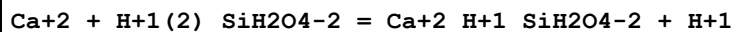
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.74(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:13.66(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:10.22(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:8.156(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:6.773(4SF)	3	ENS	6422
6	25	0	Inf. Dilution	dG:-17.6(0.1SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-18.75(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-17.6(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-18.8(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-18.9(3SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-18.9(3SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-19.0(3SF)	3	MTD	6422

Reaction No. 55790



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.7(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-19.87(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-19.87(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-19.87(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-17.30(0.01SD)	0	ENS	5156

Reaction No. 55791



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.8(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-9.08(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-9.08(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-9.08(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-8.84(0.01SD)	0	ENS	5156

Reaction No. 55792

Ca+2 + 2<H+1(2)_SiH2O4-2> = Ca+2_H+1(2)_SiH2O4-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.2(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-16.05(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-16.05(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-16.06(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-15.80(0.01SD)	0	ENS	5156

Reaction No. 55803							
Fe+2 + H+1(2)_SiH2O4-2 - H+1 - e-1 = Fe+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-13.97(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-13.97(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-13.97(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-13.20(0.01SD)	0	ENS	5156

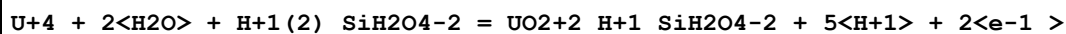
Reaction No. 55814							
Mg+2 + H+1(2)_SiH2O4-2 = Mg+2_SiH2O4-2 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.7(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-18.79(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-18.79(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-18.79(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-16.20(0.01SD)	0	ENS	5156

Reaction No. 55815							
Mg+2 + H+1(2)_SiH2O4-2 = Mg+2_H+1_SiH2O4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.9(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-8.83(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-8.83(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-8.83(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-8.59(0.01SD)	0	ENS	5156

Reaction No. 55816							
Mg+2 + 2<H+1(2)_SiH2O4-2> = Mg+2_H+1(2)_SiH2O4-2(2) + 2<H+1>							

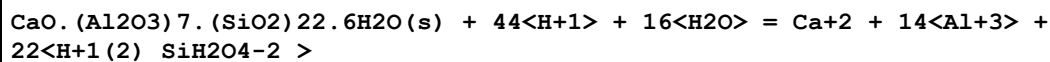
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.4 (3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-15.12 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-15.12 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-15.12 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-15.30 (0.01SD)	0	ENS	5156

Reaction No. 55927



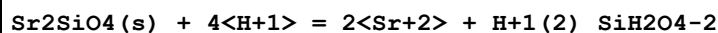
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.58 (4SF)	2	CRV	32224
2	25	0	CHEMVAL1	lgK:-11.62 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-11.62 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-11.62 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-11.00 (0.01SD)	4	ENS	5156

Reaction No. 56091



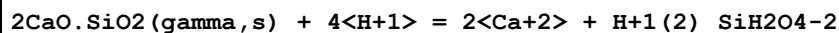
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:36.84 (0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:36.84 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:36.84 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:36.80 (0.01SD)	4	ENS	5156

Reaction No. 56097



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.61 (4SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:42.83 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:42.83 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:42.83 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:42.80 (0.01SD)	0	ENS	5156

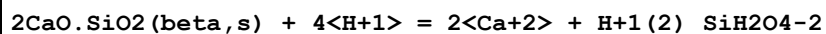
Reaction No. 56148



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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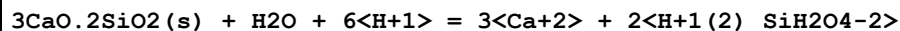
1	25	0	Inf. Dilution	lgK:38.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:37.67 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:39.01 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:39.01 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:39.01 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:40.40 (0.01SD)	0	ENS	5156

Reaction No. 56149



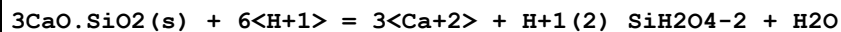
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:39.16 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:40.46 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:40.46 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:40.46 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:39.20 (0.01SD)	0	ENS	5156

Reaction No. 56150



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:48.62 (4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:51.77 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:51.77 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:51.77 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:51.80 (0.01SD)	0	ENS	5156

Reaction No. 56151



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:73.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:73.84 (4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:75.42 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:75.42 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:75.42 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:75.40 (0.01SD)	0	ENS	5156

Reaction No. 56152							
$\text{CaO.MgO.SiO}_2(\text{s}) + 4\text{H}^+ = \text{Ca}^{2+} + \text{Mg}^{2+} + \text{H}_2\text{SiO}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 31.4 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 29.85 (4\text{SF})$	0	CRV	32224
3	25	0	CHEMVAL1	$\lg K: 30.27 (0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL2	$\lg K: 30.27 (0.01\text{SD})$	0	ENS	5156
5	25	0	CHEMVAL3	$\lg K: 30.27 (0.01\text{SD})$	0	ENS	5156
6	25	0	CHEMVAL4	$\lg K: 30.30 (0.01\text{SD})$	0	ENS	5156

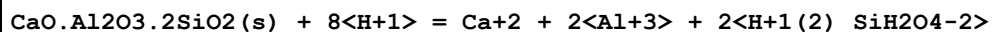
Reaction No. 56153							
$2\text{CaO.MgO.2SiO}_2(\text{s}) + \text{H}_2\text{O} + 6\text{H}^+ = 2\text{H}_2\text{SiO}_4^{2-} + 2\text{Ca}^{2+} + \text{Mg}^{2+}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 47.4 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 45.39 (4\text{SF})$	0	CRV	32224
3	25	0	CHEMVAL1	$\lg K: 48.74 (0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL2	$\lg K: 48.74 (0.01\text{SD})$	0	ENS	5156
5	25	0	CHEMVAL3	$\lg K: 48.74 (0.01\text{SD})$	0	ENS	5156
6	25	0	CHEMVAL4	$\lg K: 45.80 (0.01\text{SD})$	0	ENS	5156

Reaction No. 56154							
$3\text{CaO.MgO.2SiO}_2(\text{s}) + 8\text{H}^+ = 3\text{Ca}^{2+} + \text{Mg}^{2+} + 2\text{H}_2\text{SiO}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 69.0 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 68.55 (4\text{SF})$	0	CRV	32224
3	25	0	CHEMVAL1	$\lg K: 71.54 (0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL2	$\lg K: 71.54 (0.01\text{SD})$	0	ENS	5156
5	25	0	CHEMVAL3	$\lg K: 71.54 (0.01\text{SD})$	0	ENS	5156
6	25	0	CHEMVAL4	$\lg K: 68.70 (0.01\text{SD})$	0	ENS	5156

Reaction No. 56155							
$\text{Ca}_2\text{Al}_2\text{Si}_4\text{O}_{12} \cdot 4\text{H}_2\text{O}(\text{s}) + 8\text{H}^+ - 2\text{e}^- = 2\text{Ca}^{2+} + 2\text{Al}^{3+} + 4\text{H}_2\text{SiO}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: 11.85 (0.01\text{SD})$	0	ENS	5156
2	25	0	CHEMVAL2	$\lg K: 11.85 (0.01\text{SD})$	0	ENS	5156
3	25	0	CHEMVAL3	$\lg K: 11.85 (0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL4	$\lg K: 11.80 (0.01\text{SD})$	1	ENS	5156

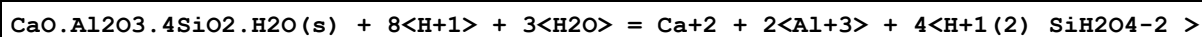
Note (4): query mineral "laumontite"

Reaction No. 56156



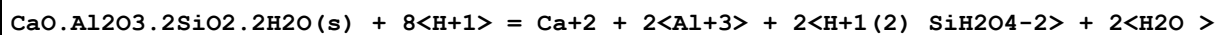
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:27.19 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:27.06 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:27.06 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:27.06 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:23.00 (0.01SD)	0	ENS	5156

Reaction No. 56157



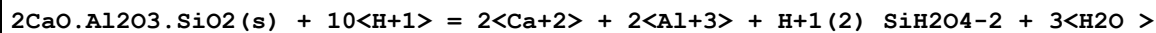
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:17.46 (0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:17.46 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:17.46 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:17.50 (0.01SD)	3	ENS	5156

Reaction No. 56158



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:22.81 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:22.06 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:22.06 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:22.06 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:17.00 (0.01SD)	0	ENS	5156

Reaction No. 56159



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:56.88 (4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:56.42 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:56.42 (0.01SD)	0	ENS	5156

5	25	0	CHEMVAL3	lgK:56.42 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:52.00 (0.01SD)	0	ENS	5156

Reaction No. 56160							
$\text{Ca}_2\text{Al}_2\text{Si}_3\text{O}_{10}(\text{OH})_2(\text{s}) + 10\langle\text{H}+1\rangle = 2\langle\text{Ca}+2\rangle + 2\langle\text{Al}+3\rangle + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:33.51 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:32.41 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:32.41 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:32.41 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:32.40 (0.01SD)	0	ENS	5156

Reaction No. 56161							
$\text{Ca}_2\text{Al}_3\text{Si}_3\text{O}_{12}(\text{OH})(\text{ortho},\text{s}) + 13\langle\text{H}+1\rangle = 2\langle\text{Ca}+2\rangle + 3\langle\text{Al}+3\rangle + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:44.20 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:39.83 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:39.83 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:39.83 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:39.80 (0.01SD)	0	ENS	5156

Reaction No. 56162							
$\text{Ca}_2\text{Al}_4\text{Si}_8\text{O}_{24}.7\text{H}_2\text{O}(\text{s}) + 16\langle\text{H}+1\rangle + \text{H}_2\text{O} = 2\langle\text{Ca}+2\rangle + 4\langle\text{Al}+3\rangle + 8\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:25.06 (0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:25.06 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:25.06 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:16.60 (0.01SD)	3	ENS	5156

Reaction No. 56164							
$3\text{CaO}. \text{Al}_2\text{O}_3. 3\text{SiO}_2(\text{s}) + 12\langle\text{H}+1\rangle = 3\langle\text{Ca}+2\rangle + 2\langle\text{Al}+3\rangle + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:52.47 (4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:57.21 (0.01SD)	0	ENS	5156

4	25	0	CHEMVAL2	lgK:57.21 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:57.21 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:52.60 (0.01SD)	0	ENS	5156

Reaction No. 56165							
$\text{CaO.FeO.2SiO}_2(\text{s}) + 2\text{<H}_2\text{O>} + 4\text{<H+1>} = \text{Ca}^{+2} + \text{Fe}^{+2} + 2\text{<H+1(2)<SiH}_2\text{O}_4\text{-2>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.40 (4SF)	0	CRV	6405
Note (2): p. 246							
3	25	0	Inf. Dilution	lgK:17.35 (4SF)	0	CRV	32224
4	25	0	CHEMVAL1	lgK:20.51 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:20.51 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:20.51 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:20.50 (0.01SD)	0	ENS	5156

Reaction No. 56166							
$3\text{CaO.Fe}_2\text{O}_3.3\text{SiO}_2(\text{s}) + 12\text{<H+1>} = 3\text{<Ca}^{+2}> + 2\text{<Fe}^{+3}> + 3\text{<H+1(2)<SiH}_2\text{O}_4\text{-2>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.7 (3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:59.55 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:59.55 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:59.55 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:59.60 (0.01SD)	0	ENS	5156

Reaction No. 56170							
$\text{CaO.Al}_2\text{O}_3.\text{SiO}_2(\text{s}) + 8\text{<H+1>} = \text{Ca}^{+2} + 2\text{<Al}^{+3}> + \text{H+1(2)<SiH}_2\text{O}_4\text{-2>} + 2\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:36.59 (4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:33.71 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:33.71 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:33.71 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:33.60 (0.01SD)	0	ENS	5156

Reaction No. 56184							
$\text{MgO.SiO}_2(\text{Clinoen.,s}) + \text{H}_2\text{O} + 2\text{<H+1>} = \text{Mg}^{+2} + \text{H+1(2)<SiH}_2\text{O}_4\text{-2>}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.64 (4SF)	0	ENS	6405
3	25	0	CHEMVAL1	lgK:12.65 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:12.65 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:12.65 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:11.40 (0.01SD)	0	ENS	5156

Reaction No. 56185							
$\text{Mg}_7\text{Si}_8\text{O}_{22}(\text{OH})_2(\text{Anth.},s) + 14\text{H}^+ + 8\text{H}_2\text{O} = 7\text{Mg}^{2+} + 8\text{H}^+(2)_{\text{SiH}_2\text{O}_4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:72.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:67.75 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:80.39 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:80.39 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:80.39 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:80.40 (0.01SD)	0	ENS	5156

Reaction No. 56186							
$\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4(\text{Chry.},s) + 6\text{H}^+ = 3\text{Mg}^{2+} + 2\text{H}^+(2)_{\text{SiH}_2\text{O}_4-2} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:32.9 (0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:32.20 (4SF)	0	ENS	6496
4	25	0	Inf. Dilution	lgK:31.52 (4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:34.50 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:34.50 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:34.50 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:34.50 (0.01SD)	0	ENS	5156
9	25	0	Inf. Dilution	dH:-46.8 (3SF)	2	ENS	6496

Reaction No. 56187							
$\text{MgAl}_4\text{Si}_2\text{O}_7\text{H}_{12}(s) + 16\text{H}_2\text{O} + 44\text{H}^+ = \text{Mg}^{2+} + 14\text{Al}^{3+} + 22\text{H}^+(2)_{\text{SiH}_2\text{O}_4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:36.24 (0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:36.24 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:36.24 (0.01SD)	0	ENS	5156

4	25	0	CHEMVAL4	lgK:36.20(0.01SD)	4	ENS	5156
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Reaction No. 56190							
$\text{Mg}_2(\text{H}_3\text{SiO}_4) \cdot 4.4\text{H}_2\text{O}(\text{s}) + 4\langle\text{H}+1\rangle = 2\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + 4\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-9.22(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-9.22(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-9.22(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-9.22(0.01SD)	4	ENS	5156

Reaction No. 56191							
$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2 \cdot 3\text{H}_2\text{O}(\text{s}) + \text{H}_2\text{O} + 6\langle\text{H}+1\rangle = 3\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:29.86(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:29.86(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:29.86(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:29.90(0.01SD)	4	ENS	5156

Reaction No. 56192							
$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2(\text{s}) + 6\langle\text{H}+1\rangle + 4\langle\text{H}_2\text{O}\rangle = 3\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:21.53(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:24.45(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:24.45(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:24.45(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:19.60(0.01SD)	0	ENS	5156

Reaction No. 56193							
$2\text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2(\text{s}) + 16\langle\text{H}+1\rangle + 2\langle\text{H}_2\text{O}\rangle = 2\langle\text{Mg}+2\rangle + 4\langle\text{Al}+3\rangle + 5\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:45.5(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:53.88(4SF)	0	CRV	32224
4	25	0	CHEMVAL1	lgK:53.77(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:53.77(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:53.77(0.01SD)	0	ENS	5156

7	25	0	CHEMVAL4	lgK:49.40(0.01SD)	0	ENS	5156
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Reaction No. 56198							
$\text{KAlSi}_3\text{O}_8(\text{Micr.}, \text{s}) + 4\langle\text{H}_2\text{O}\rangle + 4\langle\text{H}+1\rangle = \text{K}+1 + \text{Al}+3 + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.2(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:1.29(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:1.29(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:1.29(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:0.88(0.01SD)	0	ENS	5156

Reaction No. 56199							
$\text{KAl}_7\text{Si}_{11}\text{O}_{30}(\text{OH})_6(\text{s}) + 8\langle\text{H}_2\text{O}\rangle + 22\langle\text{H}+1\rangle = \text{K}+1 + 7\langle\text{Al}+3\rangle + 11\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:17.58(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:17.58(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:17.58(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:17.60(0.01SD)	4	ENS	5156

Reaction No. 56201							
$\text{KAlSi}_2\text{O}_6(\text{s}) + 4\langle\text{H}+1\rangle + 2\langle\text{H}_2\text{O}\rangle = 2\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + \text{Al}+3 + \text{K}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.2(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:6.42(3SF)	0	EOD	23102
3	25	0	CHEMVAL1	lgK:9.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:9.78(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:9.78(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:6.61(0.01SD)	0	ENS	5156

Reaction No. 56202							
$\text{KAlSiO}_4(\text{s}) + 4\langle\text{H}+1\rangle = \text{K}+1 + \text{Al}+3 + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.23(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:12.79(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:12.79(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:12.79(0.01SD)	0	ENS	5156

6	25	0	CHEMVAL4	lgK:13.00(0.01SD)	0	ENS	5156
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Reaction No. 56203							
KAl ₃ Si ₃ O ₁₀ (OH) ₂ (s) + 10<H+1> = K+1 + 3<Al+3> + 3<H+1(2)_SiH ₂ O ₄ -2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:14.56(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:13.96(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:13.96(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:13.96(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:13.00(0.01SD)	0	ENS	5156

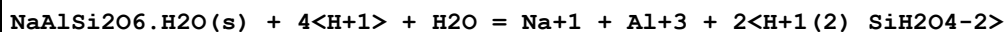
Reaction No. 56214							
NaAlSi ₃ O ₈ (AlbiteHI,s) + 4<H+1> + 4<H ₂ O> = Na+1 + Al+3 + 3<H+1(2)_SiH ₂ O ₄ -2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.41(3SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:7.02(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.02(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.02(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.02(0.01SD)	0	ENS	5156

Reaction No. 56215							
NaAl ₇ Si ₁₁ O ₃₀ (OH) ₆ (s) + 8<H ₂ O> + 22<H+1> = Na+1 + 7<Al+3> + 11<H+1(2)_SiH ₂ O ₄ -2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:18.36(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:18.36(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:18.36(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:18.70(0.01SD)	4	ENS	5156

Reaction No. 56217							
NaAlSi ₂ O ₆ (Dehyd.Anal.,s) + 2<H ₂ O> + 4<H+1> = Na+1 + Al+3 + 2<H+1(2)_SiH ₂ O ₄ -2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:15.20(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:15.20(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:15.20(0.01SD)	0	ENS	5156

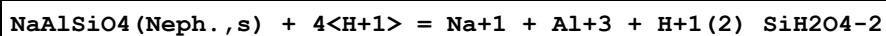
5	25	0	CHEMVAL4	lgK:15.20 (0.01SD)	0	ENS	5156
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Reaction No. 56218



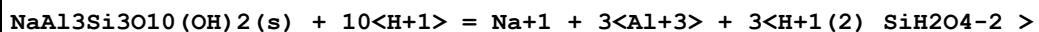
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:6.64 (3SF)	0	CRV	27383
3	25	0	Inf. Dilution	lgK:7.27 (3SF)	0	CRV	32224
4	25	0	Inf. Dilution	lgK:11.98 (4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:10.58 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:10.58 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:10.58 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:10.60 (0.01SD)	0	ENS	5156

Reaction No. 56219



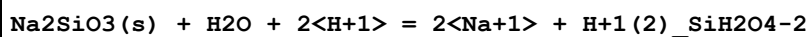
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:14.13 (4SF)	2	CRV	32224
3	25	0	CHEMVAL1	lgK:14.96 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.96 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.96 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:15.00 (0.01SD)	0	ENS	5156

Reaction No. 56220



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.49 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:23.23 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:23.23 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:23.23 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:23.20 (0.01SD)	0	ENS	5156

Reaction No. 56221



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:23.2 (3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:24.62 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:24.62 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:24.62 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:24.60 (0.01SD)	0	ENS	5156

Reaction No. 56222							
Na ₂ Si ₂ O ₅ (s) + 3<H ₂ O> + 2<H+1> = 2<Na+1> + 2<H+1(2)_SiH ₂ O ₄ -2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.6 (3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:21.66 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:21.66 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:21.66 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:21.70 (0.01SD)	0	ENS	5156

Reaction No. 56227							
FeO.SiO ₂ (s) + H ₂ O + 2<H+1> = Fe+2 + H+1(2)_SiH ₂ O ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.54 (3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:7.12 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.12 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.12 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.12 (0.01SD)	0	ENS	5156

Reaction No. 56228							
Fe ₃ Si ₂ O ₅ (OH) ₄ (s) + 6<H+1> = 3<Fe+2> + 2<H+1(2)_SiH ₂ O ₄ -2> + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.65 (4SF)	4	CRV	32224
2	25	0	CHEMVAL1	lgK:23.80 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:23.80 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:23.80 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:23.80 (0.01SD)	0	ENS	5156

Reaction No. 56229							
Fe ₃ Si ₄ O ₁₀ (OH) ₂ (s) + 4<H ₂ O> + 6<H+1> = 3<Fe+2> + 4<H+1(2)_SiH ₂ O ₄ -2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:7.34 (3SF)	4	CRV	32224
2	25	0	CHEMVAL1	lgK:13.66 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:13.66 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:13.66 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:13.70 (0.01SD)	0	ENS	5156

Reaction No. 56236							
$\text{Al}_2\text{SiO}_5(\text{Kyan., s}) + 6\text{H}^+ = 2\text{Al}^{3+} + \text{H}_2\text{SiO}_4^{2-} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.326 (5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:13.99 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:13.99 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:13.99 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.00 (0.01SD)	0	ENS	5156

Reaction No. 56237							
$\text{Al}_2\text{Si}_4\text{O}_{10}(\text{OH})_2(\text{s}) + 4\text{H}_2\text{O} = 2\text{Al}^{3+} + 4\text{H}_2\text{SiO}_4^{2-} - 6\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.8 (1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.081 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:3.46 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:3.46 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:3.46 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:3.42 (0.01SD)	0	ENS	5156

Reaction No. 56263							
$\text{UO}_2\cdot\text{SiO}_2(\text{s}) + 4\text{H}^+ = \text{U}^{4+} + \text{H}_2\text{SiO}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.5 (2SF)	0	CRV	32222
3	25	0	Inf. Dilution	lgK:-9.20 (3SF)	0	CRV	32224
Note (3): source indicates uncertainty							
4	25	0	CHEMVAL1	lgK:-9.18 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:-9.18 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:-9.18 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:-7.63 (0.01SD)	0	ENS	5156

Reaction No. 56331							
$\text{Mg}_5\text{Al}_2\text{Si}_3\text{O}_{10}(\text{OH})_8(\text{s}) + 16\langle\text{H}+1\rangle = 5\langle\text{Mg}+2\rangle + 2\langle\text{Al}+3\rangle + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + 6\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:61.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:71.92(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:57.02(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:57.02(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:57.02(0.01SD)	0	ENS	5156

Reaction No. 56334							
$\text{Al}_2\text{Si}_2\text{O}_7 \cdot 2\text{H}_2\text{O}(\text{Dick.}, \text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Al}+3\rangle + 2\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.95(3SF)	0	MGL	6498
3	25	0	CHEMVAL1	lgK:7.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.78(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.78(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.74(0.01SD)	0	ENS	5156

Reaction No. 56347							
$\text{MgO} \cdot \text{SiO}_2(\text{Enstat.}, \text{s}) + 2\langle\text{H}+1\rangle + \text{H}_2\text{O} = \text{Mg}+2 + \langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.64(4SF)	0	CRV	6405
3	25	0	Inf. Dilution	lgK:11.46(4SF)	4	CRV	32224
4	25	0	CHEMVAL1	lgK:13.04(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:13.04(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:13.04(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:12.60(0.01SD)	0	ENS	5156

Reaction No. 56348							
$2\langle\text{H}_2(\text{g})\rangle + 3\langle\text{Mg}(\text{s})\rangle + 4.5\langle\text{O}_2(\text{g})\rangle + 2\langle\text{Si}(\text{s})\rangle = \text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4(\text{Chry.}, \text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:707.4(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:706.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:702.7(4SF)	4	ENS	6422

4	27	0	Inf. Dilution	lgK:701.7 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:512.7 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:511.9 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:398.6 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:398.0 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:322.7 (4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:322.2 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-965.1 (0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-965.1 (0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-965.1 (4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-964.48 (5SF)	4	EFE	6482
15	25	0	Inf. Dilution	dG:-963.8 (4SF)	4	EFE	6494
16	25	0	GAMSPATH	dG:-4037. (4SF) kJ	3	CRV	12638
17	25	0	Inf. Dilution	dH:-1043.4 (0.1SD)	4	ENS	802
18	25	0	Inf. Dilution	dH:-1043.4 (0.1SD)	5	ENS	802
19	25	0	Inf. Dilution	dH:-1043. (4SF)	4	MTD	6422
20	25	0	Inf. Dilution	dH:-1042.8 (5SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-1042.9 (5SF)	4	EFE	6482
22	25	0	Inf. Dilution	dH:-1042. (4SF)	4	MTD	6494
23	25	0	GAMSPATH	dH:-4364.40 (6SF) kJ	3	CRV	12638
24	127	0	Inf. Dilution	dH:-1044. (4SF)	4	MTD	6422
25	127	0	Inf. Dilution	dH:-1042. (4SF)	4	MTD	6494
26	227	0	Inf. Dilution	dH:-1043. (4SF)	4	MTD	6422
27	227	0	Inf. Dilution	dH:-1042. (4SF)	4	MTD	6494
28	327	0	Inf. Dilution	dH:-1042. (4SF)	4	MTD	6422
29	327	0	Inf. Dilution	dH:-1041. (4SF)	4	MTD	6494

Reaction No. 56349							
Ba(s) + Si(s) + 1.5<O2(g)> = BaO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:269.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:268.1 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:197.4 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:155.0 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:126.8 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-368.1 (0.05SD)	4	ENS	802

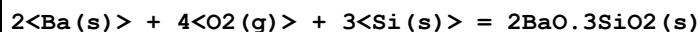
7	25	0	Inf. Dilution	dG:-368.1(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-388.0(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-388.0(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-388.0(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-387.9(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-388.1(4SF)	4	MTD	6422

Reaction No. 56350							
Ba(s) + 2<Si(s)> + 2.5<O2(g)> = BaO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:422.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:419.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:308.720(6SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:242.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:197.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-576.2(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-576.2(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-609.0(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-609.0(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-608.9(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-608.9(4SF)	4	MTD	6422

Reaction No. 56351							
2<Ba(s)> + 2<O2(g)> + Si(s) = 2BaO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:381.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:378.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:279.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:219.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:179.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-519.8(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-519.9(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-546.8(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-546.8(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-546.7(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-546.7(4SF)	4	MTD	6422

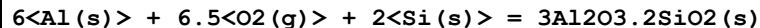
12	327	0	Inf. Dilution	dH:-547.1 (4SF)	4	MTD	6422
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Reaction No. 56352



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:694.3 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:689.8 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:507.6 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:398.4 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:325.5 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-947.2 (0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-947.2 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-1000.2 (0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-1000. (4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-1000. (4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-1000. (4SF)	2	MTD	6422
12	327	0	Inf. Dilution	dH:-1000. (4SF)	0	MTD	6422

Reaction No. 56355



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1129. (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:1129. (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1121. (4SF)	3	ENS	6422
4	27	0	Inf. Dilution	lgK:1121. (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:824.5 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:824.3 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:646.3 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:646.1 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:527.5 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:527.4 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1541.2 (0.05SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-1540. (4SF)	3	EFE	6422
13	25	0	Inf. Dilution	dG:-1540. (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-6432.7 (5SF) kJ	0	CRV	9015
15	25	0	Inf. Dilution	dG:-1811. (4SF)	0	CRV	12011

16	25	0	Inf. Dilution	dH:-1632.8(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-1637.(4SF)	2	MTD	6422
18	25	0	Inf. Dilution	dH:-1630.(4SF)	2	MTD	6494
19	25	0	Inf. Dilution	dH:-6816.2(5SF)kJ	0	CRV	9015
20	25	0	Inf. Dilution	dH:-1903.(4SF)	0	CRV	12011
21	127	0	Inf. Dilution	dH:-1631.(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-1630.(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-1629.(4SF)	0	MTD	6494

Reaction No. 56369							
2<Be(s)> + Si(s) + 2<O2(g)> = Be2SiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:355.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:355.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:353.5(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:353.0(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:260.1(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:259.7(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:204.1(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:203.8(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:166.7(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:166.4(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-485.8(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-485.5(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-484.8(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-513.7(0.1SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-512.8(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-512.2(4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-512.9(4SF)	4	MTD	6422
18	127	0	Inf. Dilution	dH:-512.4(4SF)	4	MTD	6494
19	227	0	Inf. Dilution	dH:-512.9(4SF)	4	MTD	6422
20	227	0	Inf. Dilution	dH:-512.3(4SF)	4	MTD	6494
21	327	0	Inf. Dilution	dH:-512.7(4SF)	4	MTD	6422

22	327	0	Inf. Dilution	dH:-512.0 (4SF)	4	MTD	6494
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Reaction No. 56373							
$\text{H}_2(\text{g}) + 3\text{Mg}(\text{s}) + 6\text{O}_2(\text{g}) + 4\text{Si}(\text{s}) = \text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:971.0 (4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:967.1 (4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK:964.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:960.7 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:706.8 (4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:703.8 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:552.1 (4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:549.8 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:449.0 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:447.1 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1324.8 (0.05SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-1325. (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1318.8 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1319. (4SF)	4	EFE	6494
15	25	0	GAMSPATH	dG:-5524. (4SF) kJ	0	CRV	12638
16	25	0	Inf. Dilution	dH:-1415.5 (0.1SD)	0	ENS	802
17	25	0	Inf. Dilution	dH:-1415. (4SF)	0	MTD	6422
18	25	0	Inf. Dilution	dH:-1410.3 (5SF)	4	ENS	6437
19	25	0	Inf. Dilution	dH:-1409.5 (5SF)	3	EFE	6482
20	25	0	Inf. Dilution	dH:-1410. (4SF)	4	MTD	6494
21	25	0	GAMSPATH	dH:-5903.3 (5SF) kJ	0	CRV	12638
22	127	0	Inf. Dilution	dH:-1416. (4SF)	0	MTD	6422
23	127	0	Inf. Dilution	dH:-1410. (4SF)	1	MTD	6494
24	227	0	Inf. Dilution	dH:-1416. (4SF)	0	MTD	6422
25	227	0	Inf. Dilution	dH:-1410. (4SF)	0	MTD	6494
26	327	0	Inf. Dilution	dH:-1415. (4SF)	0	MTD	6422
27	327	0	Inf. Dilution	dH:-1409. (4SF)	0	MTD	6494

Reaction No. 56374							
$2\text{Mg}(\text{s}) + 4\text{Al}(\text{s}) + 5\text{Si}(\text{s}) + 9\text{O}_2(\text{g}) = 2\text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:1516.(4SF)	5	ENS	6422
2	25	0	Inf. Dilution	lgK:1516.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1506.(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:1506.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1107.(4SF)	5	ENS	6422
6	127	0	Inf. Dilution	lgK:1107.(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:867.7(4SF)	5	ENS	6422
8	227	0	Inf. Dilution	lgK:867.5(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:708.3(4SF)	5	ENS	6422
10	327	0	Inf. Dilution	lgK:707.9(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-2055.0(0.1SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-2068.(4SF)	5	EFE	6422
13	25	0	Inf. Dilution	dG:-2067.8(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-2068.(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-2177.0(0.1SD)	0	ENS	802

Note (15): Low wgt for internal consistency

16	25	0	Inf. Dilution	dH:-2190.(4SF)	5	MTD	6422
17	25	0	Inf. Dilution	dH:-2190.(4SF)	4	ENS	6437
18	25	0	Inf. Dilution	dH:-2189.0(5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
20	127	0	Inf. Dilution	dH:-2190.(4SF)	5	MTD	6422
21	127	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
22	227	0	Inf. Dilution	dH:-2189.(4SF)	5	MTD	6422
23	227	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
24	327	0	Inf. Dilution	dH:-2187.(4SF)	5	MTD	6422
25	327	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494

Reaction No. 56390

1.5<O2(g)> + Si(s) + Sr(s) = Sr+2_SiO3-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.3(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:269.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:198.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:155.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:127.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-370.4(0.05SD)	4	ENS	802

7	25	0	Inf. Dilution	dG:-370.4(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-370.2(4SF)	4	EFE	6422
9	25	0	Inf. Dilution/m	dG:-1549.70(6SF)kJ	3	CRV	24953
10	25	0	Inf. Dilution	dH:-390.5(0.1SD)	4	ENS	802
11	25	0	Inf. Dilution	dH:-390.5(0.1SD)	5	ENS	802
12	25	0	Inf. Dilution	dH:-390.5(4SF)	4	MTD	6422
13	25	0	Inf. Dilution/m	dH:-1633.90(6SF)kJ	3	CRV	24953
14	127	0	Inf. Dilution	dH:-390.4(4SF)	2	MTD	6422
15	227	0	Inf. Dilution	dH:-390.2(4SF)	1	MTD	6422
16	327	0	Inf. Dilution	dH:-389.9(4SF)	0	MTD	6422

Reaction No. 56391



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:383.5(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:381.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:280.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:220.6(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:180.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-523.7(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-523.7(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-523.2(4SF)	3	EFE	6422
9	25	0	Inf. Dilution/m	dG:-2191.10(6SF)kJ	3	CRV	24953
10	25	0	Inf. Dilution	dH:-550.8(0.1SD)	4	ENS	802
11	25	0	Inf. Dilution	dH:-550.8(0.1SD)	5	ENS	802
12	25	0	Inf. Dilution	dH:-550.7(4SF)	4	MTD	6422
13	25	0	Inf. Dilution/m	dH:-2304.50(6SF)kJ	3	CRV	24953
14	127	0	Inf. Dilution	dH:-550.4(4SF)	2	MTD	6422
15	227	0	Inf. Dilution	dH:-549.8(4SF)	1	MTD	6422
16	327	0	Inf. Dilution	dH:-549.2(4SF)	0	MTD	6422

Reaction No. 56397



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:334.5(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:336.3(4SF)	4	ENS	6494

3	27	0	Inf. Dilution	lgK:332.3(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:334.1(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:244.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:245.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:191.4(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:192.5(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:156.2(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:157.1(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-458.7(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-458.7(0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-456.3(4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-458.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1919.7(3.1MD) kJ	6	CRV	20829
16	25	0	Inf. Dilution	dH:-486.0(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-486.0(0.1SD)	5	ENS	802
18	25	0	Inf. Dilution	dH:-483.7(4SF)	2	MTD	6422
19	25	0	Inf. Dilution	dH:-486.2(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-2034.2(3.1MD) kJ	6	CRV	20829
21	127	0	Inf. Dilution	dH:-483.7(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-486.1(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-483.4(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-485.9(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-483.1(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-485.5(4SF)	0	MTD	6494
27	25	0	Inf. Dilution	Cp:98.200(0.6MD) J/K	5	CRV	20829

Reaction No. 56398							
2<Zn(s)> + Si(s) + 2<O2(g)> = Zn2SiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:268.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:266.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:194.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:151.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:123.3(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-364.1(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-364.1(0.05SD)	5	ENS	802

8	25	0	Inf. Dilution	dG:-365.9(4SF)	4	ENS	6422
9	25	0	Inf. Dilution	dH:-391.2(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-391.2(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-393.0(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-393.0(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-392.8(4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-392.5(4SF)	4	MTD	6422

Reaction No. 56821							
$V(s) + Si(s) = VSi(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-73.(1.SD)	4	ENS	3006

Reaction No. 56822							
$2<V(s)> + Si(s) = V2Si(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-37.(1.SD)	4	ENS	3006

Reaction No. 56823							
$3<V(s)> + Si(s) = V3Si(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-26.(1.SD)	4	ENS	3006

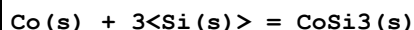
Reaction No. 56824							
$5<V(s)> + 3<Si(s)> = V5Si3(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-94.(1.SD)	4	ENS	3006

Reaction No. 56884							
$Co(s) + Si(s) = CoSi(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-23.6(0.1SD)	6	ENS	802
2	25	0	Inf. Dilution	dH:-24.0(0.1SD)	6	ENS	802

Reaction No. 56885							
$Co(s) + 2<Si(s)> = CoSi2(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

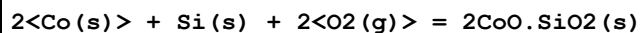
1	25	0	Inf. Dilution	dH:-24.6(0.1SD)	6	ENS	802
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Reaction No. 56886



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-25.6(0.1SD)	6	ENS	802

Reaction No. 56887



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.6(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:229.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:228.0(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:227.7(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:166.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:166.3(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:130.0(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:129.5(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:105.5(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:105.0(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-313.2(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-312.8(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-353.(0.5SD)	0	ENS	802

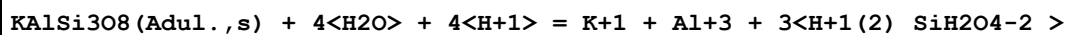
Note (13): Low wgt for internal consistency

14	25	0	Inf. Dilution	dH:-336.7(4SF)	3	MTD	6422
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Note (14): Low wgt for internal consistency

15	25	0	Inf. Dilution	dH:-337.5(4SF)	4	MTD	6494
16	127	0	Inf. Dilution	dH:-336.5(4SF)	4	MTD	6422
17	127	0	Inf. Dilution	dH:-337.2(4SF)	4	MTD	6494
18	227	0	Inf. Dilution	dH:-336.1(4SF)	4	MTD	6422
19	227	0	Inf. Dilution	dH:-336.8(4SF)	4	MTD	6494
20	327	0	Inf. Dilution	dH:-335.7(4SF)	4	MTD	6422
21	327	0	Inf. Dilution	dH:-336.2(4SF)	4	MTD	6494

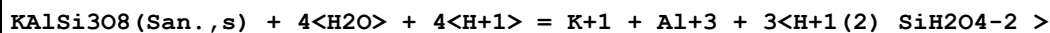
Reaction No. 57064



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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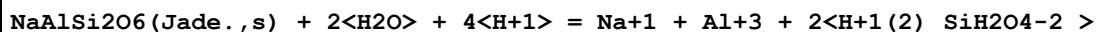
1	25	0	Inf. Dilution	lgK:3.4(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:1.26(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:1.26(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:1.26(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:1.26(0.01SD)	0	ENS	5156

Reaction No. 57065



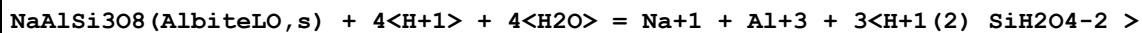
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.5(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.25(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:5.20(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:5.20(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:5.20(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:5.19(0.01SD)	0	ENS	5156

Reaction No. 57066



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.71(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:9.42(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:9.42(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:9.42(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:9.41(0.01SD)	0	ENS	5156

Reaction No. 57067



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.09(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:6.43(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:6.43(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:6.43(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:6.43(0.01SD)	0	ENS	5156

Reaction No. 57068

Al ₂ SiO ₅ (Andalu.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH ₂ O ₄ -2 + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.596 (5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:14.20 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.20 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.20 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.20 (0.01SD)	0	ENS	5156

Reaction No. 57069							
Al ₂ SiO ₅ (Sillim.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH ₂ O ₄ -2 + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.960 (5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:14.50 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.50 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.50 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.50 (0.01SD)	0	ENS	5156

Reaction No. 57414							
Co+2 + H+1_SiH ₂ O ₄ -2 = Co+2_H+1_SiH ₂ O ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:4.51 (0.06SD)	2	EOD	29656
Note (1): Same authors (in press)							

Reaction No. 57469							
H+1(2)_SiH ₂ O ₄ -2 + OH-1 = H+1_SiH ₂ O ₄ -2 + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.1 (2SF)	1	EOR	
2	25	3	Inert	lgK:5.1 (2SF)	1	EOR	
3	50	1	NaCl	lgK:3.96 (0.02SD)	7	MHY	5277
4	50	5	NaCl	lgK:4.32 (0.02SD)	7	MHY	5277
5	100	0	Inf. Dilution/m	lgK:3.17 (0.02SD)	5	MPT	4914
6	100	0.5	NaCl/m	lgK:3.29 (0.01SD)	5	MPT	4914
7	100	1	NaCl/m	lgK:3.35 (0.01SD)	5	MPT	4914
8	100	3	NaCl/m	lgK:3.52 (0.01SD)	5	MPT	4914
9	100	5	NaCl/m	lgK:3.67 (0.02SD)	5	MPT	4914

10	125	0	Inf. Dilution/m	lgK:2.94 (0.02SD)	5	MPT	4914
11	150	0	Inf. Dilution/m	lgK:2.75 (0.02SD)	5	MPT	4914
12	175	0	Inf. Dilution/m	lgK:2.59 (0.02SD)	5	MPT	4914
13	200	0	Inf. Dilution/m	lgK:2.45 (0.02SD)	5	MPT	4914
14	225	0	Inf. Dilution/m	lgK:2.33 (0.02SD)	5	MPT	4914
15	250	0	Inf. Dilution/m	lgK:2.23 (0.03SD)	5	MPT	4914
16	275	0	Inf. Dilution/m	lgK:2.15 (0.03SD)	5	MPT	4914
17	300	0	Inf. Dilution/m	lgK:2.08 (0.04SD)	5	MPT	4914
18	300	1	NaCl	lgK:2.20 (0.02SD)	7	MHY	5277
19	300	5	NaCl	lgK:2.26 (0.02SD)	7	MHY	5277
20	100	0	Inf. Dilution/m	dH:-26.48 (0.46SD) kJ	5	MPT	4914
21	100	0.5	NaCl/m	dH:-26.78 (0.46SD) kJ	5	MPT	4914
22	100	1	NaCl/m	dH:-27.11 (0.46SD) kJ	5	MPT	4914
23	100	3	NaCl/m	dH:-28.28 (0.50SD) kJ	5	MPT	4914
24	100	5	NaCl/m	dH:-29.50 (0.63SD) kJ	5	MPT	4914
25	125	0	Inf. Dilution/m	dH:-25.27 (0.38SD) kJ	5	MPT	4914
26	150	0	Inf. Dilution/m	dH:-24.02 (0.50SD) kJ	5	MPT	4914
27	175	0	Inf. Dilution/m	dH:-22.80 (0.67SD) kJ	5	MPT	4914
28	200	0	Inf. Dilution/m	dH:-21.55 (0.92SD) kJ	5	MPT	4914
29	225	0	Inf. Dilution/m	dH:-20.33 (1.17SD) kJ	5	MPT	4914
30	250	0	Inf. Dilution/m	dH:-19.08 (1.42SD) kJ	5	MPT	4914
31	275	0	Inf. Dilution/m	dH:-17.87 (1.67SD) kJ	5	MPT	4914
32	300	0	Inf. Dilution/m	dH:-16.61 (1.97SD) kJ	5	MPT	4914
33	100	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
34	100	0.5	NaCl/m	Cp:46.9 (10.9SD) J/K	5	MPT	4914
35	100	1	NaCl/m	Cp:44.4 (10.5SD) J/K	5	MPT	4914
36	100	3	NaCl/m	Cp:34.7 (10.0SD) J/K	5	MPT	4914
37	100	5	NaCl/m	Cp:25.1 (10.0SD) J/K	5	MPT	4914
38	125	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
39	150	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
40	175	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
41	200	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
42	225	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
43	250	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
44	275	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914
45	300	0	Inf. Dilution/m	Cp:49.4 (10.9SD) J/K	5	MPT	4914

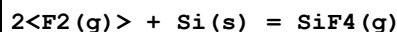
Reaction No. 57975							
Si(s) = Si(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:411.3(4SF) kJ	3	ENS	7275
2	25	0	Inf. Dilution	dG:405.5(4SF) kJ	5	CRV	8746
3	25	0	Inf. Dilution	dG:405.53(8.000SD) kJ	6	CRV	14923
4	25	0	Inf. Dilution	dH:108.(2.SD)	6	CRV	5605
5	25	0	Inf. Dilution	dH:455.6(4SF) kJ	3	ENS	7275
6	25	0	Inf. Dilution	dH:450.(8.MD) kJ	5	CRV	8746
7	25	0	Inf. Dilution	dH:450.00(8.000SD) kJ	6	CRV	14923
8	25	0	Inf. Dilution	Cp:22.251(0.001SD) J/K	5	CRV	27292

Reaction No. 57976							
O2(g) + Si(s) = SiO2(Quartz,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:150.0(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:150.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:149.1(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:149.0(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:109.4(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:109.4(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:85.62(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:85.60(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:69.77(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:69.75(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-856.25(5SF) kJ	4	CRV	440
12	25	0	Inf. Dilution	dG:-856.29(5SF) kJ	4	CRV	5619
13	25	0	Inf. Dilution	dG:-856.25(5SF) kJ	4	CRV	6421
14	25	0	Inf. Dilution	dG:-204.7(4SF)	2	EFE	6422
15	25	0	Inf. Dilution	dG:-204.66(5SF)	5	EFE	6482
16	25	0	Inf. Dilution	dG:-204.7(4SF)	2	EFE	6494
17	25	0	Inf. Dilution	dG:-856.64(5SF) kJ	1	ENS	7275
18	25	0	Inf. Dilution	dG:-856.64(5SF) kJ	1	CRV	8695
19	25	0	Inf. Dilution	dG:-856.3(4SF) kJ	5	CRV	8746
20	25	0	Inf. Dilution	dG:-856.29(5SF) kJ	4	CRV	8872

21	25	0	Inf. Dilution	dG:-856.51 (5SF) kJ	2	EOD	8872
Note (21): Data from Naumov, Ryzhenko & Khodakovsky, 1971							
22	25	0	Inf. Dilution	dG:-856.05 (5SF) kJ	2	CRV	8873
23	25	0	Inf. Dilution	dG:-856.21 (5SF) kJ	4	CRV	8874
24	25	0	Inf. Dilution	dG:-204.65 (5SF)	3	CRV	10174
25	25	0	Inf. Dilution	dG:-856.29 (1.002SD) kJ	6	CRV	14923
26	25	0	Inf. Dilution	dG:-856.28 (5SF) kJ	5	CRV	29482
27	25	0	Inf. Dilution	dG:-856.29 (5SF) kJ	5	EOD	29520
28	25	0	Inf. Dilution	dG:-204.48 (5SF)	3	CRV	29553
29	25	0	Inf. Dilution	dG:-854.79 (5SF) kJ	2	CRV	31308
30	25	0	Inf. Dilution	dG:-204.650 (6SF)	5	EFE	32152
31	25	0	Inf. Dilution/m	dG:-856.280 (6SF) kJ	3	CRV	24953
32	50	0	Inf. Dilution	dG:-204.90 (5SF)	0	CRV	10174
33	75	0	Inf. Dilution	dG:-205.18 (5SF)	0	CRV	10174
34	100	0	Inf. Dilution	dG:-205.49 (5SF)	0	CRV	10174
35	125	0	Inf. Dilution	dG:-205.81 (5SF)	0	CRV	10174
36	150	0	Inf. Dilution	dG:-206.15 (5SF)	0	CRV	10174
37	175	0	Inf. Dilution	dG:-206.51 (5SF)	0	CRV	10174
38	200	0	Inf. Dilution	dG:-206.88 (5SF)	0	CRV	10174
39	225	0	Inf. Dilution	dG:-207.28 (5SF)	0	CRV	10174
40	250	0	Inf. Dilution	dG:-207.68 (5SF)	0	CRV	10174
41	300	0	Inf. Dilution	dG:-208.54 (5SF)	0	CRV	10174
42	25	0	Inf. Dilution	dH:-217.6 (0.2SD)	2	CRV	5605
43	25	0	Inf. Dilution	dH:-217.7 (4SF)	4	MTD	6422
44	25	0	Inf. Dilution	dH:-217.66 (5SF)	5	EFE	6482
45	25	0	Inf. Dilution	dH:-217.7 (4SF)	4	MTD	6494
46	25	0	Inf. Dilution	dH:-910.94 (5SF) kJ	5	ENS	7275
47	25	0	Inf. Dilution	dH:-910.94 (5SF) kJ	5	CRV	8695
48	25	0	Inf. Dilution	dH:-910.7 (1.0MD) kJ	5	CRV	8746
49	25	0	Inf. Dilution	dH:-910.70 (1.000SD) kJ	6	CRV	14923
50	25	0	Inf. Dilution	dH:-910.70 (5SF) kJ	5	CRV	29482
51	25	0	Inf. Dilution	dH:-910.7 (4SF) kJ	5	EOD	29520
52	25	0	Inf. Dilution	dH:-217.48 (5SF)	3	CRV	29553
53	25	0	Inf. Dilution	dH:-909. (3SF) kJ	2	CRV	31308
54	25	0	GAMSPATH	dH:-910.650 (6SF) kJ	3	CRV	12638
55	25	0	Inf. Dilution/m	dH:-910.700 (6SF) kJ	3	CRV	24953

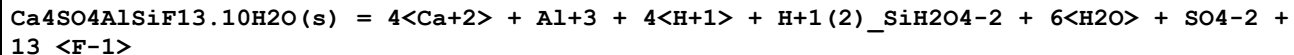
56	127	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6422
57	127	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6494
58	227	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6422
59	227	0	Inf. Dilution	dH:-217.6(4SF)	2	MTD	6494
60	327	0	Inf. Dilution	dH:-217.5(4SF)	0	MTD	6422
61	327	0	Inf. Dilution	dH:-217.4(4SF)	0	MTD	6494
62	25	0	Inf. Dilution	Cp:44.602(0.15SD)J/K	5	CRV	27292

Reaction No. 57977



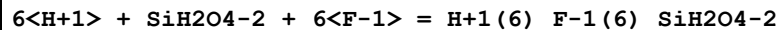
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1573.(4SF)kJ	5	CRV	8746
2	25	0	Inf. Dilution	dG:-1572.8(0.814SD)kJ	6	CRV	14923
3	25	0	Inf. Dilution	dH:-386.0(0.2SD)	5	CRV	5605
4	25	0	Inf. Dilution	dH:-1615.0(0.8MD)kJ	5	CRV	8746
5	25	0	Inf. Dilution	dH:-1615.0(0.800SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	Cp:73.622(0.25SD)J/K	5	CRV	27292

Reaction No. 58082



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:-78.77(4SF)	3	ENS	
Note (1): http://www.fipr.state.fl.us/fipr100.htm							
2	25	0	Inf. Dilution	lgK:-78.8(3SF)	2	EAE	

Reaction No. 58127



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:53.19(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:52.05(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:51.85(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:51.75(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:51.70(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:51.69(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:51.73(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:51.78(0.01SD)	4	EDH	5390

Reaction No. 58128							
$4\text{<H+1>} + \text{SiH}_2\text{O}_4\text{-2} + 2\text{<F-1>} = \text{H+1(4)}_{\text{-F-1(2)}}_{\text{-SiH}_2\text{O}_4\text{-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:34.38(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:33.42(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:33.23(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:32.13(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:32.04(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:32.01(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:32.02(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:32.02(0.01SD)	4	EDH	5390

Reaction No. 58185							
$2\text{<H+1>} + \text{SiH}_2\text{O}_4\text{-2} = \text{SiO}_2(\text{am.,s}) + 2\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:25.98(4SF)	0	CRV	9402
3	25	0	Inf. Dilution	dH:-19.40(4SF)	2	CRV	9402

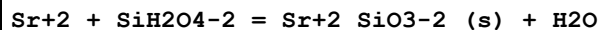
Reaction No. 58186							
$2\text{<H+1>} + \text{SiH}_2\text{O}_4\text{-2} = \text{SiO}_2(\text{Quartz,s}) + 2\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.94(4SF)	4	CRV	9402
2	25	0	Inf. Dilution	dH:-21.00(4SF)	4	CRV	9402

Reaction No. 58203							
$2\text{<Mn+2>} + \text{SiH}_2\text{O}_4\text{-2} = 2\text{MnO.SiO}_2(\text{s}) + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:10.70(4SF)	0	CRV	9402
Note (2): Low wgt for inter-reaction consistency							

Reaction No. 58204							
$2\text{<Fe+2>} + \text{SiH}_2\text{O}_4\text{-2} = 2\text{FeO.SiO}_2(\text{s}) + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0(2SF)	1	ELC	

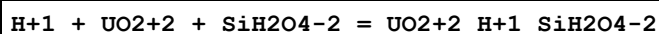
2	25	0	Inf. Dilution	lgK:5.74 (3SF)	0	CRV	9402
3	25	0	Inf. Dilution	dH:21.70 (4SF)	2	CRV	9402
4	25	0	Inf. Dilution	Cp:-34.00 (4SF)	2	CRV	9402

Reaction No. 58205



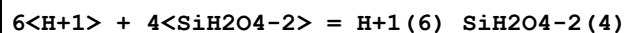
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.40 (3SF)	0	CRV	9402

Reaction No. 58219



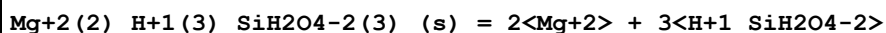
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:21.06 (0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:20.30 (0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:20.15 (0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:20.07 (0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:20.01 (0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:19.97 (0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:19.95 (0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:19.93 (0.01SD)	4	EDH	5390

Reaction No. 59987



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:79.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:78.20 (4SF)	0	CRV	9402

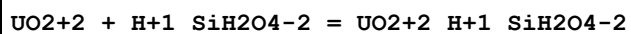
Reaction No. 65765



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0	Inf. Dilution	lgK:-38.8 (3SF)	0	CRV	552

Note (1): Solid +1 charge! Same problem in S&M hardcopy [76SmM_8]. Sepiolite?

Reaction No. 65766



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:7.97(0.10SD)	5	CRV	25812
2	25	0.1	Unknown	lgK:7.5(2SF)	6	CRV	552

Reaction No. 65767							
Fe+3 + H+1_SiH2O4-2 = Fe+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:9.72(0.04SD)	4	MSP	29280
3	25	0	Inf. Dilution/m	lgK:9.7(0.3SD)	2	EOD	29280
Note (3): Taken from Hummel (2012), The PSI/Nagra Chemical Thermo Database							
4	25	0.1	Unknown	lgK:8.9(2SF)	0	CRV	552
5	20:30	0.1	Unknown	dH:4.(1SF)	0	CRV	552

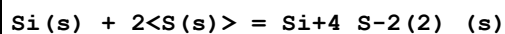
Reaction No. 65965							
H+1(2)_SiH2O4-2 + 4<H+1> + 6<F-1> = Si+4_F-1(6) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.1(0.08SD)	4	CRV	552
2	25	0	Inf. Dilution	lgK:27.01(4SF)	0	CRV	32224
3	25	0.1	Unknown	lgK:29.48(4SF)	6	CRV	552
4	25	1	Unknown	lgK:29.96(0.02SD)	6	CRV	552
5	25	2	Unknown	lgK:30.88(4SF)	6	CRV	552
6	25	3	Unknown	lgK:32.33(4SF)	6	CRV	552

Reaction No. 66082							
CaO.Al2O3.2SiO2(s) + 2<H+1> + 6<H2O> = 2<Al+3_OH-1(3)_(Gibbsite,s)> + 2<H+1(2)_SiH2O4-2> + Ca+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.16(4SF)	0	ENS	6405

Reaction No. 66239							
Zn(s) + Si(s) + 1.5<O2(g)> = ZnSiO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:206.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:205.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:150.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:117.3(4SF)	4	ENS	6422

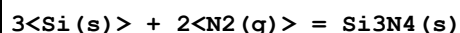
5	327	0	Inf. Dilution	lgK:95.32 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-281.9 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-301.8 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-301.9 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-301.9 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-302.0 (4SF)	4	MTD	6422

Reaction No. 66424



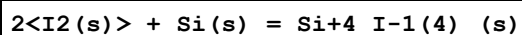
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.25 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:37.02 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:27.73 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:22.04 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.22 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-50.81 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-51.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-51.8 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-52.3 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-52.7 (3SF)	4	MTD	6422

Reaction No. 66425



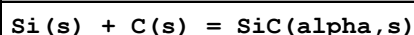
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.4 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:112.6 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:80.15 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:60.64 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:47.62 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-154.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-178.0 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-178.4 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-178.7 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-178.8 (4SF)	4	MTD	6422

Reaction No. 66426



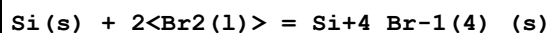
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.57 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:33.37 (4SF)	0	ENS	6422
3	121	0	Inf. Dilution	lgK:25.40 (4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-45.80 (4SF)	3	EFE	6422
5	25	0	Inf. Dilution	dG:-45.802 (5SF)	3	CRV	31291
6	25	0	Inf. Dilution	dH:-45.30 (4SF)	2	MTD	6422
7	25	0	Inf. Dilution	dH:-45.300 (5SF)	3	CRV	31291
8	121	0	Inf. Dilution	dH:-53.36 (4SF)	1	MTD	6422

Reaction No. 66427



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.41 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.33 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:9.143 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:7.228 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.952 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-16.93 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-17.5 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-17.5 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-17.5 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-17.5 (3SF)	4	MTD	6422

Reaction No. 66428



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.77 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:77.27 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:55.82 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:42.46 (4SF)	0	ENS	6422
5	25	0	Inf. Dilution	dG:-106.1 (4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-109.3 (4SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-122.8 (4SF)	0	MTD	6422
8	227	0	Inf. Dilution	dH:-121.6 (4SF)	0	MTD	6422

Reaction No. 66477

2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Kyan.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.3 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:428.2 (4SF)	2	ENS	6422
3	25	0	Inf. Dilution	lgK:428.0 (4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:425.4 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:425.2 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.4 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:312.3 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.6 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.5 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.5 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.4 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-584.1 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.98 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-583.9 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2443.90 (6SF) kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.88 (5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-620.5 (4SF)	0	CRV	12011
18	50	0	Inf. Dilution	dG:-584.41 (5SF)	0	CRV	10174
19	75	0	Inf. Dilution	dG:-585.00 (5SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-585.65 (5SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-586.36 (5SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-587.12 (5SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-587.94 (5SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-588.80 (5SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-589.72 (5SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-590.67 (5SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-592.72 (5SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-620.0 (4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-620.5 (4SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-620.03 (5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-619.9 (4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-2594.30 (6SF) kJ	0	CRV	9015
33	25	0	Inf. Dilution	dH:-656.4 (4SF)	0	CRV	12011
34	127	0	Inf. Dilution	dH:-620.2 (4SF)	0	MTD	6422

35	127	0	Inf. Dilution	dH:-620.1 (4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-620.1 (4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-620.0 (4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-619.8 (4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-619.7 (4SF)	0	MTD	6494

Reaction No. 66478							
2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Andalu.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:428.0 (4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:427.8 (4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:425.2 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:425.0 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.4 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:312.2 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.7 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.6 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.6 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.5 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-583.9 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.60 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-583.6 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2442.70 (6SF) kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.51 (5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-620.8 (4SF)	0	CRV	12011
18	25	0	Inf. Dilution	dG:-583.16 (5SF)	5	CRV	29553
19	50	0	Inf. Dilution	dG:-584.10 (5SF)	0	CRV	10174
20	75	0	Inf. Dilution	dG:-584.75 (5SF)	0	CRV	10174
21	100	0	Inf. Dilution	dG:-585.45 (5SF)	0	CRV	10174
22	125	0	Inf. Dilution	dG:-586.21 (5SF)	0	CRV	10174
23	150	0	Inf. Dilution	dG:-587.03 (5SF)	0	CRV	10174
24	175	0	Inf. Dilution	dG:-587.90 (5SF)	0	CRV	10174
25	200	0	Inf. Dilution	dG:-588.82 (5SF)	0	CRV	10174
26	225	0	Inf. Dilution	dG:-589.79 (5SF)	0	CRV	10174
27	250	0	Inf. Dilution	dG:-590.80 (5SF)	0	CRV	10174

28	300	0	Inf. Dilution	dG:-592.95 (5SF)	0	CRV	10174
29	25	0	Inf. Dilution	dH:-619.1 (4SF)	0	MTD	6422
30	25	0	Inf. Dilution	dH:-619.4 (4SF)	0	ENS	6437
31	25	0	Inf. Dilution	dH:-619.02 (5SF)	0	EFE	6482
32	25	0	Inf. Dilution	dH:-619.0 (4SF)	0	MTD	6494
33	25	0	Inf. Dilution	dH:-2590.30 (6SF) kJ	0	CRV	9015
34	25	0	Inf. Dilution	dH:-655.9 (4SF)	0	CRV	12011
35	25	0	Inf. Dilution	dH:-618.57 (5SF)	5	CRV	29553
36	127	0	Inf. Dilution	dH:-619.3 (4SF)	0	MTD	6422
37	127	0	Inf. Dilution	dH:-619.2 (4SF)	0	MTD	6494
38	227	0	Inf. Dilution	dH:-619.2 (4SF)	0	MTD	6422
39	227	0	Inf. Dilution	dH:-619.0 (4SF)	0	MTD	6494
40	327	0	Inf. Dilution	dH:-618.9 (4SF)	0	MTD	6422
41	327	0	Inf. Dilution	dH:-618.7 (4SF)	0	MTD	6494

Reaction No. 66479							
2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Sillim.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.2 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:427.7 (4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:427.3 (4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:424.9 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:424.5 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.2 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:311.9 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.6 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.4 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.5 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.3 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-583.4 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.00 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-582.9 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2441.00 (6SF) kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.02 (5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-627.6 (4SF)	0	CRV	12011
18	50	0	Inf. Dilution	dG:-583.63 (5SF)	0	CRV	10174

19	75	0	Inf. Dilution	dG:-584.30 (5SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-585.02 (5SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-585.81 (5SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-586.65 (5SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-587.54 (5SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-588.48 (5SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-589.46 (5SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-590.49 (5SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-592.68 (5SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-618.5 (4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-618.5 (4SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-618.09 (5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-618.1 (4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-2587.80 (6SF) kJ	0	CRV	9015
33	25	0	Inf. Dilution	dH:-662.6 (4SF)	0	CRV	12011
34	127	0	Inf. Dilution	dH:-618.7 (4SF)	0	MTD	6422
35	127	0	Inf. Dilution	dH:-618.3 (4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-618.6 (4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-618.2 (4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-618.4 (4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-617.9 (4SF)	0	MTD	6494

Reaction No. 66480							
2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O7.2H2O(Dick.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.0 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:665.0 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:660.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:660.6 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:481.2 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:481.3 (4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:373.6 (4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:373.7 (4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:301.9 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:302.0 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-907.2 (4SF)	4	EFE	6422

12	25	0	Inf. Dilution	dG:-907.3 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dG:-3795.9 (5SF) kJ	4	CRV	9015
14	25	0	GAMSPATH	dG:-3795. (4SF) kJ	3	CRV	12638
15	25	0	Inf. Dilution	dH:-984.3 (4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-984.3 (4SF)	4	MTD	6494
17	25	0	Inf. Dilution	dH:-4118.3 (5SF) kJ	4	CRV	9015
18	27	0	Inf. Dilution	dH:-984.4 (4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-984.8 (4SF)	3	MTD	6422
20	127	0	Inf. Dilution	dH:-984.7 (4SF)	3	MTD	6494
21	227	0	Inf. Dilution	dH:-984.7 (4SF)	2	MTD	6422
22	227	0	Inf. Dilution	dH:-984.6 (4SF)	2	MTD	6494
23	327	0	Inf. Dilution	dH:-984.2 (4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-984.0 (4SF)	0	MTD	6494

Reaction No. 66498							
4<Ca(s)> + C(s) + 6<Si(s)> + 6<Al(s)> + 13.5<O2(g)> = CaCO3.3CaO.3Al2O3.6SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2301. (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:2286. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1681. (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:1319. (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:1077. (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-3132.4 (5SF)	0	EFE	6482
Note (6): Low wgt for internal consistency							
7	25	0	Inf. Dilution	dG:-3139. (4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-3315.3 (5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-3310.2 (5SF)	0	EFE	6482
10	25	0	Inf. Dilution	dH:-3318. (4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-3318. (4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-3317. (4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-3315. (4SF)	4	MTD	6494

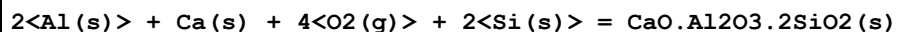
Reaction No. 66535							
2<Na(s)> + 3<Mg(s)> + 2<Al(s)> + 8<Si(s)> + H2(g) + 12<O2(g)> = Na2Mg3Al2Si8O22(OH)2(Glauc.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1968. (4SF)	4	ENS	6494

2	27	0	Inf. Dilution	lgK:1955.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1434.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:1121.(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:912.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2684.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2857.(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-2860.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-2862.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2861.(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-2860.(4SF)	4	MTD	6494

Reaction No. 66557							
Ca(s) + 2<Al(s)> + Si(s) + 3<O2(g)> = CaO.Al2O3.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:544.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:547.0(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:548.3(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:543.4(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:544.7(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:399.8(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:400.8(4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:313.7(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:314.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:256.4(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:256.9(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-746.2(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-746.25(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-748.0(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-788.3(4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-788.42(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-790.2(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-788.2(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-790.4(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-787.8(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-790.2(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-787.1(4SF)	0	MTD	6422

23	327	0	Inf. Dilution	dH:-789.7 (4SF)	0	MTD	6494
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Reaction No. 66558



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:701.2 (4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:702.2 (4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:696.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:697.6 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:512.6 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:513.2 (4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:402.1 (4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:402.7 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:328.6 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:329.0 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-956.6 (4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-956.8 (4SF)	0	EFE	6482
13	25	0	Inf. Dilution	dG:-957.9 (4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dG:-4002.10 (6SF) kJ	6	EFE	8798
Note (14): To avoid confusion note source units (J/mol) & value c.a.= -4 million							
15	25	0	Inf. Dilution	dG:-4002.1 (5SF) kJ	4	MTD	12843
16	25	0	Inf. Dilution	dH:-1010. (4SF)	4	MTD	6422
17	25	0	Inf. Dilution	dH:-1011.6 (5SF)	0	ENS	6437
18	25	0	Inf. Dilution	dH:-1010.7 (5SF)	0	EFE	6482
19	25	0	Inf. Dilution	dH:-1012. (4SF)	0	MTD	6494
20	25	0	Inf. Dilution	dH:-4227.80 (6SF) kJ	6	EFE	8798
21	25	0	Inf. Dilution	dH:-4227.8 (5SF) kJ	4	MTD	12843
22	25	0	GAMSPATH	dH:-4216.5 (5SF) kJ	0	CRV	12638
23	127	0	Inf. Dilution	dH:-1011. (4SF)	2	MTD	6422
24	127	0	Inf. Dilution	dH:-1012. (4SF)	2	MTD	6494
25	227	0	Inf. Dilution	dH:-1010. (4SF)	1	MTD	6422
26	227	0	Inf. Dilution	dH:-1012. (4SF)	1	MTD	6494
27	327	0	Inf. Dilution	dH:-1010. (4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-1011. (4SF)	0	MTD	6494

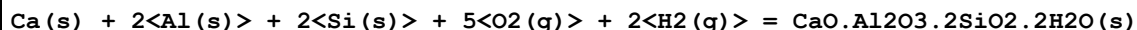
Reaction No. 66559

2<Ca(s)> + 2<Al(s)> + Si(s) + 3.5<O2(g)> = 2CaO.Al2O3.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:662.7(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:663.2(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:658.4(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:658.9(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:485.1(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:485.4(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:381.2(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:381.3(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:311.9(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:312.0(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-904.1(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-904.86(5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-904.8(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-951.6(4SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-950.7(4SF)	4	ENS	6437
16	25	0	Inf. Dilution	dH:-953.19(5SF)	4	EFE	6482
17	25	0	Inf. Dilution	dH:-952.4(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-951.5(4SF)	4	MTD	6422
19	127	0	Inf. Dilution	dH:-952.6(4SF)	4	MTD	6494
20	227	0	Inf. Dilution	dH:-951.0(4SF)	4	MTD	6422
21	227	0	Inf. Dilution	dH:-952.4(4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-950.4(4SF)	4	MTD	6422
23	327	0	Inf. Dilution	dH:-952.0(4SF)	4	MTD	6494

Reaction No. 66560							
3<Ca(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3CaO.Al2O3.3SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1098.0(5SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1100.(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:1100.(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:1093.(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:1093.(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:803.8(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:803.7(4SF)	0	ENS	6494

8	227	0	Inf. Dilution	lgK:630.2 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:630.3 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:514.6 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:514.7 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-1501. (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1498.8 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1501. (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-1588. (4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-1585.6 (5SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-1585.3 (5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dH:-1587. (4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-1589. (4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-1587. (4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-1588.0 (5SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-1587. (4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-1587. (4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-1586. (4SF)	0	MTD	6494

Reaction No. 66561



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:789.3 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:790.6 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:784.0 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:785.4 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:572.4 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:573.4 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:445.4 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:446.2 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:360.7 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:361.4 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1077. (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-1077.90 (6SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-1079. (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-1161.2 (5SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-1163.5 (5SF)	4	ENS	6437

16	25	0	Inf. Dilution	dH:-1162.9(5SF)	4	EFE	6482
17	25	0	Inf. Dilution	dH:-1164.(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-1162.(4SF)	1	MTD	6422
19	127	0	Inf. Dilution	dH:-1164.(4SF)	1	MTD	6494
20	227	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-1164.(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6494

Reaction No. 66571							
Ca(s) + Mg(s) + Si(s) + 2<O2(g)> = CaO.MgO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:374.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:375.9(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:373.6(4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:373.5(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:371.2(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:275.0(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:273.2(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:215.9(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:214.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:176.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:175.3(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-512.8(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-509.61(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-509.8(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-540.9(4SF)	0	MTD	6422
16	25	0	Inf. Dilution	dH:-536.9(4SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-537.77(5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dH:-538.0(4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-540.8(4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-538.0(4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-540.6(4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-537.7(4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-540.2(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-537.4(4SF)	0	MTD	6494

Reaction No. 66572							
Ca(s) + Mg(s) + 3<O2(g)> + 2<Si(s)> = CaO.MgO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:531.2(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:382.6(4SF)	0	ENS	6437
3	25	0	Inf. Dilution	lgK:530.3(4SF)	4	ENS	6494
4	27	0	Inf. Dilution	lgK:527.7(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:526.8(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:388.1(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:387.5(4SF)	3	ENS	6494
8	227	0	Inf. Dilution	lgK:304.4(4SF)	2	ENS	6422
9	227	0	Inf. Dilution	lgK:303.9(4SF)	2	ENS	6494
10	327	0	Inf. Dilution	lgK:248.6(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:248.1(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-724.7(4SF)	2	EFE	6422
13	25	0	Inf. Dilution	dG:-723.28(5SF)	3	EFE	6482
14	25	0	Inf. Dilution	dG:-723.4(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-3026.8(5SF) kJ	3	EOD	12843
16	25	0	Inf. Dilution	dH:-766.3(4SF)	2	MTD	6422
17	25	0	Inf. Dilution	dH:-764.96(5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-765.2(4SF)	4	MTD	6494
19	25	0	Inf. Dilution	dH:-3201.5(5SF) kJ	3	EOD	12843
20	127	0	Inf. Dilution	dH:-766.4(4SF)	2	MTD	6422
21	127	0	Inf. Dilution	dH:-765.2(4SF)	2	MTD	6494
22	227	0	Inf. Dilution	dH:-766.1(4SF)	2	MTD	6422
23	227	0	Inf. Dilution	dH:-764.9(4SF)	2	MTD	6494
24	327	0	Inf. Dilution	dH:-765.6(4SF)	0	MTD	6422
25	327	0	Inf. Dilution	dH:-764.4(4SF)	0	MTD	6494

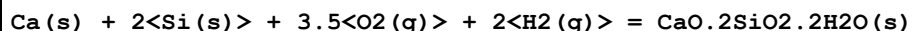
Reaction No. 66573							
2<Ca(s)> + Mg(s) + 2<Si(s)> + 3.5<O2(g)> = 2CaO.MgO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:643.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:644.7(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:642.5(4SF)	3	ENS	6494

4	27	0	Inf. Dilution	lgK:640.5 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:638.4 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:471.7 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:470.2 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:370.5 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:369.3 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:303.0 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:302.1 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-879.5 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-875.66 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-876.6 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-926.7 (4SF)	0	MTD	6422
16	25	0	Inf. Dilution	dH:-924.4 (4SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-922.67 (5SF)	0	MTD	6482
18	25	0	Inf. Dilution	dH:-923.7 (4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-926.6 (4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-923.6 (4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-926.2 (4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-923.3 (4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-925.7 (4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-923.0 (4SF)	0	MTD	6494

Reaction No. 66574							
$2\text{Ca(s)} + 5\text{Mg(s)} + 8\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = 2\text{CaO.5MgO.8SiO}_2.\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2038. (4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:2028. (4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:2025. (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:2014. (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1487. (4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:1479. (4SF)	1	ENS	6494
7	227	0	Inf. Dilution	lgK:1164. (4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:1158. (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:948.6 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:943.5 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-2780. (4SF)	0	EFE	6422

12	25	0	Inf. Dilution	dG:-2767.3 (5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-2766. (4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dH:-2954. (4SF)	0	MTD	6422
15	25	0	Inf. Dilution	dH:-2943. (4SF)	2	ENS	6437
16	25	0	Inf. Dilution	dH:-2941.1 (5SF)	3	EFE	6482
17	25	0	Inf. Dilution	dH:-2941. (4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-2954. (4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-2941. (4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-2954. (4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-2940. (4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-2952.2 (5SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-2938. (4SF)	0	MTD	6494

Reaction No. 66575



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:503.4 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:500.0 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:363.3 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:281.2 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:226.4 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-686.8 (4SF)	4	ENS	6422
7	25	0	Inf. Dilution	dH:-750.0 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-751.3 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-752.1 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-752.8 (4SF)	4	MTD	6422

Reaction No. 66576



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:795.6 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:790.3 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:576.1 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:447.5 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:361.8 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1085. (4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1177.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422

Reaction No. 66577							
$2\text{Ca(s)} + \text{Si(s)} + 2.585\text{O}_2\text{(g)} + 1.165\text{H}_2\text{(g)} = 2\text{CaO.SiO}_2.7\text{by6H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:434.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:431.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:315.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:245.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:199.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-592.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-637.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-637.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-636.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-636.5(4SF)	4	MTD	6422

Reaction No. 66578							
$3\text{Ca(s)} + 2\text{Si(s)} + 5\text{O}_2\text{(g)} + 3\text{H}_2\text{(g)} = 3\text{CaO.2SiO}_2.3\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:771.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:766.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:558.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:433.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:350.2(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1053.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1143.0(5SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1143.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1142.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1141.(4SF)	4	MTD	6422

Reaction No. 66579							
$4\text{Ca(s)} + 3\text{Si(s)} + 5.75\text{O}_2\text{(g)} + 1.5\text{H}_2\text{(g)} = 4\text{CaO.3SiO}_2.3\text{by2H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:988.6(4SF)	4	ENS	6422

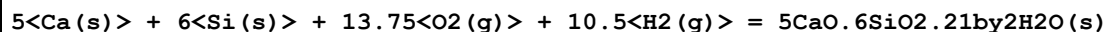
2	27	0	Inf. Dilution	lgK:982.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:720.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:562.8(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:458.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1349.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1439.0(5SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1439.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1439.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1438.(4SF)	4	MTD	6422

Reaction No. 66580							
$5\text{Ca(s)} + 6\text{Si(s)} + 10\text{O}_2\text{(g)} + 3\text{H}_2\text{(g)} = 5\text{CaO.6SiO}_2.3\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1622.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1611.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1179.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:919.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:746.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2213.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2373.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2373.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2372.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2371.(4SF)	4	MTD	6422

Reaction No. 66581							
$5\text{Ca(s)} + 6\text{Si(s)} + 11.25\text{O}_2\text{(g)} + 5.5\text{H}_2\text{(g)} = 5\text{CaO.6SiO}_2.11\text{by}_2\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1729.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1718.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1253.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:973.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:787.6(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2359.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2554.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2554.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2553.(4SF)	4	MTD	6422

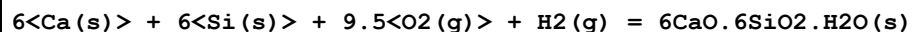
10	327	0	Inf. Dilution	dH:-2551.(4SF)	4	MTD	6422
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Reaction No. 66582



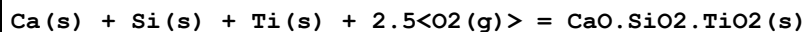
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2	27	0	Inf. Dilution	lgK:1927.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1397.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:1079.(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:866.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2647.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2911.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2911.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2910.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2906.5(5SF)	4	MTD	6422

Reaction No. 66583



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1655.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1644.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1208.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:946.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:771.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2258.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2395.(4SF)	4	MTD	6422

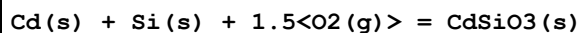
Reaction No. 66587



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:431.3(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:430.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:428.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:427.2(4SF)	0	ENS	6494

5	127	0	Inf. Dilution	lgK:315.2 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:314.2 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:247.2 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:246.4 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:201.9 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:201.2 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-588.4 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-586.79 (5SF)	5	EFE	6482
13	25	0	Inf. Dilution	dG:-586.7 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-622.2 (4SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-620.61 (5SF)	5	EFE	6482
16	25	0	Inf. Dilution	dH:-620.6 (4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-622.1 (4SF)	1	MTD	6422
18	127	0	Inf. Dilution	dH:-620.5 (4SF)	1	MTD	6494
19	227	0	Inf. Dilution	dH:-621.7 (4SF)	0	MTD	6422
20	227	0	Inf. Dilution	dH:-620.2 (4SF)	0	MTD	6494
21	327	0	Inf. Dilution	dH:-621.3 (4SF)	0	MTD	6422
22	327	0	Inf. Dilution	dH:-619.8 (4SF)	0	MTD	6494

Reaction No. 66611



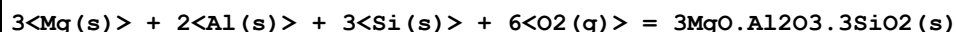
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1	25	0	Inf. Dilution	lgK:193.7 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:192.4 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:140.6 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:109.6 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:88.87 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-264.2 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1105.40 (6SF) kJ	4	CRV	9015
8	25	0	Inf. Dilution	dG:-264.20 (5SF)	4	CRV	12011
9	25	0	Inf. Dilution	dH:-284.2 (4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-1189.10 (6SF) kJ	4	CRV	9015
11	25	0	Inf. Dilution	dH:-284.20 (5SF)	4	CRV	12011
12	127	0	Inf. Dilution	dH:-284.2 (4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-284.0 (4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-285.3 (4SF)	0	MTD	6422

Reaction No. 66729							
Fe(s) + 1.5<O2(g)> + Si(s) = FeO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:195.8(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:195.9(4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK:194.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:194.6(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:142.5(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:142.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:111.3(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:111.4(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:90.51(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:90.64(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-267.1(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-267.08(5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-267.2(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-1118.0(5SF)kJ	3	EOD	12843
15	25	0	Inf. Dilution	dH:-285.6(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-285.24(5SF)	3	ENS	6437
17	25	0	Inf. Dilution	dH:-285.46(5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-285.7(4SF)	4	MTD	6494
19	25	0	Inf. Dilution	dH:-1195.2(5SF)kJ	3	EOD	12843
20	127	0	Inf. Dilution	dH:-285.5(4SF)	1	MTD	6422
21	127	0	Inf. Dilution	dH:-285.4(4SF)	1	MTD	6494
22	227	0	Inf. Dilution	dH:-285.3(4SF)	0	MTD	6422
23	227	0	Inf. Dilution	dH:-285.0(4SF)	0	MTD	6494
24	327	0	Inf. Dilution	dH:-285.1(4SF)	0	MTD	6422
25	327	0	Inf. Dilution	dH:-284.6(4SF)	0	MTD	6494

Reaction No. 66776							
3<Fe(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3FeO.Al2O3.3SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:865.8(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:860.1(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:630.9(4SF)	4	ENS	6494

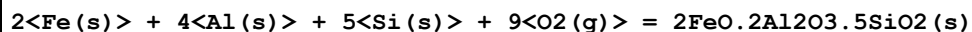
4	227	0	Inf. Dilution	lgK:493.5 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:401.9 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1181.1 (5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-1181. (4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1261.1 (5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1258.5 (5SF)	4	MTD	6482
10	25	0	Inf. Dilution	dH:-1258. (4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-1258. (4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-1258. (4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-1257. (4SF)	4	MTD	6494

Reaction No. 66777



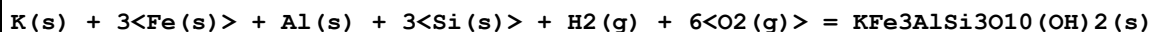
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1040. (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1033. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:759.3 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:595.1 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:485.7 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1418.7 (5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-1418. (4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1503.7 (5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1502.5 (5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-1502. (4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-1502. (4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-1502. (4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-1501. (4SF)	4	MTD	6494

Reaction No. 66778



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2018.9 (5SF)	4	ENS	6437

Reaction No. 66779



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:847.9 (4SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:840.6(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:835.1(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:610.9(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:476.5(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:386.9(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1147.(4SF)	0	EFE	6494
8	25	0	Inf. Dilution	dH:-1225.0(5SF)	0	ENS	6437
9	25	0	Inf. Dilution	dH:-1231.(4SF)	1	MTD	6494
10	127	0	Inf. Dilution	dH:-1231.(4SF)	0	MTD	6494
11	227	0	Inf. Dilution	dH:-1230.(4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1228.(4SF)	0	MTD	6494

Reaction No. 66780							
$K(s) + 3Mg(s) + Al(s) + 3Si(s) + H_2(g) + 6O_2(g) = KMg_3AlSi_3O_{10}(OH)_2(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1024.0(5SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1027.(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:1020.(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:748.0(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:584.8(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:476.0(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1392.7(5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1401.(4SF)	0	EFE	6494
9	25	0	Inf. Dilution	dH:-1482.8(5SF)	2	ENS	6437
10	25	0	Inf. Dilution	dH:-1483.6(5SF)	2	EFE	6482
11	25	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
12	127	0	Inf. Dilution	dH:-1494.(4SF)	0	MTD	6494
13	227	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
14	327	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494

Reaction No. 66781							
$2Mg(s) + 4Al(s) + 5Si(s) + H_2(g) + 9.5O_2(g) = 2MgO \cdot 2Al_2O_3 \cdot 5SiO_2 \cdot H_2O(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2253.9(5SF)	4	ENS	6437

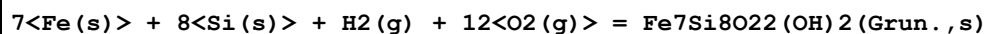
Reaction No. 66782							
$31H_2(g) + 48Mg(s) + 73.5O_2(g) + 34Si(s) = Mg_{48}Si_{34}O_{85}(OH)_2(s)$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.1597E+4 (5SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1.5792E+4 (5SF)	0	EFE	6482
3	25	0	GAMSPATH	dG:-6.614E+4 (4SF) kJ	0	CRV	12638
Note (3): Low wgt for internal consistency							
4	25	0	Inf. Dilution	dH:-1.7060E+4 (5SF)	0	ENS	6437
5	25	0	Inf. Dilution	dH:-1.7056E+4 (5SF)	0	EFE	6482
6	25	0	GAMSPATH	dH:-7.14250E+4 (6SF) kJ	0	CRV	12638

Reaction No. 66783							
3<Al(s)> + H2(g) + Na(s) + 6<O2(g)> + 3<Si(s)> = NaAl3Si3O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:975.6 (4SF)	2	ENS	6494
2	27	0	Inf. Dilution	lgK:969.1 (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:710.1 (4SF)	0	ENS	6494
4	227	0	Inf. Dilution	lgK:554.6 (4SF)	0	ENS	6494
5	327	0	Inf. Dilution	lgK:450.9 (4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1329.7 (5SF)	0	EFE	6482
7	25	0	Inf. Dilution	dG:-1331. (4SF)	2	EFE	6494
8	25	0	Inf. Dilution	dG:-1331.70 (6SF)	0	CRV	10174
9	25	0	Inf. Dilution	dG:-1332.5 (5SF)	5	CRV	29553
10	50	0	Inf. Dilution	dG:-1333.40 (6SF)	0	CRV	10174
11	75	0	Inf. Dilution	dG:-1335.30 (6SF)	0	CRV	10174
12	100	0	Inf. Dilution	dG:-1337.30 (6SF)	0	CRV	10174
13	125	0	Inf. Dilution	dG:-1339.50 (6SF)	0	CRV	10174
14	150	0	Inf. Dilution	dG:-1341.80 (6SF)	0	CRV	10174
15	175	0	Inf. Dilution	dG:-1344.30 (6SF)	0	CRV	10174
16	200	0	Inf. Dilution	dG:-1346.90 (6SF)	0	CRV	10174
17	225	0	Inf. Dilution	dG:-1349.60 (6SF)	0	CRV	10174
18	250	0	Inf. Dilution	dG:-1352.40 (6SF)	0	CRV	10174
19	300	0	Inf. Dilution	dG:-1358.30 (6SF)	0	CRV	10174
20	25	0	Inf. Dilution	dH:-1422.3 (5SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-1420.7 (5SF)	0	EFE	6482
22	25	0	Inf. Dilution	dH:-1422. (4SF)	4	MTD	6494
23	25	0	Inf. Dilution	dH:-1423.60 (6SF)	5	CRV	29553
24	127	0	Inf. Dilution	dH:-1423. (4SF)	0	MTD	6494

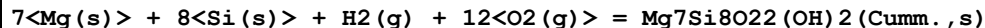
25	227	0	Inf. Dilution	dH:-1423.(4SF)	0	MTD	6494
26	327	0	Inf. Dilution	dH:-1422.(4SF)	0	MTD	6494

Reaction No. 66784



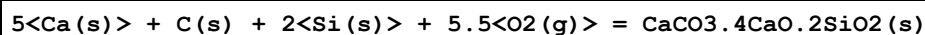
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1571.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1560.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1141.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:890.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:722.7(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2143.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2302.(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-2300.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-2300.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2298.(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-2296.(4SF)	4	MTD	6494

Reaction No. 66785



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2920.(4SF)	4	ENS	6437

Reaction No. 66786



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:968.0(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:961.7(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:707.5(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:555.1(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:453.6(4SF)	3	ENS	6494
6	25	0	Inf. Dilution	dG:-1321.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1395.8(5SF)	3	ENS	6437
8	25	0	Inf. Dilution	dH:-1396.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-1395.(4SF)	3	MTD	6494
10	227	0	Inf. Dilution	dH:-1394.(4SF)	0	MTD	6494
11	327	0	Inf. Dilution	dH:-1393.(4SF)	0	MTD	6494

Reaction No. 66787							
$\text{Mg(s)} + 2\text{Al(s)} + 2\text{Si(s)} + 5\text{O}_2\text{(g)} + 2\text{H}_2\text{(g)} = \text{MgAl}_2\text{Si}_2\text{O}_6\text{(OH)}_4\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1146.(4SF)	4	ENS	6437

Reaction No. 66788							
$\text{Mg(s)} + 2\text{Al(s)} + \text{Si(s)} + 3.5\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{MgAl}_2\text{SiO}_5\text{(OH)}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-850.4(4SF)	4	ENS	6437

Reaction No. 66789							
$4\text{Mg(s)} + 18\text{Al(s)} + 7.5\text{Si(s)} + 24\text{O}_2\text{(g)} + 2\text{H}_2\text{(g)} = \text{Mg}_4\text{Al}_{18}\text{Si}_{7.5}\text{O}_{48}\text{H}_4\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-6003.5(5SF)	4	ENS	6437

Reaction No. 66790							
$\text{Na(s)} + \text{Fe(s)} + 2\text{Si(s)} + 3\text{O}_2\text{(g)} = \text{NaFeSi}_2\text{O}_6\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:423.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:420.7(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:308.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:240.6(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:195.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-577.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-617.7(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-617.7(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-618.4(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-618.2(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-617.8(4SF)	4	MTD	6494

Reaction No. 66791							
$4\text{Mg(s)} + 4\text{Al(s)} + 2\text{Si(s)} + 9\text{O}_2\text{(g)} + 4\text{H}_2\text{(g)} = \text{Mg}_4\text{Al}_4\text{Si}_2\text{O}_{10}\text{(OH)}_8\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2162.(4SF)	4	ENS	6437

Reaction No. 66792							
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Al(s) + H2(g) + K(s) + Mg(s) + 6<O2(g)> + 4<Si(s)> = KMgAlSi4O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5464.10(6SF) kJ	3	CRV	25435
2	25	0	GAMSPATH	dG:-7971.(4SF) kJ	0	CRV	12638
3	25	0	Inf. Dilution	dH:-1394.4(5SF)	2	ENS	6437
4	25	0	Inf. Dilution	dH:-5844.50(6SF) kJ	3	CRV	25435

Reaction No. 66793							
2<Al(s)> + 5<Fe(s)> + 4<H2(g)> + 9<O2(g)> + 3<Si(s)> = Fe5Al2Si3O10(OH)8(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-6535.60(6SF) kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-1708.5(5SF)	2	ENS	6437
3	25	0	Inf. Dilution	dH:-7154.00(6SF) kJ	3	CRV	25435

Reaction No. 66794							
18<Fe(s)> + 12<Si(s)> + 25<O2(g)> + 5<H2(g)> = Fe18Si12O40(OH)10(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-4384.4(5SF)	4	ENS	6437

Reaction No. 66795							
K(s) + 2<Mg(s)> + 3<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = KMg2Al3Si2O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1514.5(5SF)	4	ENS	6437

Reaction No. 66796							
Na(s) + 2<Ca(s)> + 5<Mg(s)> + 7<Si(s)> + Al(s) + 12<O2(g)> + H2(g) = NaCa2Mg5Si7AlO22(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3006.8(5SF)	4	ENS	6437

Reaction No. 66797							
4<Al(s)> + 4<Fe(s)> + 4<H2(g)> + 9<O2(g)> + 2<Si(s)> = Fe4Al4Si2O10(OH)8(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-7167.(4SF) kJ	3	CRV	12638
2	25	0	Inf. Dilution	dH:-1823.5(5SF)	4	ENS	6437
3	25	0	GAMSPATH	dH:-7784.4(5SF) kJ	0	CRV	12638
Note (3): Called '14A-Chamosite'							

Reaction No. 66798							
$7\text{Fe(s)} + 8\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Fe}_7\text{Si}_8\text{O}_{22}(\text{OH})_2(\text{Anth.,s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2300.6(5SF)	4	ENS	6437

Reaction No. 66799							
$\text{Fe(s)} + 2\text{Al(s)} + 2\text{Si(s)} + 5\text{O}_2\text{(g)} + 2\text{H}_2\text{(g)} = \text{FeAl}_2\text{Si}_2\text{O}_6(\text{OH})_4\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1060.9(5SF)	4	ENS	6437

Reaction No. 66800							
$\text{K(s)} + \text{Fe(s)} + \text{Al(s)} + 4\text{Si(s)} + 6\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{KFeAlSi}_4\text{O}_{10}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5127.80(6SF) kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-1310.9(5SF)	2	ENS	6437
3	25	0	Inf. Dilution	dH:-5497.30(6SF) kJ	3	CRV	25435

Reaction No. 66801							
$2\text{Na(s)} + 3\text{Fe(s)} + 2\text{Al(s)} + 8\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Na}_2\text{Fe}_3\text{Al}_2\text{Si}_8\text{O}_{22}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2605.4(5SF)	4	ENS	6437

Reaction No. 66802							
$2\text{Ca(s)} + 4\text{Fe(s)} + 2\text{Al(s)} + 7\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Ca}_2\text{Fe}_4\text{Al}_2\text{Si}_7\text{O}_{22}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2629.(4SF)	4	ENS	6437

Reaction No. 66803							
$2\text{Ca(s)} + 4\text{Mg(s)} + 2\text{Al(s)} + 7\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Ca}_2\text{Mg}_4\text{Al}_2\text{Si}_7\text{O}_{22}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2968.5(5SF)	4	ENS	6437

Reaction No. 66804							
$4\text{Fe(s)} + 18\text{Al(s)} + 7.5\text{Si(s)} + 24\text{O}_2\text{(g)} + 2\text{H}_2\text{(g)} = \text{Fe}_4\text{Al}_{18}\text{Si}_{7.5}\text{O}_{48}\text{H}_4\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-5675.2(5SF)	4	ENS	6437

Reaction No. 66805							
$2\text{Ca(s)} + 5\text{Fe(s)} + 8\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = 2\text{CaO} \cdot 5\text{FeO} \cdot 8\text{SiO}_2 \cdot \text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2520.(3SF)	4	ENS	6437

Reaction No. 66806							
$2\text{Na(s)} + 3\text{Mg(s)} + 2\text{Al(s)} + 8\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Na}_2\text{Mg}_3\text{Al}_2\text{Si}_8\text{O}_{22}(\text{OH})_2\text{(Rieb.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2650.(4SF)	4	ENS	6437

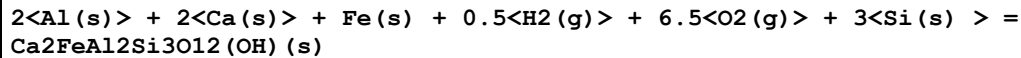
Reaction No. 66807							
$\text{Na(s)} + 2\text{Ca(s)} + 4\text{Mg(s)} + 3\text{Al(s)} + 6\text{Si(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{NaCa}_2\text{Mg}_4\text{Al}_3\text{Si}_6\text{O}_{22}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3040.1(5SF)	4	ENS	6437

Reaction No. 66808							
$\text{K(s)} + 2\text{Fe(s)} + 3\text{Al(s)} + 2\text{Si(s)} + 6\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{KFe}_2\text{Al}_3\text{Si}_2\text{O}_{10}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1345.2(5SF)	4	ENS	6437

Reaction No. 66809							
$\text{Ca(s)} + 4\text{Al(s)} + 2\text{Si(s)} + 6\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{CaAl}_4\text{Si}_2\text{O}_{10}(\text{OH})_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1026.(4SF)	1	ENS	6494
2	27	0	Inf. Dilution	lgK:1020.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:747.8(4SF)	0	ENS	6494
4	227	0	Inf. Dilution	lgK:584.7(4SF)	0	ENS	6494
5	327	0	Inf. Dilution	lgK:476.0(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1398.70(6SF)	1	EFE	6482
7	25	0	Inf. Dilution	dG:-1400.(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1492.5(5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1490.6(5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-1492.(4SF)	1	MTD	6494
11	127	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
12	227	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494

13	327	0	Inf. Dilution	dH:-1492.(4SF)	0	MTD	6494
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Reaction No. 66810



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-6072.(4SF) kJ	1	CRV	12638
2	25	0	Inf. Dilution	dH:-1544.5(5SF)	3	ENS	6437
3	25	0	GAMSPATH	dH:-6461.90(6SF) kJ	2	CRV	12638

Note (3): c.f. dG = -6072 & dH = -6462 kJ for 'Ordered Epidote'

Reaction No. 66811



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:760.4(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:754.7(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:755.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:749.8(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:556.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:552.3(4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:437.4(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:433.9(4SF)	0	ENS	6494
9	327	0	Inf. Dilution	lgK:358.0(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:355.0(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1037.(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-1030.00(6SF)	0	EFE	6482
13	25	0	Inf. Dilution	dG:-1030.(4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dH:-1092.(4SF)	2	MTD	6422
15	25	0	Inf. Dilution	dH:-1084.4(5SF)	2	ENS	6437
16	25	0	Inf. Dilution	dH:-1084.5(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-1092.(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-1091.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-1090.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-1083.(4SF)	0	MTD	6494

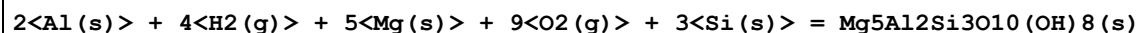
Reaction No. 66812							
$\text{Fe(s)} + 2\text{Al(s)} + \text{Si(s)} + 3.5\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{FeAl}_2\text{SiO}_5\text{(OH)}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-767.54 (5SF)	4	ENS	6437

Reaction No. 66814							
$\text{Mg(s)} + 1.5\text{O}_2\text{(g)} + \text{Si(s)} = \text{MgO.SiO}_2\text{(Enstat.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.5 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:253.8 (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:186.5 (4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:146.2 (4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:119.3 (4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-348.51 (5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-348.5 (4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dG:-1458.3 (5SF) kJ	3	EOD	12843
9	25	0	GAMSPATH	dG:-1480. (4SF) kJ	0	CRV	12638
10	25	0	Inf. Dilution	dH:-395.57 (5SF)	0	ENS	6437
11	25	0	Inf. Dilution	dH:-369.40 (5SF)	4	EFE	6482
12	25	0	Inf. Dilution	dH:-369.4 (4SF)	4	MTD	6494
13	25	0	Inf. Dilution	dH:-1549.00 (6SF) kJ	2	CRV	8695
14	25	0	Inf. Dilution	dH:-1545.6 (5SF) kJ	3	EOD	12843
15	25	0	GAMSPATH	dH:-1546.80 (6SF) kJ	3	CRV	12638
16	127	0	Inf. Dilution	dH:-369.4 (4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-369.2 (4SF)	0	MTD	6494
18	327	0	Inf. Dilution	dH:-369.0 (4SF)	0	MTD	6494

Reaction No. 66815							
$\text{K(s)} + \text{Al(s)} + 2\text{Si(s)} + 3\text{O}_2\text{(g)} = \text{KAlSi}_2\text{O}_6\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:503.1 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:503.7 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:499.8 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:500.4 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:367.6 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:368.1 (4SF)	3	ENS	6494

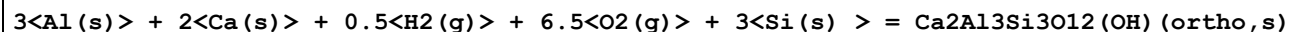
7	227	0	Inf. Dilution	lgK:288.2 (4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:288.7 (4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:235.3 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:235.7 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-686.3 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-687.2 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-725.2 (4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-722. (3SF)	4	ENS	6437
15	25	0	Inf. Dilution	dH:-726.1 (4SF)	4	MTD	6494
16	127	0	Inf. Dilution	dH:-726.2 (4SF)	3	MTD	6422
17	127	0	Inf. Dilution	dH:-726.9 (4SF)	3	MTD	6494
18	227	0	Inf. Dilution	dH:-726.6 (4SF)	2	MTD	6422
19	227	0	Inf. Dilution	dH:-726.8 (4SF)	2	MTD	6494
20	327	0	Inf. Dilution	dH:-726.8 (4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-726.4 (4SF)	0	MTD	6494

Reaction No. 66816



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1446.3 (5SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-1971.9 (5SF)	3	EFE	6482
3	25	0	Inf. Dilution	dG:-8263.40 (6SF) kJ	1	CRV	25435
4	25	0	GAMSPATH	dG:-8186. (4SF) kJ	0	CRV	12638
Note (4): Low wgt for internal consistency							
5	25	0	Inf. Dilution	dH:-2133.9 (5SF)	1	ENS	6437
6	25	0	Inf. Dilution	dH:-2129.4 (5SF)	3	EFE	6482
7	25	0	Inf. Dilution	dH:-8929.90 (6SF) kJ	1	CRV	25435
8	25	0	GAMSPATH	dH:-8841.6 (5SF) kJ	0	CRV	12638

Reaction No. 66817



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1140. (4SF)	1	EDH	
2	25	0	Inf. Dilution	lgK:1140. (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1132. (4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:831.7 (4SF)	2	ENS	6494

5	227	0	Inf. Dilution	lgK:651.4 (4SF)	1	ENS	6494
6	327	0	Inf. Dilution	lgK:531.3 (4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1552.1 (5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1555. (4SF)	4	EFE	6494
9	25	0	GAMSPATH	dG:-6502. (4SF) kJ	0	CRV	12638
10	25	0	Inf. Dilution	dH:-1648.9 (5SF)	2	ENS	6437
11	25	0	Inf. Dilution	dH:-1646.6 (5SF)	0	EFE	6482
12	25	0	Inf. Dilution	dH:-1649. (4SF)	4	MTD	6494
13	25	0	GAMSPATH	dH:-6879.00 (6SF) kJ	0	CRV	12638
14	127	0	Inf. Dilution	dH:-1650. (4SF)	1	MTD	6494
15	227	0	Inf. Dilution	dH:-1649. (4SF)	0	MTD	6494
16	327	0	Inf. Dilution	dH:-1649. (4SF)	0	MTD	6494

Reaction No. 66818							
$3\text{Fe(s)} + \text{H}_2(\text{g}) + 6\text{O}_2(\text{g}) + 4\text{Si(s)} = \text{Fe}_3\text{Si}_4\text{O}_{10}(\text{OH})_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:791.3 (4SF)	1	ELC	
2	25	0	GAMSPATH	dG:-4513. (4SF) kJ	0	CRV	12638
Note (2): Low wgt for inter-reaction consistency							
3	25	0	Inf. Dilution	dH:-1161.8 (5SF)	2	ENS	6437

Reaction No. 66819							
$2\text{Al(s)} + 2\text{Ca(s)} + \text{H}_2(\text{g}) + 6\text{O}_2(\text{g}) + 3\text{Si(s)} = \text{Ca}_2\text{Al}_2\text{Si}_3\text{O}_{10}(\text{OH})_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1020. (4SF)	3	ENS	6494
2	27	0	Inf. Dilution	lgK:1014. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:743.7 (4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:581.7 (4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:473.8 (4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1391.1 (5SF)	5	EFE	6482
7	25	0	Inf. Dilution	dG:-1392. (4SF)	3	EFE	6494
8	25	0	GAMSPATH	dG:-5813. (4SF) kJ	2	CRV	12638
9	25	0	Inf. Dilution	dH:-1481.8 (5SF)	3	ENS	6437
10	25	0	Inf. Dilution	dH:-1481.5 (5SF)	5	EFE	6482
11	25	0	Inf. Dilution	dH:-1483. (4SF)	3	MTD	6494
12	127	0	Inf. Dilution	dH:-1483. (4SF)	1	MTD	6494

13	227	0	Inf. Dilution	dH:-1482.(4SF)	0	MTD	6494
14	327	0	Inf. Dilution	dH:-1481.(4SF)	0	MTD	6494

Reaction No. 66820							
Na(s) + Al(s) + Si(s) + 2<O2(g)> = NaAlSiO4(Neph.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:347.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:346.9(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:346.1(4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:344.6(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:343.9(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:253.4(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:252.8(4SF)	1	ENS	6494
8	194	0	Inf. Dilution	lgK:214.1(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:198.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:161.8(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-473.2(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-472.2(4SF)	0	EFE	6494
13	25	0	Inf. Dilution	dG:-475.50(5SF)	0	CRV	10174
14	50	0	Inf. Dilution	dG:-476.27(5SF)	0	CRV	10174
15	75	0	Inf. Dilution	dG:-477.11(5SF)	0	CRV	10174
16	100	0	Inf. Dilution	dG:-478.00(5SF)	0	CRV	10174
17	125	0	Inf. Dilution	dG:-478.94(5SF)	0	CRV	10174
18	150	0	Inf. Dilution	dG:-479.94(5SF)	0	CRV	10174
19	175	0	Inf. Dilution	dG:-480.98(5SF)	0	CRV	10174
20	200	0	Inf. Dilution	dG:-482.07(5SF)	0	CRV	10174
21	225	0	Inf. Dilution	dG:-483.21(5SF)	0	CRV	10174
22	250	0	Inf. Dilution	dG:-484.38(5SF)	0	CRV	10174
23	300	0	Inf. Dilution	dG:-486.84(5SF)	0	CRV	10174
24	25	0	Inf. Dilution	dH:-500.6(4SF)	0	MTD	6422
25	25	0	Inf. Dilution	dH:-503.21(5SF)	0	ENS	6437
26	25	0	Inf. Dilution	dH:-499.6(4SF)	0	MTD	6494
27	127	0	Inf. Dilution	dH:-501.4(4SF)	0	MTD	6422
28	127	0	Inf. Dilution	dH:-500.4(4SF)	0	MTD	6494
29	194	0	Inf. Dilution	dH:-501.1(4SF)	0	MTD	6422
30	227	0	Inf. Dilution	dH:-500.6(4SF)	0	MTD	6494

31	327	0	Inf. Dilution	dH:-499.9 (4SF)	0	MTD	6494
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Reaction No. 66822							
K(s) + Al(s) + Si(s) + 2<O2(g)> = KAlSiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:351.3 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:351.3 (4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:351.9 (4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:349.0 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:349.6 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:256.6 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:257.1 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:201.1 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:201.5 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:164.1 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:164.5 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-479.3 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-480.1 (4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dG:-482.98 (5SF)	0	CRV	10174
15	50	0	Inf. Dilution	dG:-483.81 (5SF)	0	CRV	10174
16	75	0	Inf. Dilution	dG:-484.69 (5SF)	0	CRV	10174
17	100	0	Inf. Dilution	dG:-485.63 (5SF)	0	CRV	10174
18	125	0	Inf. Dilution	dG:-486.62 (5SF)	0	CRV	10174
19	150	0	Inf. Dilution	dG:-487.66 (5SF)	0	CRV	10174
20	175	0	Inf. Dilution	dG:-488.75 (5SF)	0	CRV	10174
21	200	0	Inf. Dilution	dG:-489.89 (5SF)	0	CRV	10174
22	225	0	Inf. Dilution	dG:-491.06 (5SF)	0	CRV	10174
23	250	0	Inf. Dilution	dG:-492.28 (5SF)	0	CRV	10174
24	300	0	Inf. Dilution	dG:-494.81 (5SF)	0	CRV	10174
25	25	0	Inf. Dilution	dH:-507.0 (4SF)	0	MTD	6422
26	25	0	Inf. Dilution	dH:-505.4 (4SF)	0	ENS	6437
27	25	0	Inf. Dilution	dH:-507.8 (4SF)	0	MTD	6494
28	127	0	Inf. Dilution	dH:-507.9 (4SF)	0	MTD	6422
29	127	0	Inf. Dilution	dH:-508.6 (4SF)	0	MTD	6494
30	227	0	Inf. Dilution	dH:-508.2 (4SF)	0	MTD	6422
31	227	0	Inf. Dilution	dH:-508.5 (4SF)	0	MTD	6494

32	327	0	Inf. Dilution	dH:-508.4 (4SF)	0	MTD	6422
33	327	0	Inf. Dilution	dH:-508.2 (4SF)	0	MTD	6494

Reaction No. 66823							
2<Al(s)> + H2(g) + 6<O2(g)> + 4<Si(s)> = Al2Si4O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:853.0 (4SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:922.6 (4SF)	2	ENS	6494
3	27	0	Inf. Dilution	lgK:916.5 (4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:670.9 (4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:523.6 (4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:425.4 (4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1258.8 (5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1259. (4SF)	2	EFE	6494
9	25	0	Inf. Dilution	dG:-1259.00 (6SF)	0	CRV	10174
10	25	0	Inf. Dilution	dG:-1258.4 (5SF)	3	CRV	29553
11	25	0	GAMSPATH	dG:-5253. (4SF) kJ	0	CRV	12638
12	50	0	Inf. Dilution	dG:-1260.50 (6SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-1262.20 (6SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-1264.00 (6SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-1265.90 (6SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-1267.90 (6SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-1270.10 (6SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-1272.40 (6SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-1274.80 (6SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-1277.30 (6SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-1282.50 (6SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-1348.2 (5SF)	0	ENS	6437
23	25	0	Inf. Dilution	dH:-1348.2 (5SF)	0	EFE	6482
24	25	0	Inf. Dilution	dH:-1348. (4SF)	0	MTD	6494
25	25	0	Inf. Dilution	dH:-1347.7 (5SF)	3	CRV	29553
26	25	0	GAMSPATH	dH:-5626.60 (6SF) kJ	0	CRV	12638
27	127	0	Inf. Dilution	dH:-1348. (4SF)	0	MTD	6494
28	227	0	Inf. Dilution	dH:-1348. (4SF)	0	MTD	6494
29	327	0	Inf. Dilution	dH:-1347. (4SF)	0	MTD	6494

Reaction No. 66825							
7<Mg(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) = Mg7Si8O22(OH)2(Anth.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1991.(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:1987.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1978.(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:1974.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1452.(4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:1449.(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:1136.(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:1134.(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:926.0(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:923.7(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-2717.(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-2710.9(5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-2711.(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-2889.(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	dH:-2885.(4SF)	4	ENS	6437
16	25	0	Inf. Dilution	dH:-2884.6(5SF)	3	EFE	6482
17	25	0	Inf. Dilution	dH:-2885.(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-2890.(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-2885.(4SF)	4	MTD	6494
20	227	0	Inf. Dilution	dH:-2889.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-2883.(4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-2888.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-2881.(4SF)	4	MTD	6494

Reaction No. 66826							
3<Al(s)> + H2(g) + K(s) + 6<O2(g)> + 3<Si(s)> = KAl3Si3O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:983.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:982.6(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:982.6(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:976.1(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:976.1(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:715.5(4SF)	0	ENS	6422

7	127	0	Inf. Dilution	lgK:715.2 (4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:559.1 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:558.7 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:454.8 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:454.4 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-1340. (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1337.6 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1340. (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-1340.30 (6SF)	0	CRV	10174
16	25	0	Inf. Dilution	dG:-1339.0 (5SF)	5	CRV	29553
17	25	0	GAMSPATH	dG:-5588. (4SF) kJ	0	CRV	12638
18	50	0	Inf. Dilution	dG:-1342.10 (6SF)	0	CRV	10174
19	75	0	Inf. Dilution	dG:-1344.10 (6SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-1346.20 (6SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-1348.50 (6SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-1350.90 (6SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-1353.40 (6SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-1356.00 (6SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-1358.80 (6SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-1361.70 (6SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-1367.80 (6SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-1430. (4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-1428.4 (5SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-1428.5 (5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-1432. (4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-1429.9 (5SF)	5	CRV	29553
33	25	0	GAMSPATH	dH:-5969.40 (6SF) kJ	0	CRV	12638
34	127	0	Inf. Dilution	dH:-1431. (4SF)	0	MTD	6422
Note (34): Low Wgt for internal consistency							
35	127	0	Inf. Dilution	dH:-1433. (4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-1431. (4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-1432. (4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-1430. (4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-1431. (4SF)	0	MTD	6494

Reaction No. 66828

Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(San.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:657.4 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:655.2 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:650.9 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:478.5 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:375.0 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:306.1 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-893.9 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-893.60 (5SF)	0	EFE	6482
9	25	0	Inf. Dilution	dG:-3746.00 (6SF) kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-895.43 (5SF)	0	CRV	10174
11	25	0	GAMSPATH	dG:-3739. (4SF) kJ	0	CRV	12638
12	50	0	Inf. Dilution	dG:-896.85 (5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-898.36 (5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-899.97 (5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-901.67 (5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-903.46 (5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-905.33 (5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-907.29 (5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-909.31 (5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-911.41 (5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-915.78 (5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-946.4 (4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-946.4 (4SF)	0	ENS	6437
24	25	0	Inf. Dilution	dH:-945.91 (5SF)	2	EFE	6482
25	25	0	Inf. Dilution	dH:-3965.70 (6SF) kJ	2	EFE	8798
26	25	0	GAMSPATH	dH:-3960.30 (6SF) kJ	0	CRV	12638
27	127	0	Inf. Dilution	dH:-947.1 (4SF)	0	MTD	6422
28	227	0	Inf. Dilution	dH:-946.6 (4SF)	0	MTD	6422
29	327	0	Inf. Dilution	dH:-945.7 (4SF)	0	MTD	6422

Reaction No. 66829							
3<Ca(s)> + 2<Fe(s)> + 3<Si(s)> + 6<O2(g)> = 3CaO.Fe2O3.3SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:955.5 (4SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:950.8 (4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:944.5 (4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:693.3 (4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:542.7 (4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:442.3 (4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1297. (4SF)	0	EFE	6494
8	25	0	Inf. Dilution	dH:-1379.0 (5SF)	2	ENS	6437
9	25	0	Inf. Dilution	dH:-1379. (4SF)	0	MTD	6494
10	127	0	Inf. Dilution	dH:-1379. (4SF)	0	MTD	6494
11	227	0	Inf. Dilution	dH:-1378. (4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1377. (4SF)	0	MTD	6494

Reaction No. 66830							
Na(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = NaAlSi2O6(Jade.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:499.6 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:500.0 (4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:499.4 (4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:496.7 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:496.1 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:364.7 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:364.2 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:285.4 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:285.0 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:232.5 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:232.2 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-682.1 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-680.33 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-681.3 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-682.07 (5SF)	0	CRV	10174
16	50	0	Inf. Dilution	dG:-682.91 (5SF)	0	CRV	10174
17	75	0	Inf. Dilution	dG:-683.82 (5SF)	0	CRV	10174
18	100	0	Inf. Dilution	dG:-684.82 (5SF)	0	CRV	10174
19	125	0	Inf. Dilution	dG:-685.88 (5SF)	0	CRV	10174
20	150	0	Inf. Dilution	dG:-687.01 (5SF)	0	CRV	10174
21	175	0	Inf. Dilution	dG:-688.21 (5SF)	0	CRV	10174

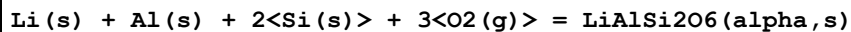
22	200	0	Inf. Dilution	dG:-689.47 (5SF)	0	CRV	10174
23	225	0	Inf. Dilution	dG:-690.79 (5SF)	0	CRV	10174
24	250	0	Inf. Dilution	dG:-692.17 (5SF)	0	CRV	10174
25	300	0	Inf. Dilution	dG:-695.07 (5SF)	0	CRV	10174
26	25	0	Inf. Dilution	dH:-725.8 (4SF)	0	MTD	6422
27	25	0	Inf. Dilution	dH:-724. (3SF)	0	ENS	6437
28	25	0	Inf. Dilution	dH:-723.02 (5SF)	0	EFE	6482
29	25	0	Inf. Dilution	dH:-724.0 (4SF)	0	MTD	6494
30	127	0	Inf. Dilution	dH:-725.7 (4SF)	0	MTD	6422
31	127	0	Inf. Dilution	dH:-724.9 (4SF)	0	MTD	6494
32	227	0	Inf. Dilution	dH:-725.6 (4SF)	0	MTD	6422
33	227	0	Inf. Dilution	dH:-724.8 (4SF)	0	MTD	6494
34	327	0	Inf. Dilution	dH:-725.4 (4SF)	0	MTD	6422
35	327	0	Inf. Dilution	dH:-724.6 (4SF)	0	MTD	6494

Reaction No. 66831							
Ca(s) + Fe(s) + 3<O2(g)> + 2<Si(s)> = CaO.FeO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:468.9 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:465.8 (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:342.2 (4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:268.1 (4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:218.7 (4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-639.7 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dG:-2676.3 (5SF) kJ	3	EOD	12843
8	25	0	Inf. Dilution	dH:-339.2 (4SF)	0	ENS	6437
9	25	0	Inf. Dilution	dH:-678.8 (4SF)	4	MTD	6494
10	25	0	Inf. Dilution	dH:-2839.9 (5SF) kJ	3	EOD	12843
11	127	0	Inf. Dilution	dH:-678.6 (4SF)	1	MTD	6494
12	227	0	Inf. Dilution	dH:-678.2 (4SF)	0	MTD	6494
13	327	0	Inf. Dilution	dH:-677.7 (4SF)	0	MTD	6494

Reaction No. 66948							
Li(s) + Al(s) + Si(s) + 2<O2(g)> = LiAlSiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:352.2 (4SF)	4	ENS	6422

2	25	0	Inf. Dilution	lgK:352.0 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:349.9 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:349.7 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:257.4 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:257.3 (4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:201.9 (4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:201.8 (4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:164.8 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:164.8 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-480.4 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-480.2 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-507.7 (4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-507.5 (4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-507.8 (4SF)	3	MTD	6422
16	127	0	Inf. Dilution	dH:-507.6 (4SF)	3	MTD	6494
17	227	0	Inf. Dilution	dH:-508.4 (4SF)	2	MTD	6422
18	227	0	Inf. Dilution	dH:-508.2 (4SF)	2	MTD	6494
19	327	0	Inf. Dilution	dH:-508.1 (4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-508.0 (4SF)	0	MTD	6494

Reaction No. 66949



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:504.8 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:504.6 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:501.5 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:501.3 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:368.5 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:368.4 (4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:288.7 (4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:288.6 (4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:235.4 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:235.3 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-688.7 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-688.4 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-730.1 (4SF)	3	MTD	6422

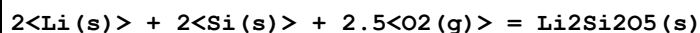
14	25	0	Inf. Dilution	dH:-729.8(4SF)	3	MTD	6494
15	127	0	Inf. Dilution	dH:-730.3(4SF)	1	MTD	6422
16	127	0	Inf. Dilution	dH:-730.0(4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-730.9(4SF)	0	MTD	6422
18	227	0	Inf. Dilution	dH:-730.7(4SF)	0	MTD	6494
19	327	0	Inf. Dilution	dH:-730.7(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-730.5(4SF)	0	MTD	6494

Reaction No. 66950							
Li(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = LiAlSi2O6(beta,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:501.2(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:501.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:498.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:497.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:366.2(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:366.0(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:287.1(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:286.9(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:234.4(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:234.2(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-683.8(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-683.4(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-723.4(4SF)	3	MTD	6422
14	25	0	Inf. Dilution	dH:-723.1(4SF)	3	MTD	6494
15	127	0	Inf. Dilution	dH:-723.5(4SF)	1	MTD	6422
16	127	0	Inf. Dilution	dH:-723.2(4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-724.0(4SF)	0	MTD	6422
18	227	0	Inf. Dilution	dH:-723.8(4SF)	0	MTD	6494
19	327	0	Inf. Dilution	dH:-723.6(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-732.4(4SF)	0	MTD	6494

Reaction No. 66955							
2<Li(s)> + 1.5<O2(g)> + Si(s) = Li2SiO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:273.1(4SF)	4	ENS	6422

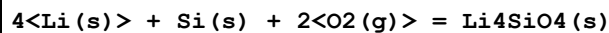
2	27	0	Inf. Dilution	lgK:271.3 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:199.5 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:156.3 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:127.4 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-1557.2 (5SF) kJ	5	EFE	6330
7	25	0	Inf. Dilution	dG:-372.6 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-1648.1 (5SF) kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-394.2 (4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-394.5 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-396.0 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-396.0 (4SF)	0	MTD	6422

Reaction No. 66956



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:423.4 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:420.7 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:309.2 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:242.2 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:197.5 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-577.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-612.1 (4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-612.3 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-613.7 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-613.3 (4SF)	0	MTD	6422

Reaction No. 66957



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:386.1 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:383.5 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:282.1 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:220.9 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:180.0 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-526.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-556.9 (4SF)	3	MTD	6422

8	127	0	Inf. Dilution	dH:-557.7 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-561.5 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-562.3 (4SF)	0	MTD	6422

Reaction No. 66974



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:360.5 (4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:359.8 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:358.2 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:357.4 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:263.4 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:262.8 (4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:206.6 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:206.1 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:168.7 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:168.3 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-491.8 (4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-491.16 (5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-490.8 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-2053.6 (5SF) kJ	5	CRV	8695
Note (14): p. 169. Sign changed							
15	25	0	Inf. Dilution	dG:-2055.1 (5SF) kJ	5	CRV	8695
Note (15): p. 167. Sign changed							
16	25	0	Inf. Dilution	dG:-2053.6 (5SF) kJ	3	EOD	12843
17	25	0	Inf. Dilution	dG:-2053.5 (5SF) kJ	5	CRV	29482
18	25	0	Inf. Dilution	dH:-520.3 (4SF)	4	MTD	6422
19	25	0	Inf. Dilution	dH:-519.70 (5SF)	4	EFE	6482
20	25	0	Inf. Dilution	dH:-519.4 (4SF)	4	MTD	6494
21	25	0	Inf. Dilution	dH:-2174.0 (5SF) kJ	5	CRV	8695
22	25	0	Inf. Dilution	dH:-2173.0 (5SF) kJ	3	EOD	12843
23	25	0	Inf. Dilution	dH:-2172.6 (5SF) kJ	5	CRV	29482
24	127	0	Inf. Dilution	dH:-520.4 (4SF)	2	MTD	6422
25	127	0	Inf. Dilution	dH:-519.4 (4SF)	2	MTD	6494
26	227	0	Inf. Dilution	dH:-520.2 (4SF)	1	MTD	6422
27	227	0	Inf. Dilution	dH:-519.3 (4SF)	1	MTD	6494

28	327	0	Inf. Dilution	dH:-519.9(4SF)	0	MTD	6422
29	327	0	Inf. Dilution	dH:-519.0(4SF)	0	MTD	6494

Reaction No. 67042							
Al(s) + Na(s) + 3<O2(g)> + 2<Si(s)> = NaAlSi2O6(Dehyd.Anal.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:493.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:493.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:490.7(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:490.1(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:360.7(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:360.2(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:282.6(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:282.3(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:230.6(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:230.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-673.8(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-673.0(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-713.5(4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-713.0(4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-714.3(4SF)	3	MTD	6422
16	127	0	Inf. Dilution	dH:-713.7(4SF)	3	MTD	6494
17	227	0	Inf. Dilution	dH:-714.0(4SF)	2	MTD	6422
18	227	0	Inf. Dilution	dH:-713.5(4SF)	2	MTD	6494
19	327	0	Inf. Dilution	dH:-713.7(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-713.0(4SF)	0	MTD	6494

Reaction No. 67043							
Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteLO,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:651.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:650.0(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:645.7(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:474.7(4SF)	0	ENS	6422
Note (4): Low Wgt for internal consistency							
5	227	0	Inf. Dilution	lgK:372.1(4SF)	0	ENS	6422

6	327	0	Inf. Dilution	lgK:303.7 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-886.7 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-887.12 (5SF)	0	EFE	6482
9	25	0	Inf. Dilution	dG:-3713.00 (6SF) kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-888.94 (5SF)	0	CRV	10174
11	25	0	Inf. Dilution	dG:-887.37 (5SF)	5	CRV	29553
12	50	0	Inf. Dilution	dG:-890.22 (5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-891.61 (5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-893.10 (5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-894.67 (5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-896.34 (5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-898.09 (5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-899.91 (5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-901.81 (5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-903.78 (5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-907.90 (5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-938.7 (4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-940.51 (5SF)	0	MTD	6482
24	25	0	Inf. Dilution	dH:-3936.20 (6SF) kJ	0	EFE	8798
25	25	0	Inf. Dilution	dH:-940.77 (5SF)	5	CRV	29553
26	127	0	Inf. Dilution	dH:-939.5 (4SF)	0	MTD	6422
Note (26): Low Wgt for internal consistency							
27	227	0	Inf. Dilution	dH:-939.1 (4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-938.4 (4SF)	0	MTD	6422

Reaction No. 67044							
Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteHI,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:649. (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:650.6 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:646.3 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:474.9 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:372.0 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:303.4 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-887.6 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-885.11 (5SF)	0	EFE	6482

9	25	0	Inf. Dilution	dG:-3705.10 (6SF) kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-887.14 (5SF)	0	CRV	10174
11	25	0	Inf. Dilution	dG:-3705.1 (5SF) kJ	3	EOD	12843
12	50	0	Inf. Dilution	dG:-888.50 (5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-889.95 (5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-891.51 (5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-893.15 (5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-894.89 (5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-896.71 (5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-898.60 (5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-900.57 (5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-902.61 (5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-906.87 (5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-940.9 (4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-937.29 (5SF)	0	MTD	6482
24	25	0	Inf. Dilution	dH:-3923.40 (6SF) kJ	0	EFE	8798
25	25	0	Inf. Dilution	dH:-3923.4 (5SF) kJ	3	EOD	12843
26	127	0	Inf. Dilution	dH:-941.8 (4SF)	0	MTD	6422
Note (26): Low Wgt for internal consistency							
27	227	0	Inf. Dilution	dH:-941.7 (4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-941.4 (4SF)	0	MTD	6422

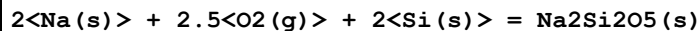
Reaction No. 67045							
Al(s) + H2(g) + Na(s) + 3.5<O2(g)> + 2<Si(s)> = NaAlSi2O6.H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:539.2 (4SF)	1	ENS	6422
2	25	0	Inf. Dilution	lgK:541.3 (4SF)	1	ENS	6494
3	27	0	Inf. Dilution	lgK:535.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:537.8 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:391.8 (4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:393.6 (4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:305.5 (4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:307.1 (4SF)	0	ENS	6494
9	327	0	Inf. Dilution	lgK:248.0 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:249.5 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-735.5 (4SF)	1	EFE	6422

12	25	0	Inf. Dilution	dG:-738.5 (4SF)	1	EFE	6494
13	25	0	Inf. Dilution	dG:-740.73 (5SF)	0	CRV	10174
14	25	0	Inf. Dilution	dG:-3078.90 (6SF) kJ	3	CRV	25435
15	25	0	Inf. Dilution	dG:-3084. (4SF) kJ	3	EFE	29327
16	50	0	Inf. Dilution	dG:-742.18 (5SF)	0	CRV	10174
17	75	0	Inf. Dilution	dG:-743.73 (5SF)	0	CRV	10174
18	100	0	Inf. Dilution	dG:-745.38 (5SF)	0	CRV	10174
19	125	0	Inf. Dilution	dG:-747.13 (5SF)	0	CRV	10174
20	150	0	Inf. Dilution	dG:-748.96 (5SF)	0	CRV	10174
21	175	0	Inf. Dilution	dG:-750.87 (5SF)	0	CRV	10174
22	200	0	Inf. Dilution	dG:-752.86 (5SF)	0	CRV	10174
23	225	0	Inf. Dilution	dG:-754.93 (5SF)	0	CRV	10174
24	250	0	Inf. Dilution	dG:-757.06 (5SF)	0	CRV	10174
25	300	0	Inf. Dilution	dG:-761.51 (5SF)	0	CRV	10174
26	25	0	Inf. Dilution	dH:-789.4 (4SF)	3	MTD	6422
27	25	0	Inf. Dilution	dH:-791.1 (4SF)	1	MTD	6494
28	25	0	Inf. Dilution	dH:-3296.9 (5SF) kJ	3	CRV	25435
29	25	0	Inf. Dilution	dH:-3308.00 (6SF) kJ	3	CRV	27383
30	25	0	Inf. Dilution	dH:-3303.2 (2.9SD) kJ	4	ETR	29327
31	127	0	Inf. Dilution	dH:-790.0 (4SF)	0	MTD	6422
32	127	0	Inf. Dilution	dH:-791.9 (4SF)	0	MTD	6494
33	227	0	Inf. Dilution	dH:-789.4 (4SF)	0	MTD	6422
34	227	0	Inf. Dilution	dH:-791.6 (4SF)	0	MTD	6494
35	327	0	Inf. Dilution	dH:-788.5 (4SF)	0	MTD	6422
36	327	0	Inf. Dilution	dH:-790.9 (4SF)	0	MTD	6494

Reaction No. 67047							
2<Na(s)> + 1.5<O2(g)> + Si(s) = Na2SiO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:257.1 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:255.4 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:187.4 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:146.4 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:119.1 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1462.8 (5SF) kJ	5	EFE	6330
7	25	0	Inf. Dilution	dG:-350.7 (4SF)	4	EFE	6422

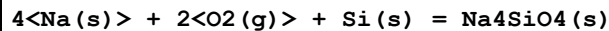
8	25	0	Inf. Dilution	dH:-1554.9 (5SF) kJ	5	CRV	6330
9	25	0	Inf. Dilution	dH:-373.2 (4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-374.6 (4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-374.5 (4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-374.2 (4SF)	4	MTD	6422

Reaction No. 67048



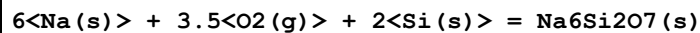
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:407.2 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:404.5 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:296.9 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:232.3 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:189.2 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-555.5 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-590.4 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-591.8 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-591.5 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-591.0 (4SF)	4	MTD	6422

Reaction No. 67049



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:346.1 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:343.9 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:252.0 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:196.7 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:159.8 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-472.2 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-503.5 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-506.3 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-506.6 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-506.7 (4SF)	4	MTD	6422

Reaction No. 67050



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:596.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:593.0 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:434.7 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:339.4 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:275.9 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-814.3 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-868.1 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-872.0 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-871.9 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-871.5 (4SF)	4	MTD	6422

Reaction No. 67106



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:226.1 (4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:224.6 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:163.5 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:126.9 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:102.6 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-308.4 (4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dG:-1288.40 (2.5SD) kJ	5	CRV	20828
8	25	0	Inf. Dilution	dH:-335.4 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-1396.00 (2.5SD) kJ	5	CRV	20828
10	127	0	Inf. Dilution	dH:-335.1 (4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-334.6 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-334.2 (4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:123.200 (0.25SD) J/K	5	CRV	20828

Reaction No. 67146



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:185.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:184.7 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:134.8 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:104.9 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:85.04 (4SF)	0	ENS	6422

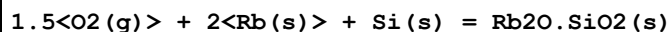
6	25	0	Inf. Dilution	dG:-253.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-253.86(5SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-273.7(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-273.83(5SF)	4	CRV	12011
10	127	0	Inf. Dilution	dH:-273.6(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-273.3(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-272.9(4SF)	0	MTD	6422

Reaction No. 67147							
2<O2(g)> + 2<Pb(s)> + Si(s) = 2PbO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:219.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:218.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:158.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:123.1(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:99.38(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-299.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-299.4(4SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-325.9(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-325.8(4SF)	4	CRV	12011
10	127	0	Inf. Dilution	dH:-325.6(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-325.2(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-324.7(4SF)	0	MTD	6422

Reaction No. 67148							
3<O2(g)> + 4<Pb(s)> + Si(s) = 4PbO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:286.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:284.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:206.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:159.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:128.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-391.39(5SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-431.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-431.0(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-430.4(4SF)	4	MTD	6422

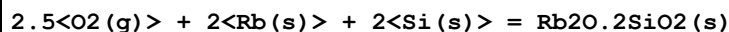
10	327	0	Inf. Dilution	dH:-429.6(4SF)	4	MTD	6422
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Reaction No. 67209



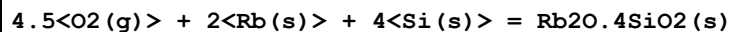
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:252.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:250.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:183.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:143.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:116.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-344.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-367.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-368.4(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-368.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-368.1(4SF)	4	MTD	6422

Reaction No. 67210



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:404.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:296.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:231.6(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:188.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-555.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-591.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-592.5(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-592.2(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-591.7(4SF)	4	MTD	6422

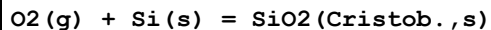
Reaction No. 67211



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:710.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:706.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:518.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:405.1(4SF)	4	ENS	6422

5	327	0	Inf. Dilution	lgK:329.9 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-969.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1032. (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1033. (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1033. (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1032. (4SF)	4	MTD	6422

Reaction No. 67229



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.7 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:149.7 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:148.7 (4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:148.7 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:109.2 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:109.2 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:85.46 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:85.46 (4SF)	4	ENS	6494
9	270	0	Inf. Dilution	lgK:77.94 (4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:69.71 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-204.2 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-204.09 (5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-204.3 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-855.43 (5SF) kJ	3	CRV	8695
15	25	0	GAMSPATH	dG:-842.2 (4SF) kJ	0	CRV	12638
Note (15): dH not in GAMSPATH database; dH = -217kcal/mol assumed							
16	25	0	Inf. Dilution/m	dG:-854.020 (6SF) kJ	3	CRV	24953
17	25	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-216.96 (5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-909.48 (5SF) kJ	3	CRV	8695
21	25	0	Inf. Dilution/m	dH:-907.860 (6SF) kJ	3	CRV	24953
22	127	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6422
23	127	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6494
24	227	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6422
25	227	0	Inf. Dilution	dH:-217.1 (4SF)	4	MTD	6494

26	270	0	Inf. Dilution	dH:-217.0 (4SF)	4	MTD	6422
27	327	0	Inf. Dilution	dH:-216.7 (4SF)	4	MTD	6494

Reaction No. 67383							
Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Micr.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:656.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:655.7 (4SF)	2	ENS	6422
3	25	0	Inf. Dilution	lgK:656.8 (4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:651.4 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:652.5 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:478.6 (4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:479.4 (4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:374.9 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:375.5 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:305.8 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:306.3 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-894.5 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-895.18 (5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-896.1 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-3749.50 (6SF) kJ	0	EFE	8798
16	25	0	Inf. Dilution	dG:-897.07 (5SF)	0	CRV	10174
17	50	0	Inf. Dilution	dG:-898.40 (5SF)	0	CRV	10174
18	75	0	Inf. Dilution	dG:-899.82 (5SF)	0	CRV	10174
19	100	0	Inf. Dilution	dG:-901.35 (5SF)	0	CRV	10174
20	125	0	Inf. Dilution	dG:-902.97 (5SF)	0	CRV	10174
21	150	0	Inf. Dilution	dG:-904.67 (5SF)	0	CRV	10174
22	175	0	Inf. Dilution	dG:-906.46 (5SF)	0	CRV	10174
23	200	0	Inf. Dilution	dG:-908.33 (5SF)	0	CRV	10174
24	225	0	Inf. Dilution	dG:-910.27 (5SF)	0	CRV	10174
25	250	0	Inf. Dilution	dG:-912.28 (5SF)	0	CRV	10174
26	300	0	Inf. Dilution	dG:-916.48 (5SF)	0	CRV	10174
27	25	0	Inf. Dilution	dH:-948.4 (4SF)	0	MTD	6422
28	25	0	Inf. Dilution	dH:-949.04 (5SF)	0	EFE	6482
29	25	0	Inf. Dilution	dH:-950.0 (4SF)	0	MTD	6494
30	25	0	Inf. Dilution	dH:-3974.90 (6SF) kJ	0	EFE	8798

31	127	0	Inf. Dilution	dH:-949.1 (4SF)	0	MTD	6422
Note (31): Low Wgt for internal consistency							
32	127	0	Inf. Dilution	dH:-950.9 (4SF)	0	MTD	6494
33	227	0	Inf. Dilution	dH:-948.7 (4SF)	0	MTD	6422
34	227	0	Inf. Dilution	dH:-950.9 (4SF)	0	MTD	6494
35	327	0	Inf. Dilution	dH:-947.9 (4SF)	0	MTD	6422
36	327	0	Inf. Dilution	dH:-950.6 (4SF)	0	MTD	6494

Reaction No. 67384							
Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Adul.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:654.3 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:650.0 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:477.8 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:374.4 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:305.5 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-892.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-945.1 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-946.1 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-945.9 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-945.4 (4SF)	4	MTD	6422

Reaction No. 67388							
2<K(s)> + 1.5<O2(g)> + Si(s) = K2O.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.0 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:253.4 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:185.9 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:145.3 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:118.3 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-347.9 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-370.0 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-371.1 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-370.9 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-370.4 (4SF)	4	MTD	6422

Reaction No. 67389							
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2<K(s)> + 2.5<O2(g)> + 2<Si(s)> = K2O.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:413.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:410.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:301.6(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:235.9(4SF)	4	ENS	6422
5	250	0	Inf. Dilution	lgK:224.4(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-564.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-599.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-600.6(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-600.2(4SF)	4	MTD	6422
10	250	0	Inf. Dilution	dH:-600.0(4SF)	4	MTD	6422

Reaction No. 67390							
2<K(s)> + 4.5<O2(g)> + 4<Si(s)> = K2O.4SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:711.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:706.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:518.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:405.8(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:330.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-970.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1031.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1030.(4SF)	4	MTD	6422

Reaction No. 67397							
0.5<O2(g)> + Si(s) = SiO(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.30(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:22.19(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:17.81(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:15.17(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:13.39(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-30.43(4SF)	4	EFE	6494

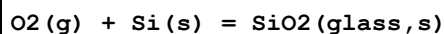
7	25	0	Inf. Dilution	dH:-24.0 (3SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-24.14 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-24.28 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-24.45 (4SF)	4	MTD	6494

Reaction No. 67398							
O2(g) + Si(s) = SiO2(Coesite,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.4 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:148.4 (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:108.8 (4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:85.14 (4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:69.34 (4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-203.77 (5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-203.8 (4SF)	4	EFE	6494
8	25	0	Inf. Dilution/m	dG:-850.850 (6SF) kJ	3	CRV	24953
9	25	0	Inf. Dilution	dH:-216.92 (5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-217.0 (4SF)	4	MTD	6494
11	25	0	Inf. Dilution/m	dH:-905.580 (6SF) kJ	3	CRV	24953
12	127	0	Inf. Dilution	dH:-217.0 (4SF)	2	MTD	6494
13	227	0	Inf. Dilution	dH:-216.9 (4SF)	1	MTD	6494
14	327	0	Inf. Dilution	dH:-216.8 (4SF)	0	MTD	6494

Reaction No. 67399							
O2(g) + Si(s) = SiO2(Stish.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:140.7 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:139.7 (4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:102.2 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:79.72 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:64.73 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-191.9 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution/m	dG:-802.830 (6SF) kJ	3	CRV	24953
8	25	0	Inf. Dilution	dH:-205.9 (4SF)	4	MTD	6494
9	25	0	Inf. Dilution/m	dH:-861.320 (6SF) kJ	3	CRV	24953
10	127	0	Inf. Dilution	dH:-205.9 (4SF)	4	MTD	6494

11	227	0	Inf. Dilution	dH:-205.8 (4SF)	4	MTD	6494
12	327	0	Inf. Dilution	dH:-205.7 (4SF)	4	MTD	6494

Reaction No. 67400



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:148.8 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:147.8 (4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:108.6 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:85.00 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:69.31 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-203.0 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-215.5 (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-215.6 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-215.5 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-215.3 (4SF)	4	MTD	6494

Reaction No. 67407



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1056. (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1050. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:771.9 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:605.4 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:494.4 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1441. (4SF)	4	EFE	6494
7	25	0	GAMSPATH	dG:-6031. (4SF) kJ	3	CRV	12638
8	25	0	Inf. Dilution	dH:-1524. (4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-1525. (4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-1524. (4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-1523. (4SF)	4	MTD	6494

Reaction No. 67408



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:346.4 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:344.2 (4SF)	0	ENS	6494

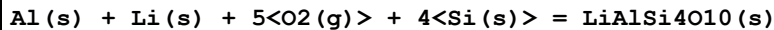
3	127	0	Inf. Dilution	lgK:253.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:198.6(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:162.2(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-472.6(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-499.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-500.0(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-499.9(4SF)	3	MTD	6494
10	327	0	Inf. Dilution	dH:-499.6(4SF)	0	MTD	6494

Reaction No. 67409							
Al(s) + Na(s) + 2<O2(g)> + Si(s) = NaAlSiO4(Carn.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:345.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:343.4(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:252.4(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:197.8(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:161.3(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-471.6(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dG:-1958.(4SF)kJ	2	EFE	24802
8	25	0	Inf. Dilution	dH:-499.4(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-500.1(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-500.0(4SF)	3	MTD	6494
11	327	0	Inf. Dilution	dH:-499.8(4SF)	0	MTD	6494

Reaction No. 67410							
Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(glass,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:642.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:637.9(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:469.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:367.9(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:300.4(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-876.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-926.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-927.1(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-926.8(4SF)	4	MTD	6494

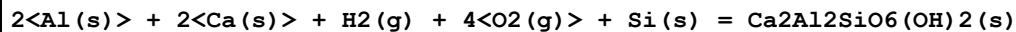
10	327	0	Inf. Dilution	dH:-926.3(4SF)	4	MTD	6494
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Reaction No. 67411



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:807.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:802.5(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:589.7(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:462.1(4SF)	2	ENS	6494
5	327	0	Inf. Dilution	lgK:376.9(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1102.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1168.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1168.(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-1169.(4SF)	2	MTD	6494
10	327	0	Inf. Dilution	dH:-1168.(4SF)	0	MTD	6494

Reaction No. 67412



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:713.6(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:708.9(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:519.9(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:406.5(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:330.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-973.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1037.(4SF)	4	MTD	6494

Reaction No. 67413



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:691.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:687.2(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:506.0(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:397.3(4SF)	4	ENS	6494

5	327	0	Inf. Dilution	lgK:324.8 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-943.7 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-995.0 (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-995.2 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-994.9 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-994.4 (4SF)	4	MTD	6494

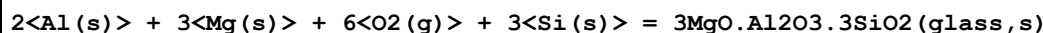
Reaction No. 67414							
Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Ilmen.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:244.8 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:243.2 (4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:178.5 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:139.8 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:113.9 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-334.0 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-355.3 (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-355.0 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-354.7 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-354.4 (4SF)	4	MTD	6494

Reaction No. 67415							
Ca(s) + Mg(s) + 3<O2(g)> + 2<Si(s)> = CaO.MgO.2SiO2(glass,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:523.5 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:520.1 (4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:382.7 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:300.3 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:245.4 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-714.2 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-754.3 (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-754.3 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-754.0 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-753.5 (4SF)	4	MTD	6494

Reaction No. 67416							
Ca(s) + 1.5<O2(g)> + Si(s) = CaO.SiO2(glass,s)							

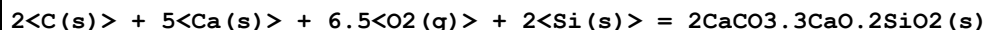
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:267.5 (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:265.8 (4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:195.7 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:153.7 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:125.7 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-364.9 (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-384.5 (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-384.4 (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-384.3 (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-384.1 (4SF)	4	MTD	6494

Reaction No. 67417



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1022. (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1016. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:747.6 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:586.8 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:479.7 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1395. (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1473. (4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1472. (4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1471. (4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1470. (4SF)	4	MTD	6494

Reaction No. 67418



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1054. (4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1047. (4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:769.3 (4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:603.0 (4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:492.2 (4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1437. (4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1523. (4SF)	4	MTD	6494

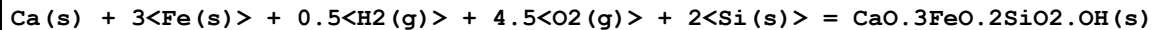
8	127	0	Inf. Dilution	dH:-1522.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1521.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1520.(4SF)	4	MTD	6494

Reaction No. 67419



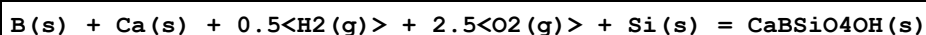
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:855.3(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:849.7(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:623.4(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:487.7(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:397.2(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1167.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1243.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494

Reaction No. 67420



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:602.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:598.1(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:437.5(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:341.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:277.0(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-821.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-882.6(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-881.8(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-880.9(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-879.8(4SF)	4	MTD	6494

Reaction No. 67421



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:403.4(4SF)	0	ENS	6494

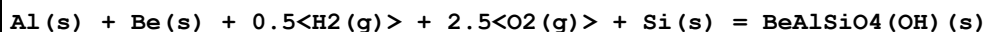
3	127	0	Inf. Dilution	lgK:295.6(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:230.8(4SF)	2	ENS	6494
5	327	0	Inf. Dilution	lgK:187.7(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-554.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-592.2(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-592.4(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-592.3(4SF)	2	MTD	6494
10	327	0	Inf. Dilution	dH:-592.0(4SF)	0	MTD	6494

Reaction No. 67422							
$4\text{Be(s)} + \text{H}_2(\text{g}) + 4.5\text{O}_2(\text{g}) + 2\text{Si(s)} = 4\text{BeO} \cdot 2\text{SiO}_2 \cdot \text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:752.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:747.5(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:548.1(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:428.4(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:348.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1027.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494

Reaction No. 67423							
$2\text{Al(s)} + 3\text{Be(s)} + 9\text{O}_2(\text{g}) + 6\text{Si(s)} = 3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1489.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1479.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1087.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:852.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:695.2(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2032.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494
8	27	0	Inf. Dilution	dH:-2153.(4SF)	0	MTD	6494
9	127	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494

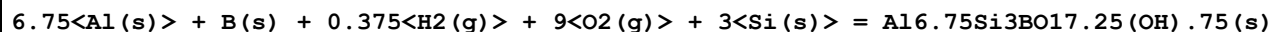
11	327	0	Inf. Dilution	dH:-2152.(4SF)	4	MTD	6494
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Reaction No. 67424



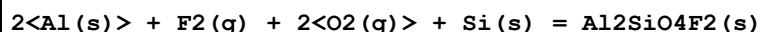
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:415.2(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:412.5(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:302.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:236.0(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:191.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-566.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-605.4(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-605.6(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-605.5(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-605.1(4SF)	4	MTD	6494

Reaction No. 67425



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1501.(4SF)	3	ENS	6494
2	27	0	Inf. Dilution	lgK:1491.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1095.(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:856.7(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:698.1(4SF)	3	ENS	6494
6	25	0	Inf. Dilution	dG:-2048.(4SF)	3	EFE	6494
7	25	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494
8	127	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
10	327	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494

Reaction No. 67426



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:509.9(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:506.6(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:372.3(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:291.8(4SF)	4	ENS	6494

5	327	0	Inf. Dilution	lgK:238.1(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-695.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-737.2(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-737.3(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-737.0(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-736.5(4SF)	4	MTD	6494

Reaction No. 67429							
Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Proto.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-348.2(4SF)	4	EFE	6482
2	25	0	Inf. Dilution	dH:-369.0(4SF)	4	EFE	6482

Reaction No. 67430							
Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Feld.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:658.(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-895.18(5SF)	0	EFE	6482
3	25	0	Inf. Dilution	dG:-897.07(5SF)	0	CRV	10174
4	25	0	Inf. Dilution	dG:-896.44(5SF)	5	CRV	29553
5	25	0	GAMSPATH	dG:-3737.(4SF) kJ	0	CRV	12638
6	50	0	Inf. Dilution	dG:-898.39(5SF)	0	CRV	10174
7	75	0	Inf. Dilution	dG:-899.81(5SF)	0	CRV	10174
8	100	0	Inf. Dilution	dG:-901.33(5SF)	0	CRV	10174
9	125	0	Inf. Dilution	dG:-902.93(5SF)	0	CRV	10174
10	150	0	Inf. Dilution	dG:-904.63(5SF)	0	CRV	10174
11	175	0	Inf. Dilution	dG:-906.42(5SF)	0	CRV	10174
12	200	0	Inf. Dilution	dG:-908.28(5SF)	0	CRV	10174
13	225	0	Inf. Dilution	dG:-910.23(5SF)	0	CRV	10174
14	250	0	Inf. Dilution	dG:-912.25(5SF)	0	CRV	10174
15	300	0	Inf. Dilution	dG:-916.48(5SF)	0	CRV	10174
16	25	0	Inf. Dilution	dH:-949.04(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-950.30(5SF)	5	CRV	29553
18	25	0	GAMSPATH	dH:-3962.40(6SF) kJ	0	CRV	12638

Reaction No. 67431							
O2(g) + Si(s) = SiO2(Tridy.,s)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-204.11 (5SF)	4	EFE	6482
2	25	0	Inf. Dilution	dG:-855.26 (5SF) kJ	3	CRV	8695
3	25	0	Inf. Dilution/m	dG:-853.610 (6SF) kJ	3	CRV	24953
4	25	0	Inf. Dilution	dH:-216.96 (5SF)	4	EFE	6482
5	25	0	Inf. Dilution	dH:-909.06 (5SF) kJ	5	CRV	8695
6	25	0	Inf. Dilution/m	dH:-907.490 (6SF) kJ	3	CRV	24953

Reaction No. 67432							
3<Al(s)> + 2<Ca(s)> + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s)> = Ca2Al3Si3O12(OH) (clino,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1137. (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1552.7 (5SF)	0	EFE	6482
3	25	0	GAMSPATH	dG:-6484. (4SF) kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH:-1647.9 (5SF)	2	EFE	6482
5	25	0	GAMSPATH	dH:-6879.40 (6SF) kJ	0	CRV	12638

Reaction No. 67434							
Al2Si2O5(OH)4(Kaol.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH2O4-2> + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:7.428 (4SF)	0	CRV	440
3	25	0	Inf. Dilution	lgK:5.706 (4SF)	0	CRV	5619
4	25	0	Inf. Dilution	lgK:11.122 (5SF)	0	CRV	6421
5	25	0	Inf. Dilution	lgK:7.435 (4SF)	0	ENS	6496
6	25	0	Inf. Dilution	lgK:5.45 (3SF)	0	MGL	6498
Note (6): Possibly for reaction with halved stoichiometry, possibly not!							
7	25	0	Inf. Dilution	lgK:7.41 (3SF)	0	CRV	8872
8	25	0	Inf. Dilution	lgK:11.122 (5SF)	0	EOD	8872
Note (8): Data for Naumov, Ryzhenko & KKhodakovsky, 1971							
9	25	0	Inf. Dilution	lgK:7.028 (4SF)	0	CRV	8873
10	25	0	Inf. Dilution	lgK:7.604 (4SF)	0	CRV	8874
11	25	0	Inf. Dilution	lgK:7.435 (4SF)	3	EOD	29487
Note (11): Presumed reference to Kaolinite							
12	25	0	Inf. Dilution	lgK:6.5 (2SF)	3	MSL	31760
Note (12): Slightly impure solid & average							

13	25	0	Inf. Dilution	lgK:7.446(4SF)	0	CRV	32224
14	40	0	Inf. Dilution	lgK:6.(1SF)	5	MSL	31760
Note (14): Slightly impure solid & average							
15	90	0	Inf. Dilution	lgK:2.80(3SF)	3	EOD	29487
Note (15): Presumed reference to Kaolinite							
16	25	0	Inf. Dilution	dH:-35.3(3SF)	2	ENS	6496
17	25	0	Inf. Dilution	dH:-147.7(4SF)kJ	3	EOD	29487
Note (17): Presumed reference to Kaolinite							

Reaction No. 67435							
$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2 \cdot \text{H}_2\text{O}(\text{s}) + 6\langle\text{H}+1\rangle + 3\langle\text{H}_2\text{O}\rangle = 3\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.79(4SF)	1	ENS	6496
2	25	0	Inf. Dilution	lgK:23.54(4SF)	1	CRV	32224
Note (2): trihydrate							

Reaction No. 67455							
$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Hall.}, \text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Al}+3\rangle + 2\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.72(3SF)	0	MGL	6498

Reaction No. 67456							
$\text{Al}_2\text{O}_3\text{SiOH}(\text{OH})_3(\text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Al}+3\rangle + \text{H}+1(2)\text{SiH}_2\text{O}_4-2 + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:12.00(4SF)	0	MGL	6498

Reaction No. 69044							
$4\langle\text{H}+1\rangle + 4\langle\text{SiH}_2\text{O}_4-2\rangle = \text{H}+1(4)\text{SiH}_2\text{O}_4-2(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.90(4SF)	5	CRV	9402

Reaction No. 70514							
$\text{C}(\text{s}) + \text{Si}(\text{s}) = \text{SiC}(\text{beta}, \text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-62.8(3SF)kJ	4	EFE	7381

2	25	0	Inf. Dilution	dH: -65.3 (3SF) kJ	4	ENS	7381
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Reaction No. 70515							
$2\text{Cl}_2(\text{g}) + \text{Si}(\text{s}) = \text{SiCl}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG: -617.0 (4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG: -616.98 (5SF) kJ	4	EFE	7381
3	25	0	Inf. Dilution	dH: -657.0 (4SF) kJ	3	CRV	6330
4	25	0	Inf. Dilution	dH: -657.01 (5SF) kJ	4	ENS	7381

Reaction No. 71000							
$\text{SiO}_3^{2-} + 3\text{H}_2\text{O} + 4\text{e}^- = \text{Si}(\text{s}) + 6\text{OH}^-$							
Note: The species SiO_3^{2-} is an artefact; it should be $\text{SiH}_2\text{O}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E: -1.697 (4SF)	0	CRV	6330
2	25	0	Inf. Dilution	E: -1.697 (4SF)	0	CRV	22519

Reaction No. 71001							
$\text{SiO}_2(\text{Quartz}, \text{s}) + 4\text{e}^- + 4\text{H}^+ = \text{Si}(\text{s}) + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: -66.9 (3SF)	1	EOR	
2	25	0	Inf. Dilution	E: -0.857 (3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E: -0.990 (3SF)	5	CRV	7761
4	25	0	Inf. Dilution	E: 0.857 (3SF)	0	CRV	22519
Note (4): Wrong sign							
5	25	0	Inf. Dilution	E: -0.909 (3SF)	0	CRV	22551
6	25	0	Inf. Dilution	dH: 339.0 (4SF) kJ	5	CRV	7761

Reaction No. 71002							
$\text{SiO}(\text{s}) + 2\text{H}^+ + 2\text{e}^- = \text{Si}(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: -28.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E: -0.8 (1SF)	0	CRV	6330
3	25	0	Inf. Dilution	E: -0.8 (1SF)	0	CRV	7761
4	25	0	Inf. Dilution	E: -0.8 (1SF)	0	CRV	22519
5	25	0	Inf. Dilution	dH: 131.4 (4SF) kJ	0	CRV	7761

Reaction No. 71003							
$\text{SiF}_6\text{-2} + 4\text{e-1} = \text{Si(s)} + 6\text{F-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-91.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.24(3SF)	0	CRV	6330
Note (2): Low wgt for inter-reaction consistency							

Reaction No. 71432							
$\text{SiO}_2(\text{Quartz},\text{s}) + 8\text{H+1} + 8\text{e-1} = \text{SiH}_4(\text{g}) + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-76.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.569(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:373.6(4SF) kJ	2	CRV	7761

Reaction No. 71433							
$\text{H+1(2)}_{\text{SiH}_2\text{O}_4\text{-2}} + 4\text{e-1} + 4\text{H+1} = \text{Si(s)} + 4\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.931(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	E:-0.848(3SF)	0	CRV	22551
4	25	0	Inf. Dilution	dH:313.9(4SF) kJ	2	CRV	7761

Reaction No. 71434							
$\text{SiO}_2(\text{am.},\text{s}) + 4\text{H+1} + 4\text{e-1} = \text{Si(s)} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-65.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.973(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:329.5(4SF) kJ	2	CRV	7761

Reaction No. 71435							
$\text{SiO}_2(\text{Quartz},\text{s}) + 2\text{H+1} + 2\text{e-1} = \text{SiO(s)} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-38.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.2(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:214.3(4SF) kJ	2	CRV	7761

Reaction No. 71448							
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Si(s) + 4<H2O> + 4<e-1> = SiH4(g) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Inert	lgK:-68.9(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.975(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	dH:257.5(4SF) kJ	5	CRV	7761

Reaction No. 71449							
SiH2O4-2 + 6<H2O> + 8<e-1> = SiH4(g) + 10<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-189.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.405(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:847.5(4SF) kJ	2	CRV	7761

Reaction No. 71450							
SiO(s) + H2O + 2<e-1> = Si(s) + 2<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.6(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:239.7(4SF) kJ	2	CRV	7761

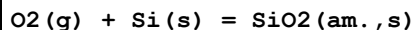
Reaction No. 71451							
H+1_SiH2O4-2 + H2O + 4<e-1> = Si(s) + 5<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-123.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.820(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:565.5(4SF) kJ	2	CRV	7761

Reaction No. 71452							
SiH2O4-2 + 2<H2O> + 4<e-1> = Si(s) + 6<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-124.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.834(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:589.3(4SF) kJ	2	CRV	7761

Reaction No. 71453							
SiH2O4-2 + H2O + 2<e-1> = SiO(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	3	Inert	lgK:-69.(2SF)	2	EES	
2	25	0	Inf. Dilution	E:-2.0(2SF)	3	CRV	7761
3	25	0	Inf. Dilution	dH:339.9(4SF)kJ	3	CRV	7761

Reaction No. 71739



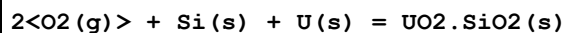
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-849.9(4SF)kJ	0	EOD	23101
3	25	0	Inf. Dilution	dG:-848.90(5SF)kJ	5	CRV	31308
4	25	0	GAMSPATH	dG:-848.9(4SF)kJ	0	CRV	12638
5	25	0	Inf. Dilution/m	dG:-849.230(6SF)kJ	0	CRV	24953
6	25	0	Inf. Dilution	dH:-903.(3SF)kJ	0	EOD	23101
7	25	0	Inf. Dilution	dH:-903.(3SF)kJ	5	CRV	31308
8	25	0	GAMSPATH	dH:-897.750(6SF)kJ	0	CRV	12638
9	25	0	Inf. Dilution/m	dH:-901.550(6SF)kJ	2	CRV	24953

Reaction No. 71746



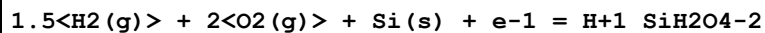
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-4040.(4SF)kJ	3	CRV	12638
2	25	0	GAMSPATH	dH:-4367.70(6SF)kJ	3	CRV	12638

Reaction No. 71748



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:329.7(4SF)	1	ELC	
2	25	0	GAMSPATH	dG:-1084.(4SF)kJ	0	CRV	12638
3	25	0	GAMSPATH	dH:-1991.30(6SF)kJ	0	CRV	12638

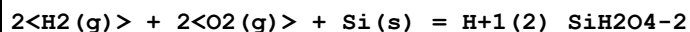
Reaction No. 71754



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:219.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1251.7(1.162SD)kJ	0	CRV	14923
3	25	0	GAMSPATH	dG:-1252.(4SF)kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH:-1431.4(3.743SD)kJ	1	CRV	14923

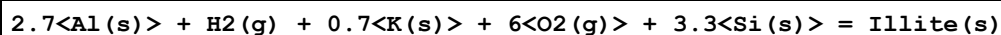
5	25	0	GAMSPATH	dH:-1430.70 (6SF) kJ	1	CRV	12638
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Reaction No. 71819



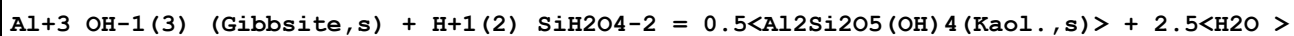
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1307.5 (5SF) kJ	0	CRV	6494
3	25	0	Inf. Dilution	dG:-1309.20 (6SF) kJ	0	CRV	8798
4	25	0	Inf. Dilution	dG:-1316.6 (5SF) kJ	0	CRV	9015
5	25	0	Inf. Dilution	dG:-1309.2 (6SF) kJ	0	CRV	9033
6	25	0	Inf. Dilution	dG:-1309.3 (5SF) kJ	4	EOD	12843
7	25	0	Inf. Dilution	dG:-1307.7 (1.156SD) kJ	0	CRV	14923
8	25	0	Inf. Dilution	dG:-312.92 (5SF)	5	CRV	29553
9	25	0	Inf. Dilution	dG:-1239.70 (0.6SD) kJ	0	CRV	30883
10	25	0	Inf. Dilution	dH:-1460.0 (5SF) kJ	0	CRV	6494
11	25	0	Inf. Dilution	dH:-1460.20 (6SF) kJ	0	CRV	8798
12	25	0	Inf. Dilution	dH:-1468.6 (5SF) kJ	0	CRV	9015
13	25	0	Inf. Dilution	dH:-1460.9 (6SF) kJ	0	MSL	9033
14	25	0	Inf. Dilution	dH:-1458.9 (5SF) kJ	4	MTD	12843
15	25	0	Inf. Dilution	dH:-1457.0 (3.163SD) kJ	0	CRV	14923
16	25	0	Inf. Dilution	dH:-348.68 (5SF)	5	CRV	29553
17	25	0	GAMSPATH	dH:-1456.30 (6SF) kJ	0	CRV	12638

Reaction No. 71820



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-5491. (4SF) kJ	3	CRV	12638
2	25	0	GAMSPATH	dH:-5870. (4SF) kJ	3	CRV	12638

Reaction No. 71872



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.1 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.25 (3SF)	0	CRV	6405

Note (2): p. 246

Reaction No. 71873

$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + 2\langle\text{K}+1\rangle + 6\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle - \text{H}_2\text{O} = 2$ $\langle\text{KMg}_3\text{AlSi}_3\text{O}_{10}(\text{OH})_2(s)\rangle + 14\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-69.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:70.14(4SF)	0	CRV	6405
Note (2): p. 246							

Reaction No. 71874							
$\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2(s) + \text{H}+1 + 1.5\langle\text{H}_2\text{O}\rangle = 1.5\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + \text{K}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.51(3SF)	2	CRV	6405

Reaction No. 71875							
$\text{CaO}.\text{Al}_2\text{O}_3.2\text{SiO}_2(s) + 2\langle\text{H}+1\rangle + \text{H}_2\text{O} = \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + \text{Ca}+2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:19.66(4SF)	0	CRV	6405

Reaction No. 71876							
$2\langle\text{KAlSi}_3\text{O}_8(\text{Feld.},s)\rangle + 2\langle\text{H}+1\rangle + 9\langle\text{H}_2\text{O}\rangle = \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + 4\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + 2\langle\text{K}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.0(2SF)	3	CRV	6405

Reaction No. 71877							
$\text{Na}_2\text{O}.\text{Al}_2\text{O}_3.7.(\text{SiO}_2)22.6\text{H}_2\text{O}(s) + 2\langle\text{H}+1\rangle + 23\langle\text{H}_2\text{O}\rangle = 7\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + 8$ $\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + 2\langle\text{Na}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.22(4SF)	5	CRV	6405
Note (1): p. 246 - comment re inconsistency with Stumm & Morgan data							

Reaction No. 71878							
$\text{CaO}.\text{Al}_2\text{O}_3.7.(\text{SiO}_2)22.6\text{H}_2\text{O}(s) + 2\langle\text{H}+1\rangle + 23\langle\text{H}_2\text{O}\rangle = 7\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + 8$ $\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + \text{Ca}+2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-15.4(3SF)	0	CRV	6405
Note (2): p. 246 - comment re inconsistency with Stumm & Morgan data							

Reaction No. 71947							
$\text{H}+1(2)\text{SiH}_2\text{O}_4-2 + 3\text{H}+1(2)\text{45DiClCat}-2 = \text{Si}+4\text{45DiClCat}-2(3) + 2\text{H}+1 + 4\text{H}_2\text{O} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.27(3SF)	0	ENS	7271
Note (1): Incorrect citation of value for 3,4-Dinitrocatechol							
2	25	0.1	Unknown	lgK:-8.5(2SF)	3	MMR	8688
Note (2): Suspect other values							

Reaction No. 71949							
$\text{H}+1(2)\text{SiH}_2\text{O}_4-2 + 3\text{H}+1(2)\text{34DiNitrCat}-2 = \text{Si}+4\text{34DiNitrCat}-2(3) + 2\text{H}+1 + 4\text{H}_2\text{O} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.27(3SF)	4	ENS	8688

Reaction No. 71951							
$\text{H}+1(2)\text{SiH}_2\text{O}_4-2 + 3\text{H}+1(2)\text{4NitrCat}-2 = \text{Si}+4\text{4NitrCat}-2(3) + 2\text{H}+1 + 4\text{H}_2\text{O} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-7.74(3SF)	4	MMR	8688
Note (1): Suspect other values							

Reaction No. 73370							
$\text{Al}+3 + \text{H}+1(2)\text{SiH}_2\text{O}_4-2 = \text{Al}+3\text{H}+1\text{SiH}_2\text{O}_4-2 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.50(0.05SD)	4	MGL	7023

Reaction No. 73538							
$2\text{H}_2(\text{g}) + 3.5\text{O}_2(\text{g}) + 2\text{Si}(\text{s}) + 2\text{e}-1 = \text{Si}_2\text{O}_3(\text{OH})_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:398.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2269.9(2.878SD)kJ	0	CRV	14923

Reaction No. 73539							
$2.5\text{H}_2(\text{g}) + 3.5\text{O}_2(\text{g}) + 2\text{Si}(\text{s}) + \text{e}-1 = \text{Si}_2\text{O}_2(\text{OH})_5-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:409.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2332.1(2.878SD)kJ	0	CRV	14923

Reaction No. 73540							
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1.5<H2(g)> + 4.5<O2(g)> + 3<Si(s)> + 3<e-1> = Si3O6(OH)3-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:535.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3048.5(3.870SD)kJ	0	CRV	14923

Reaction No. 73541							
2.5<H2(g)> + 5<O2(g)> + 3<Si(s)> + 3<e-1> = Si3O5(OH)5-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:576.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3292.0(3.896SD)kJ	0	CRV	14923

Reaction No. 73542							
2<H2(g)> + 6<O2(g)> + 4<Si(s)> + 4<e-1> = Si4O8(OH)4-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:715.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-4075.2(5.437SD)kJ	0	CRV	14923

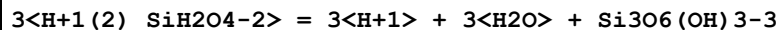
Reaction No. 73543							
2.5<H2(g)> + 6<O2(g)> + 4<Si(s)> + 3<e-1> = Si4O7(OH)5-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:726.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-4136.8(4.934SD)kJ	0	CRV	14923

Reaction No. 73546							
2<H+1(2)_SiH2O4-2> = 2<H+1> + H2O + Si2O3(OH)4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.000(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-19.000(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:108.450(0.9SD)kJ	2	CRV	27292

Reaction No. 73547							
2<H+1(2)_SiH2O4-2> = H+1 + Si2O2(OH)5-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.100(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-8.100(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	lgK:-8.50(3SF)	4	EOD	30201
4	50	0	Inf. Dilution	lgK:-8.14(0.25SD)	4	EOD	30201

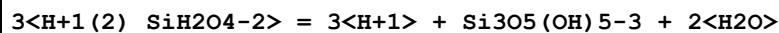
5	75	0	Inf. Dilution	lgK:-7.86(0.25SD)	4	EOD	30201
6	100	0	Inf. Dilution	lgK:-7.65(0.25SD)	4	EOD	30201
7	25	0	Inf. Dilution	dG:46.235(0.9SD)kJ	2	CRV	27292

Reaction No. 73548



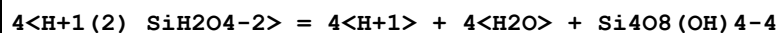
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.600(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-28.600(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:163.250(0.9SD)kJ	2	CRV	27292

Reaction No. 73549



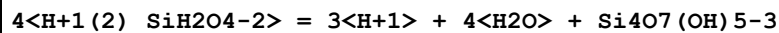
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-27.500(0.300SD)	2	CRV	14923
2	25	0	Inf. Dilution	lgK:-27.500(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	lgK:-29.40(4SF)	3	EOD	30201
4	50	0	Inf. Dilution	lgK:-28.75(0.25SD)	4	EOD	30201
5	75	0	Inf. Dilution	lgK:-28.34(4SF)	4	EOD	30201
6	100	0	Inf. Dilution	lgK:-28.11(0.25SD)	4	EOD	30201
7	25	0	Inf. Dilution	dG:156.970(0.9SD)kJ	1	CRV	27292

Reaction No. 73550



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.300(0.500SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-36.300(0.25SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:207.200(1.4SD)kJ	2	CRV	27292

Reaction No. 73551



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.500(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-25.500(0.15SD)	3	CRV	27292
3	25	0	Inf. Dilution	dG:145.560(0.9SD)kJ	3	CRV	27292

Reaction No. 74156

Si+4_F-1(6) + 4<e-1> = Si(s) + 6<F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-93.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.24(3SF)	0	CRV	22519

Reaction No. 74236							
SiO2(Cristob.,s) + 4<e-1> + 4<H+1> = Si(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-66.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.900(3SF)	0	CRV	22551

Reaction No. 74237							
H+1_SiH2O4-2 + 4<e-1> + 5<H+1> = Si(s) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-53.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.712(3SF)	0	CRV	22551

Reaction No. 74238							
SiO(g) + 2<e-1> + 2<H+1> = Si(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.808(3SF)	0	CRV	22551

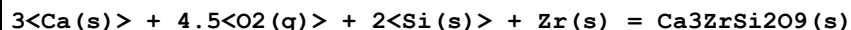
Reaction No. 74239							
H+1(2)_SiH2O4-2 + 8<e-1> + 8<H+1> = SiH4(g) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-73.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.495(3SF)	0	CRV	22551

Reaction No. 74240							
H+1(2)_SiO3-2_(s) = H+1_SiO3-2 + H+1							
Note: The species SiO3-2 is an artefact; it should be SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.(2SF)	0	CRV	22551

Reaction No. 74597							
2<Ca(s)> + 6<O2(g)> + 3<Si(s)> + Zr(s) = Ca2ZrSi3O12(s)							

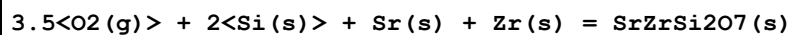
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5951.5 (15.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6283.0 (15.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:252.500 (2.5MD) J/K	6	CRV	20829

Reaction No. 74598



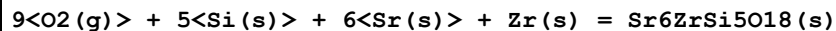
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4769.3 (10.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-5029.0 (10.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:238.100 (1.5MD) J/K	6	CRV	20829

Reaction No. 74599



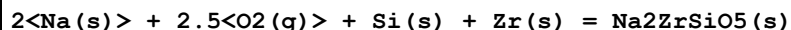
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3444.0 (4.2MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3640.8 (4.2MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:187.810 (0.4MD) J/K	6	CRV	20829

Reaction No. 74600



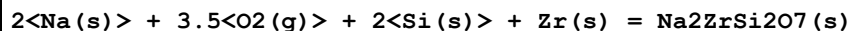
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7800. (4SF) kJ	1	EES	
2	25	0	Inf. Dilution	dH:-9867.9 (8.6MD) kJ	6	CRV	20829

Reaction No. 74601



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2523.5 (20.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2670.0 (20.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:172.400 (2.0MD) J/K	6	CRV	20829

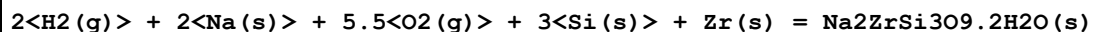
Reaction No. 74602



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3412.4 (10.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3606.0 (10.0MD) kJ	6	CRV	20829

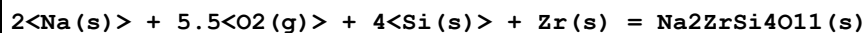
3	25	0	Inf. Dilution	Cp:211.000 (5.0MD) J/K	6	CRV	20829
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Reaction No. 74603



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4654.0 (20.0MD) kJ	6	CRV	20829

Reaction No. 74605



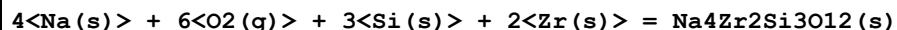
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5181.0 (20.000MD) kJ	6	CRV	20829

Reaction No. 74606



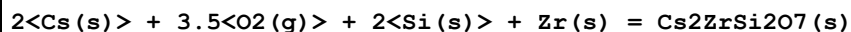
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7931.0 (30.000MD) kJ	6	CRV	20829

Reaction No. 74607



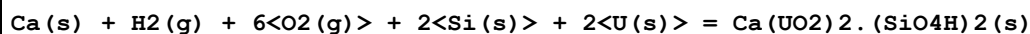
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5944.0 (20.057MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6258.0 (20.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:365.000 (5.0MD) J/K	6	CRV	20829

Reaction No. 74609



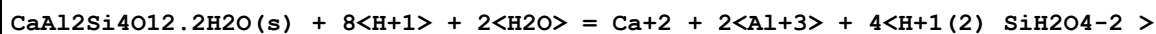
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3587.0 (10.0MD) kJ	6	CRV	20829

Reaction No. 75172



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1189. (4SF)	5	EOD	13532

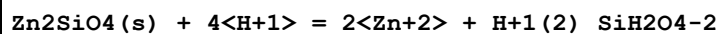
Reaction No. 75220



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.87 (4SF)	0	EOD	23102

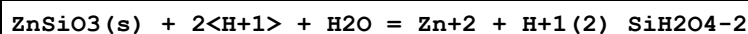
2	25	0	Inf. Dilution	lgK:14.42 (4SF)	3	CRV	27383
3	25	0	Inf. Dilution	lgK:18.68 (4SF)	0	CRV	32224

Reaction No. 75223



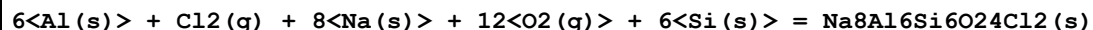
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.40 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:15.33 (4SF)	0	EOD	23102
3	25	0	Inf. Dilution	lgK:13.89 (4SF)	4	CRV	32224

Reaction No. 75224



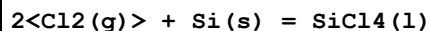
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.93 (3SF)	0	EOD	23102
3	25	0	Inf. Dilution	lgK:-19.85 (4SF)	0	CRV	32224

Reaction No. 75559



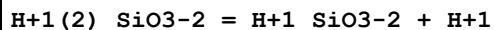
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1.2697E+4 (5SF) kJ	2	EFE	24802

Reaction No. 75627



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-619.8 (4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-687.0 (4SF) kJ	3	CRV	6330

Reaction No. 75630

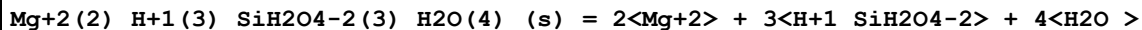


Note: The species SiO₃-2 is an artefact; it should be SiH₂O₄-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:9.86 (3SF)	0	CRV	6405

Note (1): p. 285

Reaction No. 75631



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0	Inf. Dilution	lgK:-38.8(3SF)	0	CRV	8
Note (1): Solid +1 charge! Same problem in S&M database [95SMM_552]. Sepiolite?							

Reaction No. 75633							
1.7<Al(s)> + 0.7<Ca(s)> + 0.3<Na(s)> + 4<O2(g)> + Si(s) = Plagioclase(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3906.5(5SF)kJ	5	CRV	25435
2	25	0	Inf. Dilution	dH:-4108.2(5SF)kJ	5	CRV	25435

Reaction No. 75634							
0.35<Ca(s)> + 0.23<Fe(s)> + 0.42<Mg(s)> + 1.5<O2(g)> + Si(s) = Augite(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2835.3(5SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-2888.5(5SF)kJ	3	CRV	25435

Reaction No. 75635							
0.4<Fe(s)> + 1.6<Mg(s)> + 2<O2(g)> + Si(s) = Olivine(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1919.7(5SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-2022.9(5SF)kJ	3	CRV	25435

Reaction No. 75636							
2<Al(s)> + 2<H2(g)> + 3.72<O2(g)> + 1.22<Si(s)> = Allophane(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3227.90(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-3512.20(6SF)kJ	3	CRV	25435

Reaction No. 75637							
O2(g) + Si(s) = SiO2(Moganite,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-851.310(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-900.720(6SF)kJ	3	CRV	25435

Reaction No. 75638							
SiO2(Chalcedony,s) + 2<H2O> = H+1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	0	0	Inf. Dilution	lgK:-3.857 (4SF)	0	CRV	25435
2	25	0	Inf. Dilution	lgK:-3.472 (4SF)	0	CRV	25435
3	25	0	Inf. Dilution	lgK:-3.73 (3SF)	4	CRV	32224
4	60	0	Inf. Dilution	lgK:-3.106 (4SF)	0	CRV	25435
5	100	0	Inf. Dilution	lgK:-2.790 (4SF)	0	CRV	25435
6	150	0	Inf. Dilution	lgK:-2.470 (4SF)	0	CRV	25435
7	200	0	Inf. Dilution	lgK:-2.205 (4SF)	0	CRV	25435
8	250	0	Inf. Dilution	lgK:-1.988 (4SF)	0	CRV	25435
9	300	0	Inf. Dilution	lgK:-1.835 (4SF)	0	CRV	25435

Reaction No. 76501							
$2\text{Al(s)} + 2\text{H}_2\text{(g)} + 2\text{Na(s)} + 6\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{Na}_2\text{Al}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O (Natr., s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:931.4 (4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-5316.5 (5SF) kJ	3	CRV	6494
3	25	0	Inf. Dilution	dG:-5316.6 (5.0SD) kJ	5	MCL	29317
4	25	0	Inf. Dilution	dH:-5718.6 (5SF) kJ	3	CRV	6494
5	25	0	Inf. Dilution	dH:-5718.6 (5.0SD) kJ	5	CRV	29317

Reaction No. 76502							
$2\text{B(s)} + \text{Ca(s)} + 4\text{O}_2\text{(g)} + 2\text{Si(s)} = \text{CaB}_2\text{Si}_2\text{O}_8\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:644.2 (4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-3677.0 (5SF) kJ	3	CRV	6494
3	25	0	Inf. Dilution	dH:-3902.8 (5SF) kJ	3	CRV	6494

Reaction No. 76503							
$2\text{Al(s)} + \text{Ca(s)} + 3\text{H}_2\text{(g)} + 6.5\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{CaAl}_2\text{Si}_3\text{O}_{10} \cdot 3\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:980.7 (4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-5597.6 (5SF) kJ	3	CRV	6494
3	25	0	Inf. Dilution	dG:-5597.9 (5.0SD) kJ	5	MCL	29317
4	25	0	Inf. Dilution	dH:-6049.0 (5SF) kJ	3	CRV	6494
5	25	0	Inf. Dilution	dH:-6049.00 (5.5SD) kJ	4	CRV	27383
6	25	0	Inf. Dilution	dH:-6049.0 (5.0SD) kJ	5	CRV	29317

Reaction No. 77352							
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$\text{Am(s)} + 1.5\text{<H}_2\text{(g)>} + 2\text{<O}_2\text{(g)>} + \text{Si(s)} - 2\text{<e-1>} = \text{Am+3_H+1_SiH}_2\text{O}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1896.80 (6SF) kJ	5	CRV	20827

Reaction No. 77366							
$\text{Am+3} + \text{H+1(2)_SiH}_2\text{O}_4\text{-2} = \text{Am+3_H+1_SiH}_2\text{O}_4\text{-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.680 (4SF)	5	CRV	20827
2	25	0	Inf. Dilution	dG:9.590 (4SF) kJ	5	CRV	20827

Reaction No. 77619							
$2\text{<SiH}_2\text{O}_4\text{-2>} + 2\text{<H+1>} = \text{H+1(2)_SiH}_2\text{O}_4\text{-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.32 (4SF)	1	ELC	

Reaction No. 79081							
$\text{SiH}_2\text{O}_4\text{-2} + \text{H+1} + \text{Am+3} = \text{Am+3_H+1_SiH}_2\text{O}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.47 (4SF)	1	ELC	

Reaction No. 79429							
$\text{SiH}_2\text{O}_4\text{-2} + \text{H+1} + \text{Al+3} = \text{Al+3_H+1_SiH}_2\text{O}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.65 (4SF)	1	ELC	

Reaction No. 79464							
$1.5\text{<O}_2\text{(g)>} + \text{Si(s)} + 2\text{<e-1>} = \text{SiO}_3\text{-2}$							
Note: The species SiO3-2 is an artefact; it should be SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-887. (3SF) kJ	0	CRV	7421

Reaction No. 79503							
$0.5\text{<H}_2\text{(g)>} + 1.5\text{<O}_2\text{(g)>} + \text{Si(s)} + \text{e-1} = \text{H+1_SiO}_3\text{-2}$							
Note: The species SiO3-2 is an artefact; it should be SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1015.2 (5SF) kJ	0	EFE	28693

Reaction No. 79504							
$0.5\text{H}_2(\text{g}) + \text{Na}(\text{s}) + 1.5\text{O}_2(\text{g}) + \text{Si}(\text{s}) = \text{H}_{+1}\text{Na}_{+1}\text{SiO}_3\text{-2}$							
Note: The species SiO ₃ -2 is an artefact; it should be SiH ₂ O ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1283.1 (5SF) kJ	0	EFE	28693

Reaction No. 79505							
$\text{Al}(\text{s}) + 1.5\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) + \text{Si}(\text{s}) - 2\text{e-1} = \text{Al}_{+3}\text{H}_{+1}\text{SiH}_2\text{O}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1782.5 (5SF) kJ	3	EFE	28693

Reaction No. 79668							
$1.99\text{Al}(\text{s}) + 0.657\text{Ca}(\text{s}) + 2.147\text{H}_2(\text{g}) + 0.676\text{Na}(\text{s}) + 6.3235\text{O}_2(\text{g}) + 3.01\text{Si}(\text{s}) = \text{Mesolite}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5513.2 (5.4SD) kJ	5	MCL	29317
2	25	0	Inf. Dilution	dH:-5947.1 (5.4SD) kJ	5	CRV	29317

Reaction No. 79669							
$\text{CaAl}_2\text{Si}_4\text{O}_{12}\cdot 6\text{H}_2\text{O}(\text{s}) + 8\text{H}_{+1} = \text{Ca}_{+2} + 2\text{Al}_{+3} + 4\text{H}_{+1}(2)\text{SiH}_2\text{O}_4\text{-2} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.51 (4SF)	4	CRV	27383

Reaction No. 79670							
$\text{Ca}_2\text{Al}_4\text{Si}_4\text{O}_{16}\cdot 9\text{H}_2\text{O}(\text{s}) + 16\text{H}_{+1} = 2\text{Ca}_{+2} + 4\text{Al}_{+3} + 4\text{H}_{+1}(2)\text{SiH}_2\text{O}_4\text{-2} + 9\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.96 (4SF)	4	CRV	27383

Reaction No. 79674							
$\text{Fe}_{+3} + \text{H}_{+1}(2)\text{SiH}_2\text{O}_4\text{-2} = \text{Fe}_{+3}\text{H}_{+1}\text{SiH}_2\text{O}_4\text{-2} + \text{H}_{+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO ₄ /m	lgK:-0.521 (3SF)	4	MSP	29280
2	25	0.1	NaClO ₄ /m	lgK:-0.527 (3SF)	4	MSP	29280
3	25	0.3	NaClO ₄ /m	lgK:-0.526 (3SF)	4	MSP	29280
4	25	0.3	NaClO ₄ /m	lgK:-0.532 (3SF)	4	MSP	29280
5	25	0.7	NaClO ₄ /m	lgK:-0.418 (3SF)	4	MSP	29280
6	25	0.7	NaClO ₄ /m	lgK:-0.564 (3SF)	4	MSP	29280

Reaction No. 79675							
$\text{Eu}+3 + 2\text{H}+1\text{SiH}_2\text{O}_4-2 = \text{Eu}+3\text{H}+1(2)\text{SiH}_2\text{O}_4-2(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO ₄ /m	lgK:11.73(4SF)	4	MSP	29280

Reaction No. 79676							
$\text{Eu}+3 + \text{H}+1\text{SiH}_2\text{O}_4-2 = \text{Eu}+3\text{H}+1\text{SiH}_2\text{O}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO ₄ /m	lgK:6.17(0.10SD)	4	MSP	29280
2	25	0.7	NaClO ₄	lgK:6.18(0.18SD)	5	MGL	29968
Note (2): Paper casts doubt on earlier results by various other authors							

Reaction No. 79849							
$\text{H}+1\text{SiH}_2\text{O}_4-2 + \text{Am}+3 = \text{Am}+3\text{H}+1\text{SiH}_2\text{O}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.13(0.18SD)	5	CRV	25812

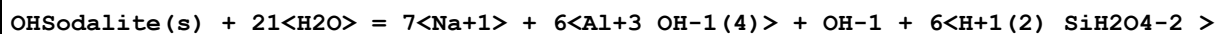
Reaction No. 80285							
$2\text{CaO.SiO}_2(\text{beta},s) + 2\text{CO}_2(\text{g}) = 2\text{Ca}+2\text{CO}_3-2(\text{Calc.},s) + \text{SiO}_2(\text{am.},s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-142.06(5SF)kJ	0	CRV	30199

Reaction No. 80306							
$\text{Si}_2\text{O}_2(\text{OH})_5-1 = \text{H}+1 + \text{Si}_2\text{O}_3(\text{OH})_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.90(4SF)	4	EOD	30201
2	50	0	Inf. Dilution	lgK:-10.59(0.25SD)	4	EOD	30201
3	75	0	Inf. Dilution	lgK:-10.38(0.25SD)	4	EOD	30201
4	100	0	Inf. Dilution	lgK:-10.24(0.25SD)	4	EOD	30201

Reaction No. 80307							
$\text{Al}+3\text{OH}-1(4) + \text{SiH}_2\text{O}_4-2 = \text{Al}(\text{OH})_3.\text{HSiO}_4-3 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-0.42(0.10SD)	4	EOD	30201
2	25	0	Inf. Dilution	lgK:-0.42(0.15SD)	4	EOD	30201
3	50	0	Inf. Dilution	lgK:-0.38(0.25SD)	4	EOD	30201

4	75	0	Inf. Dilution	lgK:-0.35(0.25SD)	4	EOD	30201
5	100	0	Inf. Dilution	lgK:-0.33(0.25SD)	4	EOD	30201

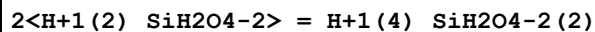
Reaction No. 80308



Note: Stoichiometric analog of actual reported solid

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-60.34(4SF)	4	EOD	30201
2	30	0	Inf. Dilution	lgK:-61.18(4SF)	4	EOD	30201
3	45	0	Inf. Dilution	lgK:-64.20(4SF)	4	EOD	30201
4	60	0	Inf. Dilution	lgK:-66.69(4SF)	4	EOD	30201
5	75	0	Inf. Dilution	lgK:-68.89(4SF)	3	EOD	30201
6	90	0	Inf. Dilution	lgK:-71.26(4SF)	4	EOD	30201
7	100	0	Inf. Dilution	lgK:-72.68(4SF)	4	EOD	30201

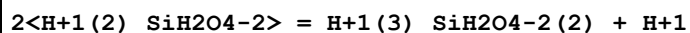
Reaction No. 80408



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:1.2(2SF)	3	CRV	30909

Note (1): Corrected misprint in source

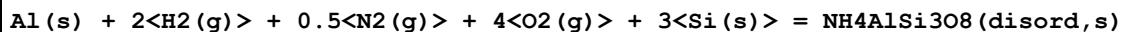
Reaction No. 80409



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-7.75(3SF)	3	CRV	30909

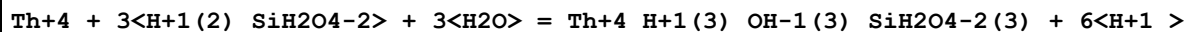
Note (1): Corrected misprint in source

Reaction No. 80410



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3525.1(5SF)kJ	3	ETR	30953

Reaction No. 80588



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-27.8(0.35SD)	5	CRV	31301

Reaction No. 80595							
$\text{U}+4 + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}(\text{s}) = \text{UO}_2.\text{SiO}_2(\text{s}) + 4\text{H}+1>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 2.98(1.\text{SD})$	5	CRV	31301

Reaction No. 80660							
$\text{C}(\text{s}) + 15\text{H}_2(\text{g}) + 3\text{Ca}(\text{s}) + 12.5\text{O}_2(\text{g}) + \text{S}(\text{s}) + \text{Si}(\text{s}) = \text{Ca}_3\text{Si}(\text{OH})_6\text{CO}_3\text{SO}_4.12\text{H}_2\text{O}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -7564.5(5\text{SF}) \text{ kJ}$	5	CRV	31308
2	25	0	Inf. Dilution	$\text{dH}: -8700.(4\text{SF}) \text{ kJ}$	5	CRV	31308

Reaction No. 80662							
$\text{Ca}_3\text{Si}(\text{OH})_6\text{CO}_3\text{SO}_4.12\text{H}_2\text{O}(\text{s}) = 3\text{Ca}+2> + \text{H}+1_{\text{SiH}_2\text{O}_4-2} + \text{SO}_4-2 + \text{CO}_3-2 + \text{OH}-1 + 13\text{H}_2\text{O} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -24.75(4\text{SF})$	5	CRV	31308

Reaction No. 80666							
$\text{Na}_2\text{Al}_2\text{Si}_3\text{O}_{10}.2\text{H}_2\text{O}(\text{Natr.}, \text{s}) + 8\text{H}_2\text{O} = 2\text{Na}+1> + 2\text{Al}+3_{\text{OH}-1(4)}> + 3\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -25.19(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -7.20(3\text{SF})$	0	CRV	31308
Note (2): For species $\text{SiO}_2(\text{aq})$!							

Reaction No. 80880							
$\text{Illite}(\text{s}) + 8.8\text{H}+1> + 1.2\text{H}_2\text{O} = 2.7\text{Al}+3> + 3.3\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + 0.7\text{K}+1 >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 11.5(3\text{SF})$	5	MSL	31760
Note (1): Slightly impure solid							
2	40	0	Inf. Dilution	$\lg K: 9.6(2\text{SF})$	5	MSL	31760
Note (2): Slightly impure solid							

Reaction No. 80886							
$\text{NaAlSiO}_4.2\text{H}_2\text{O}(\text{s}) + 4\text{H}+1> = \text{Na}+1 + \text{Al}+3 + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 12.1(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 11.1(0.2\text{SD})$	0	MSL	31772

Reaction No. 80887							
$\text{NaAlSiO}_4 \cdot 2\text{H}_2\text{O}(\text{s}) + 2\langle\text{OH}^{-1}\rangle = \text{Na}^{+1} + \text{Al}^{+3}\text{OH}^{-1}(4) + \text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -5.74(0.2\text{SD})$	5	MSL	31772

Reaction No. 80888							
$\text{NaAlSiO}_4 \cdot 2\text{H}_2\text{O}(\text{am.,s}) + 2\langle\text{OH}^{-1}\rangle = \text{Na}^{+1} + \text{Al}^{+3}\text{OH}^{-1}(4) + \text{SiH}_2\text{O}_4^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.15(3\text{SF})$	5	MSL	31772
Note (1): Table 4							

Reaction No. 81040							
$\text{Ni}^{+2} + \text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2} = \text{NiSiO}_3(\text{s}) + 2\langle\text{H}^{+1}\rangle + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 2.09(3\text{SF})$	4	CRV	32222

Reaction No. 81041							
$2\langle\text{Ni}^{+2}\rangle + \text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2} = 2\text{NiO} \cdot \text{SiO}_2(\text{s}) + 4\langle\text{H}^{+1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -19.24(1.32\text{SD})$	4	CRV	32222

Reaction No. 81104							
$\text{UO}_2^{+2} + \text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2} = \text{UO}_2^{+2}\text{H}^{+1}\text{SiH}_2\text{O}_4^{-2} + \text{H}^{+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.84(0.1\text{SD})$	4	CRV	32222

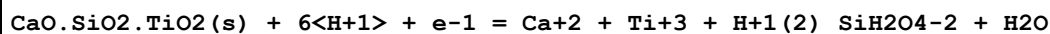
Reaction No. 81105							
$\text{UO}_2^{+2} \cdot 2\text{SiO}_4 \cdot 2\text{H}_2\text{O}(\text{s}) + 4\langle\text{H}^{+1}\rangle = 2\langle\text{UO}_2^{+2}\rangle + \text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2} + 2\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 5.74(0.21\text{SD})$	4	CRV	32222

Reaction No. 81346							
$3\text{CaO} \cdot \text{Fe}_2\text{O}_3 \cdot 3\text{SiO}_2(\text{s}) + 12\langle\text{H}^{+1}\rangle + 2\langle\text{e}^{-1}\rangle = 3\langle\text{Ca}^{+2}\rangle + 2\langle\text{Fe}^{+2}\rangle + 3\langle\text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 54.81(4\text{SF})$	4	CRV	32224

Reaction No. 81347							
$\text{CaO} \cdot \text{SiO}_2(\text{Pseudo.,s}) + 2\langle\text{H}^{+1}\rangle + \text{H}_2\text{O} = \text{Ca}^{+2} + \text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2}$							

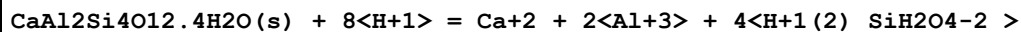
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.87 (4SF)	4	CRV	32224

Reaction No. 81348



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.4 (2SF)	1	ELC	32224
2	25	0	Inf. Dilution	lgK:-9.140 (4SF)	0	CRV	32224

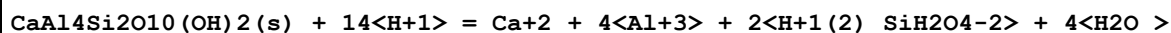
Reaction No. 81349



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.66 (4SF)	6	CRV	27383
2	25	0	Inf. Dilution	lgK:14.25 (4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:11.85 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:11.85 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:11.85 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:11.80 (0.01SD)	1	ENS	5156

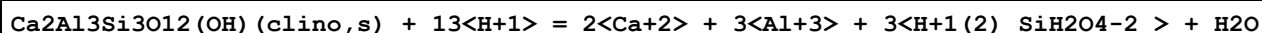
Note (6): query mineral name "laumontite"

Reaction No. 81350



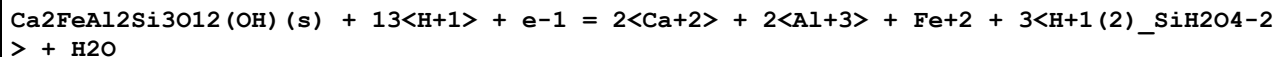
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.33 (4SF)	4	CRV	32224

Reaction No. 81351



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.16 (4SF)	4	CRV	32224

Reaction No. 81352



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.29 (4SF)	4	CRV	32224

Reaction No. 81358

Ca(UO ₂) ₂ .(SiO ₄ H) ₂ (s) + 14<H+1> + 4<e-1> = Ca+2 + 2<U+4> + 2<H+1(2)_SiH ₂ O ₄ -2> + 4<H ₂ O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.85(4SF)	4	CRV	32224

Reaction No. 81414							
CuSiO ₃ .2H ₂ O(s) + 2<H+1> = Cu+2 + H+1(2)_SiH ₂ O ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.92(3SF)	4	CRV	32224

Reaction No. 81444							
FeSi(s) + 4<H ₂ O> = Fe+2 + H+1(2)_SiH ₂ O ₄ -2 + 4<H+1> + 6<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:65.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:63.84(4SF)	0	CRV	32224

Reaction No. 81445							
FeSi ₂ (s) + 8<H ₂ O> = Fe+2 + 2<H+1(2)_SiH ₂ O ₄ -2> + 8<H+1> + 10<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:128.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:125.97(5SF)	0	CRV	32224

Reaction No. 81446							
FeSi _{2.33} (s) + 9.32<H ₂ O> = Fe+2 + 2.33<H+1(2)_SiH ₂ O ₄ -2> + 9.32<H+1> + 11.32<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:152.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:150.18(5SF)	0	CRV	32224

Reaction No. 81447							
Fe ₃ Si(s) + 4<H ₂ O> = 3<Fe+2> + H+1(2)_SiH ₂ O ₄ -2 + 4<H+1> + 10<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:94.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:87.82(4SF)	0	CRV	32224

Reaction No. 81452							
KAlSi ₃ O ₈ (Feld.,s) + 4<H+1> + 4<H ₂ O> = K+1 + Al+3 + 3<H+1(2)_SiH ₂ O ₄ -2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.05(1SF)	4	CRV	32224

Reaction No. 81453							
$\text{KAlSi}_3\text{O}_8(\text{Mur.}, \text{s}) + 4\langle\text{H}_2\text{O}\rangle + 4\langle\text{H}+1\rangle = \text{K}+1 + \text{Al}+3 + 3\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.25(3SF)	4	CRV	32224

Reaction No. 81454							
$\text{KFe}_3\text{AlSi}_3\text{O}_{10}(\text{OH})_2(\text{s}) + 10\langle\text{H}+1\rangle = \text{K}+1 + 3\langle\text{Fe}+2\rangle + \text{Al}+3 + 3\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.13(4SF)	4	CRV	32224

Reaction No. 81455							
$\text{KMg}_3\text{AlSi}_3\text{O}_{10}(\text{OH})_2(\text{s}) + 10\langle\text{H}+1\rangle = \text{K}+1 + 3\langle\text{Mg}+2\rangle + \text{Al}+3 + 3\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.17(4SF)	4	CRV	32224

Reaction No. 81467							
$\text{Mg}_2\text{Si}(\text{s}) + 4\langle\text{H}_2\text{O}\rangle = 2\langle\text{Mg}+2\rangle + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + 4\langle\text{H}+1\rangle + 8\langle\text{e}-1\rangle >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:209.12(5SF)	4	CRV	32224

Reaction No. 81473							
$2\text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2 \cdot \text{H}_2\text{O}(\text{s}) + 16\langle\text{H}+1\rangle + \text{H}_2\text{O} = 2\langle\text{Mg}+2\rangle + 4\langle\text{Al}+3\rangle + 5\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.39(4SF)	4	CRV	32224

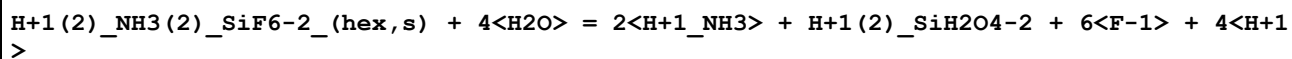
Reaction No. 81474							
$\text{Mg}_{48}\text{Si}_{34}\text{O}_{85}(\text{OH})_{62}(\text{s}) + 96\langle\text{H}+1\rangle = 48\langle\text{Mg}+2\rangle + 34\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} + 11\langle\text{H}_2\text{O}\rangle >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:483.58(5SF)	4	CRV	32224

Reaction No. 81475							
$\text{Mg}_4\text{Si}_6\text{O}_{15}(\text{OH})_{26}\text{H}_2\text{O}(\text{s}) + 8\langle\text{H}+1\rangle + \text{H}_2\text{O} = 4\langle\text{Mg}+2\rangle + 6\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} >$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.94(4SF)	4	CRV	32224

Reaction No. 81485							
$\text{Mo}_3\text{Si}(\text{s}) + 16\langle\text{H}_2\text{O}\rangle = 3\langle\text{MoO}_4-2\rangle + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + 28\langle\text{H}+1\rangle + 22\langle\text{e}-1\rangle >$							

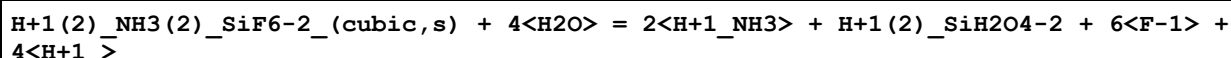
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.08(4SF)	4	CRV	32224

Reaction No. 81504



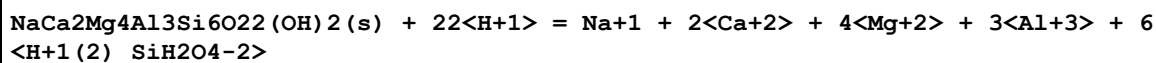
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.26(4SF)	4	CRV	32224

Reaction No. 81505



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.27(4SF)	4	CRV	32224

Reaction No. 81506



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:103.45(5SF)	4	CRV	32224

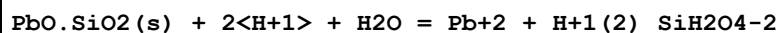
Reaction No. 81508



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.36(4SF)	2	CRV	32224

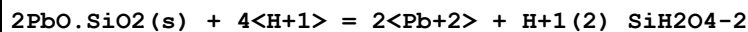
Note (1): H2O number changed (assumed corrected)

Reaction No. 81543



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.75(3SF)	4	CRV	32224

Reaction No. 81544



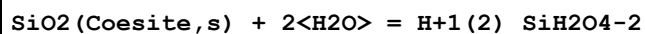
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.19(4SF)	4	CRV	32224

Reaction No. 81599



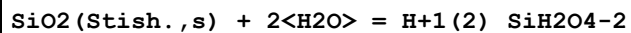
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.58 (3SF)	4	CRV	32224

Reaction No. 81600



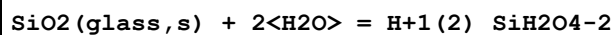
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.19 (3SF)	4	CRV	32224

Reaction No. 81601



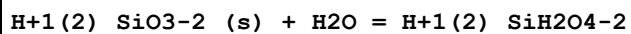
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.36 (3SF)	4	CRV	32224

Reaction No. 81602



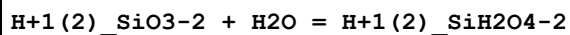
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.00 (3SF)	4	CRV	32224

Reaction No. 81603



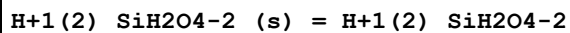
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.84 (3SF)	4	CRV	32224

Reaction No. 81604



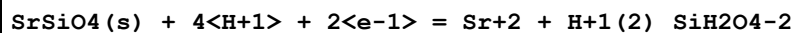
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.56 (3SF)	4	CRV	32224

Reaction No. 81605



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.44 (3SF)	4	CRV	32224

Reaction No. 81613



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.78 (4SF)	4	CRV	32224

Reaction No. 81709							
$\text{ZrSiO}_4(\text{s}) + 4\text{H}^+ = \text{Zr}^{4+} + 2\text{H}_2\text{O} + 2\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -14.8(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -5.64(3\text{SF})$	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
ENT	Nearest enthalpy & entropy changes
EOD	Calculated from data of others

EOR	Other Reaction Constants
ESC	Standard conditions anchor
ETR	Thermodynamic relationships
ETS	Ternary (statistical)
EVH	Van't Hoff Equation (dH = constant)
MAX	Anion exchange
MCL	Calorimetry
MCO	Colorimetry (often for measuring pH)
MGC	Gas chromatography
MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPT	Potentiometry
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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